PrepFusion

Electrical and Electronic Measurements

Detailed Solutions • 1991–2026

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Electrical and Electronic Measurements (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

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Contents

SOLUTIONS

Error Analysis

Topics

1. Static Characteristics of Measuring Device
2. Types of Error and Uncertainties of error

Q1.25.1 NAT 2025 1M Ans: 5

Given:

$$R_1 = 100\,\Omega, \qquad R_2 = 900\,\Omega$$

Each resistor has a maximum error of 5%.
We need the maximum percentage error in the equivalent resistance when the resistors are connected in parallel.

Method 1

For two resistors in parallel,

$$R = \frac{R_1 R_2}{R_1 + R_2}$$

Taking logarithms,

$$\ln R = \ln R_1 + \ln R_2 - \ln(R_1 + R_2)$$

Differentiating,

$$\frac{dR}{R} = \frac{dR_1}{R_1} + \frac{dR_2}{R_2} - \frac{dR_1 + dR_2}{R_1 + R_2}$$

For maximum error, use absolute values:

$$\frac{\Delta R}{R} = \frac{\Delta R_1}{R_1} + \frac{\Delta R_2}{R_2} - \frac{\Delta R_1 + \Delta R_2}{R_1 + R_2}$$

Now,

$$\Delta R_1 = 0.05 \times 100 = 5\,\Omega, \qquad \Delta R_2 = 0.05 \times 900 = 45\,\Omega$$

Substituting,

$$\frac{\Delta R}{R} = \frac{5}{100} + \frac{45}{900} - \frac{5 + 45}{100 + 900}$$
$$= 0.05 + 0.05 - 0.05 = 0.05$$

Hence, the maximum percentage error is

$$\boxed{5}$$

Method 2

Nominal equivalent resistance:

$$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{100 \times 900}{100 + 900} = \frac{90000}{1000} = 90\,\Omega$$

For maximum resistance, take both resistors at their maximum values:

$$R_{1,\text{max}} = 100 + 5 = 105\,\Omega$$

$$R_{2,\text{max}} = 900 + 45 = 945\,\Omega$$

Then,

$$R_{\text{max}} = \frac{105 \times 945}{105 + 945} = \frac{99225}{1050} = 94.5\,\Omega$$

So,

$$\Delta R = 94.5 - 90 = 4.5\,\Omega$$

Percentage error:

$$\frac{\Delta R}{R} \times 100 = \frac{4.5}{90} \times 100 = 5$$

Thus,

$$\boxed{5}$$

Key Points

- Logarithmic differentiation is a convenient method for error propagation in composite expressions.

Mistakes to be avoided

- While using logarithmic differentiation, ensure the negative sign for the denominator term is retained.
- Do not directly add 5% + 5%; the denominator term also contributes to the error expression.

Q1.22.1 MCQ 2022 1M Ans: B

For a balanced Wheatstone bridge,

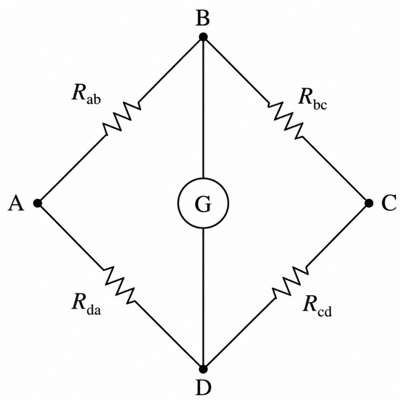

Fig. Solution Q1.22.1

$$\frac{R_{BC}}{R_{AB}} = \frac{R_{CD}}{R_{DA}}$$

Substituting the given nominal values,

$$R_{AB} = 1\,\text{k}\Omega = 1000\,\Omega, \quad R_{BC} = 100\,\Omega, \quad R_{DA} = 300\,\Omega$$

$$R_{CD} = \frac{100 \times 300}{1000} = 30\,\Omega$$

Now,

$$R_{CD} = \frac{R_{BC}R_{DA}}{R_{AB}}$$

For multiplication/division,

$$\frac{\Delta R_{CD}}{R_{CD}} = \frac{\Delta R_{BC}}{R_{BC}} + \frac{\Delta R_{DA}}{R_{DA}} + \frac{\Delta R_{AB}}{R_{AB}}$$

Given percentage errors:

$$R_{AB} : 2.1\%, \quad R_{BC} : 0.5\%, \quad R_{DA} : 0.4\%$$

Therefore,

$$\%\ \text{error} = 2.1 + 0.5 + 0.4 = 3.0\%$$

Absolute error:

$$\Delta R_{CD} = \frac{3}{100} \times 30 = 0.9\,\Omega$$

Hence,

$$R_{CD} = 30\,\Omega \pm 0.9\,\Omega$$

Therefore, the correct option is

$$\boxed{\textbf{B. } 30\,\Omega \pm 0.9\,\Omega}$$

Key Points

- For multiplication/division operations, percentage errors add directly.

Q1.20.1 NAT 2020 2M Ans: 5 to 6

The diode current equation is, taking $\eta = 1$,

$$I = I_0 e^{V_D/V_T}$$

We need the percentage uncertainty in current due to uncertainties in V_T and V_D.
Step 1: Take logarithm
Taking natural logarithm,

$$\ln I = \ln I_0 + \frac{V_D}{V_T}$$

Step 2: Sensitivity with respect to V_T
Differentiating w.r.t V_T,

$$\frac{1}{I}\frac{\partial I}{\partial V_T} = -\frac{V_D}{V_T^2}$$

Thus,

$$\left|\frac{\partial I}{\partial V_T}\right| = \frac{V_D}{V_T^2} I$$

Step 3: Sensitivity with respect to V_D
Differentiating w.r.t V_D,

$$\frac{1}{I}\frac{\partial I}{\partial V_D} = \frac{1}{V_T}$$

Hence,

$$\left|\frac{\partial I}{\partial V_D}\right| = \frac{I}{V_T}$$

Step 4: Use uncertainty propagation formula
For independent uncertainties,

$$W_I = \sqrt{\left(\frac{\partial I}{\partial V_T} W_{V_T}\right)^2 + \left(\frac{\partial I}{\partial V_D} W_{V_D}\right)^2}$$

Dividing by I,

$$\frac{W_I}{I} = \sqrt{\left(\frac{V_D}{V_T^2} W_{V_T}\right)^2 + \left(\frac{1}{V_T} W_{V_D}\right)^2}$$

Step 5: Substitute values

$$V_T = 29\,\text{mV} = 0.029\,\text{V}$$
$$W_{V_T} = 2\,\text{mV} = 0.002\,\text{V}$$
$$V_D = 0.02\,\text{V}$$
$$W_{V_D} = 1\,\text{mV} = 0.001\,\text{V}$$

Therefore,

$$\frac{W_I}{I} = \sqrt{\left(\frac{0.02}{(0.029)^2}(0.002)\right)^2 + \left(\frac{0.001}{0.029}\right)^2}$$

$$\frac{W_I}{I} = \sqrt{0.002262 + 0.001189}$$

$$= \sqrt{0.003451} = 0.0587$$

Percentage uncertainty:

$$0.0587 \times 100 = 5.87\%$$

$$\boxed{5.87}$$

Key Points

- If measured quantities x,y have uncertainties/errors, then how to calculate the uncertainty in a derived quantity q:
- **Addition:**

$$q = mx + ny \;\rightarrow\; \delta q = \sqrt{(m\delta x)^2 + (n\delta y)^2}$$

- **Subtraction:**

$$q = mx - ny \;\rightarrow\; \delta q = \sqrt{(m\delta x)^2 + (n\delta y)^2}$$

- **Multiplication:**

$$q = mxy \;\rightarrow\; \frac{\delta q}{q} = \sqrt{\left(\frac{\delta x}{x}\right)^2 + \left(\frac{\delta y}{y}\right)^2}$$

- **Division:**

$$q = m\frac{x}{y} \;\rightarrow\; \frac{\delta q}{q} = \sqrt{\left(\frac{\delta x}{x}\right)^2 + \left(\frac{\delta y}{y}\right)^2}$$

- **Power:**

$$q = x^m y^n \;\rightarrow\; \frac{\delta q}{q} = \sqrt{\left(m\frac{\delta x}{x}\right)^2 + \left(n\frac{\delta y}{y}\right)^2}$$

- **Logarithm:**

$$q = \ln(mx) \;\rightarrow\; \delta q = \frac{\delta x}{|x|}$$

- **Exponential:**

$$q = e^{mx} \;\rightarrow\; \frac{\delta q}{q} = m\delta x$$

Q1.17.1 NAT 2017 1M Ans: 4.0 to 4.1 ● ○ ○

Given measurements:

$$V = 220\,\text{V} \pm 1\%, \qquad I = 5\,\text{A} \pm 1\%, \qquad W = 555\,\text{W} \pm 2\%$$

For a single-phase load,

$$\cos\phi = \frac{W}{VI}$$

For multiplication/division,add the worst case errors of measured value:

$$\%\Delta(\cos\phi) = \%\Delta W + \%\Delta V + \%\Delta I$$

Substituting the given values,

$$\%\Delta(\cos\phi) = 2 + 1 + 1 = 4\%$$

Hence, the worst-case percentage error in power factor is

$$\boxed{4.0}$$

Q1.17.2 MCQ 2017 1M Ans: A ● ● ○

Let the equivalent resistance of two parallel resistors be

$$R = \frac{R_1 R_2}{R_1 + R_2}$$

where R_1, R_2 are measured resistances and $\Delta R_1, \Delta R_2$ are their uncertainties.
If a quantity depends on multiple variables,

$$R = f(R_1, R_2),$$

then the small change in R is

$$dR = \frac{\partial R}{\partial R_1} dR_1 + \frac{\partial R}{\partial R_2} dR_2.$$

For uncertainties,

$$\Delta R = \frac{\partial R}{\partial R_1} \Delta R_1 + \frac{\partial R}{\partial R_2} \Delta R_2.$$

But since uncertainties are independent random quantities, we use RMS combination:

$$(\Delta R)^2 = \left(\frac{\partial R}{\partial R_1}\Delta R_1\right)^2 + \left(\frac{\partial R}{\partial R_2}\Delta R_2\right)^2.$$

Thus,

$$\Delta R = \pm\sqrt{\left(\frac{\partial R}{\partial R_1}\Delta R_1\right)^2 + \left(\frac{\partial R}{\partial R_2}\Delta R_2\right)^2}.$$

Therefore, the correct option is

$$\boxed{\textbf{A.}\ \pm\sqrt{\left(\frac{\partial R}{\partial R_1}\Delta R_1\right)^2 + \left(\frac{\partial R}{\partial R_2}\Delta R_2\right)^2}}$$

Key Points

- **Why RMS method is used:**

  Suppose,

  $$q = x + y$$

  and both measurements have random uncertainties:

  $$\delta x, \quad \delta y$$

  If we directly add uncertainties,

  $$\delta q = \delta x + \delta y$$

  it assumes both errors always occur in the same direction simultaneously.
- Direct addition gives the **worst-case error**, but practical measurement errors are random in nature.
- One uncertainty may be positive while another may be negative, causing partial cancellation. Hence, uncertainties are combined statistically using RMS method.

Q1.15.1 NAT 2015 1M Ans: 11.75 to 12.25 ● ● ○

Given,

$$V_{bus} = 100\,\text{kV rms}, \qquad C_1 = 1\,\mu\text{F} \pm 10\%, \qquad C_2 = 9\,\mu\text{F} \pm 10\%$$

The output is taken across C_2.

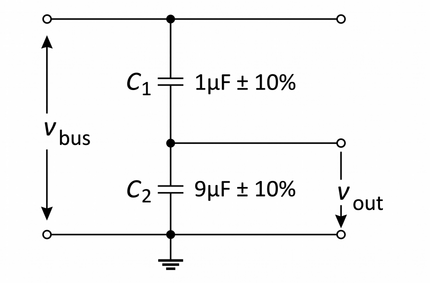

Fig. Solution Q1.15.1

For series capacitors,

$$V_{out} = V_{bus}\frac{X_{C2}}{X_{C1} + X_{C2}}$$

Since $X_C = \frac{1}{\omega C}$,

$$V_{out} = V_{bus}\frac{\frac{1}{C_2}}{\frac{1}{C_1} + \frac{1}{C_2}} = V_{bus}\frac{C_1}{C_1 + C_2}$$

To maximize V_{out}, take C_1 maximum and C_2 minimum:

$$C_{1,max} = 1.1\,\mu\text{F}, \qquad C_{2,min} = 8.1\,\mu\text{F}$$

Thus,

$$V_{out,max} = 100\frac{1.1}{1.1 + 8.1} = 100\frac{1.1}{9.2} = 11.9565\,\text{kV}$$

$$\boxed{11.96}$$

Mistakes to be avoided

- While writing voltage division expression across capacitor keep in mind that it is capacitive reactance and not capacitance.

Q1.06.1 MCQ 2006 2M Ans: D ● ○ ○

Given,

$$w = \frac{z}{xy}$$

For multiplication/division, maximum relative uncertainties add:

$$\frac{\Delta w}{w} = \frac{\Delta x}{x} + \frac{\Delta y}{y} + \frac{\Delta z}{z}$$

For x (Given x=80):

$$\Delta x = 0.5\% \times 80 = 0.4$$

$$\frac{\Delta x}{x} = \frac{0.4}{80} = 0.005 = 0.5\%$$

For y (Given y=20):

$$\Delta y = 1\% \times 100 = 1$$

$$\frac{\Delta y}{y} = \frac{1}{20} = 0.05 = 5\%$$

For z (Given z=50):

$$\Delta z = 1.5\% \times 50 = 0.75$$
$$\frac{\Delta z}{z} = \frac{0.75}{50} = 0.015 = 1.5\%$$

Thus,

$$\frac{\Delta w}{w} = 0.5\% + 5\% + 1.5\% = 7\%$$

Therefore, the correct option is

$$\boxed{\textbf{D.}\ \pm 7.0\%}$$

Mistakes to be avoided

- For y, the uncertainty is specified as **percentage of full-scale value**, not percentage of the measured reading.

Q1.99.1 MCQ 1999 2M Ans: A ● ● ○

For a PMMC voltmeter, the **figure of merit (sensitivity)** is

$$S = \frac{1}{I_{fs}}$$

$$S = \text{sensitivity in } \Omega/\text{V}, \qquad I_{fs} = \text{full-scale deflection current}$$

Given:
Both meters are

$$100\,\mu\text{A}$$

Converting into ampere,

$$100\,\mu\text{A} = 100 \times 10^{-6}\,\text{A} = 10^{-4}\,\text{A}$$

Step 1: Calculate sensitivity

$$S = \frac{1}{10^{-4}} = 10^4\,\Omega/\text{V} = 10\,\text{k}\Omega/\text{V}$$

both use the same meter movement:

$$100\,\mu\text{A}$$

Therefore, the correct option is

$$\boxed{\textbf{A.}\ 10\,\text{k}\Omega/\text{V and } 10\,\text{k}\Omega/\text{V}}$$

Key Points

- Sensitivity depends only on the **full-scale deflection current** I_{fs}, not on the voltage range.
- Even if voltmeters have different ranges such as:

$$10\,\text{V} \quad \text{and} \quad 100\,\text{V}$$

they will have the same sensitivity if the meter movement current I_{fs} is same.

Q1.97.1 NAT 1997 2M Ans: 111.11, 10 ● ○ ○

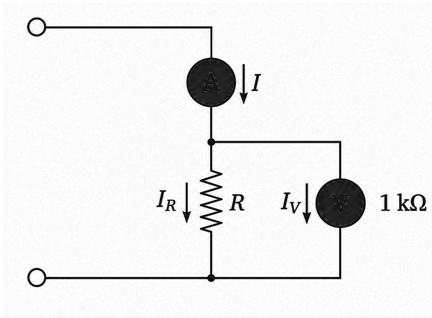

Fig. Solution Q1.97.1

Method 1

Given:

$$I = 1\,\text{A}, \quad V = 100\,\text{V}, \quad R_V(Voltmeter\ resistance) = 1\,\text{k}\Omega = 1000\,\Omega$$

The ammeter measures total current:

$$I = I_R + I_V$$

Voltmeter current:

$$I_V = \frac{V}{R_V} = \frac{100}{1000} = 0.1\,\text{A} \Rightarrow I_R = 1 - 0.1 = 0.9\,\text{A}$$

Actual resistance:

$$R = \frac{V}{I_R} = \frac{100}{0.9} = 111.11\,\Omega$$

Measured resistance:

$$R_m = \frac{V}{I} = \frac{100}{1} = 100\,\Omega$$

Percentage error:

$$\%\ \text{error} = \frac{111.11 - 100}{111.11} \times 100 = 10\%$$

Method 2

Given:

$$V = 100\,\text{V}, \quad I = 1\,\text{A}, \quad R_V = 1000\,\Omega$$

Using current division,

$$I_R = I \cdot \frac{R_V}{R + R_V}$$

Also,

$$V = I_R R$$

Substituting:

$$100 = \left(1 \cdot \frac{1000}{R + 1000}\right) R \Rightarrow 100(R + 1000) = 1000R$$

$$100R + 100000 = 1000R \Rightarrow 900R = 100000 \Rightarrow R = 111.11\,\Omega$$

Measured resistance:

$$R_m = \frac{V}{I} = \frac{100}{1} = 100\,\Omega$$

Percentage error:

$$\% \text{ error} = \frac{111.11 - 100}{111.11} \times 100 = 10\%$$

$$\boxed{R = 111.11\,\Omega, \qquad \% \text{ error} = 10\%}$$

Q1.95.1 NAT | 1995 | 1M | Ans: Guard Ring | ● ○ ○

In an ideal parallel plate capacitor, electric field lines are perfectly straight and confined between the plates.

Ideally,

$$E = \frac{V}{d}, \qquad C = \frac{\varepsilon A}{d}$$

These relations assume all electric field lines remain inside the plate area.

$$\boxed{\text{Fringing Effect}}$$

At the edges of capacitor plates, electric field lines spread outward instead of remaining straight.

How guard ring helps

A guard ring is a conducting ring placed around the sensing electrode and kept at the same potential as the main electrode. With guard ring, field lines terminate on the guard ring instead of spreading into space.

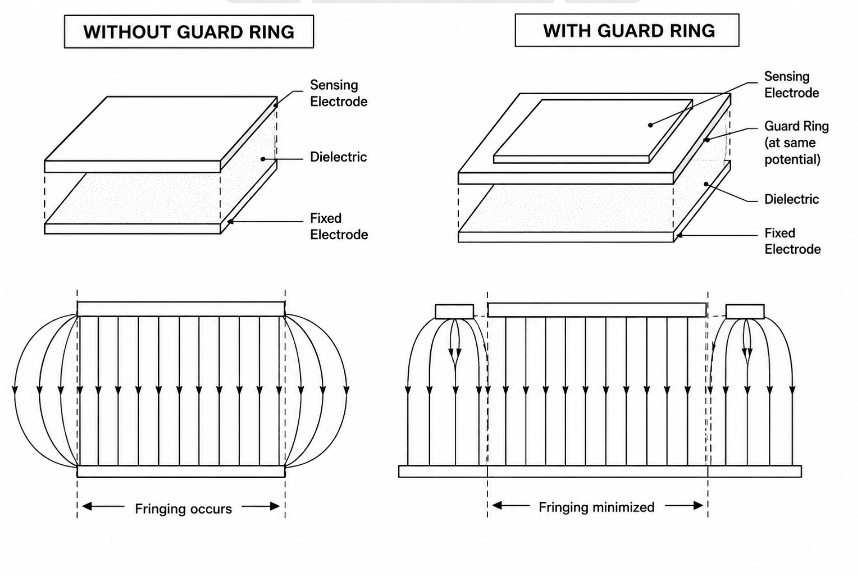

Fig. Solution Q1.95.1

Therefore answer is:

$$\boxed{\text{Fringing Effect}}$$

Key Points

- **Simple intuition:** Guard ring "captures" stray field lines before they spread outward, so the capacitor behaves more like an ideal parallel plate capacitor.
- Due to fringing:
  - effective plate area increases,
  - capacitance changes slightly,
  - measurement becomes inaccurate.

Q1.95.2 MCQ 1995 1M Ans: D ● ○ ○

Accuracy is specified as

$$\pm(\text{\% of full scale})$$

We compare percentage error while measuring 1 A.
For M_1:

$$\Delta I = \frac{0.10}{100} \times 20 = 0.02\text{ A} \Rightarrow \text{\% error} = \frac{0.02}{1} \times 100 = 2\%$$

For M_2:

$$\Delta I = \frac{0.20}{100} \times 10 = 0.02\text{ A} \Rightarrow \text{\% error} = 2\%$$

For M_3:

$$\Delta I = \frac{0.50}{100} \times 5 = 0.025\text{ A} \Rightarrow \text{\% error} = \frac{0.025}{1} \times 100 = 2.5\%$$

For M_4:

$$\Delta I = \frac{1.00}{100} \times 1 = 0.01\text{ A} \Rightarrow \text{\% error} = 1\%$$

Minimum error occurs for M4
Therefore, the correct option is

$$\boxed{\textbf{D.}\text{ M4}}$$

Key Points

- When accuracy is specified as $\pm(\text{\% of full scale})$, the absolute error depends on the full-scale range, not on the measured reading.
- If accuracy is specified as $\pm(\text{\% of full scale})$, then absolute error is:

$$\Delta I = \frac{\text{\% accuracy}}{100} \times (\text{full scale value})$$

- Lower full-scale range generally gives better accuracy for small measurements.

Mistakes to be avoided

- Do not calculate error using percentage of the measured value; it is given as percentage of full scale.
- Do not directly give answer on basis of accuracy data, sometimes it can be a trap.

Q1.94.1 MCQ 1994 1M Ans: B ● ○ ○

Precision means repeated measurements are close to each other, i.e., good repeatability.
Example:

$$10.1,\ 10.1,\ 10.1,\ 10.1$$

These measurements are highly precise.
Accuracy means the measured value is close to the true value.
Example:

$$\text{True value} = 12, \qquad \text{Measured value} = 12.01$$

This measurement is highly accurate.
A measurement can be precise but inaccurate.
Example:

$$\text{True value} = 12$$

$$\text{Measured Values} = 10.1,\ 10.1,\ 10.1$$

These readings are **highly precise**, but **inaccurate**.
because they are far from the true value.
Therefore,

$$\boxed{\text{Precision does NOT guarantee accuracy}}$$

So the answer is:

$$\boxed{\textbf{B. } False}$$

Q1.92.1 MCQ 1992 1M Ans: D ● ○ ○

Resistance measured using voltmeter-ammeter method:

$$R = \frac{V}{I}$$

For maximum possible error in division:

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

Given:

$$\frac{\Delta V}{V} = 2.4\%, \qquad \frac{\Delta I}{I} = 1.0\%$$

Therefore,

$$\frac{\Delta R}{R} = 2.4\% + 1.0\% = 3.4\%$$

Hence, maximum possible percentage error is

$$\boxed{3.4\%}$$

Therefore, the correct option is

$$\boxed{\textbf{D.}\ 3.4\%}$$

Key Points

- For multiplication/division, maximum fractional errors add directly:

$$\frac{\Delta q}{q} = \frac{\Delta x}{x} + \frac{\Delta y}{y}$$

Mistakes to be avoided

- Do not use RSS method when the question asks for maximum possible error.

DEBUG INFO

Chapter I

Error Analysis

Total Questions : 13

MCQ : 7

MSQ : 0

NAT : 6

SOLUTIONS

11

Analog Measuring Instruments

Topics

1. PMMC Type Instruments
2. Moving Iron & Dynamometer Type Instruments
3. Dynamometer Type Wattmeter
4. LPF Wattmeter
5. Two Wattmeter Measurement
6. Energy Meter
7. Hot wire Instruments & Electrostatic Voltmeters

Q2.24.1 NAT | 2024 | 2M | Ans: 692.80

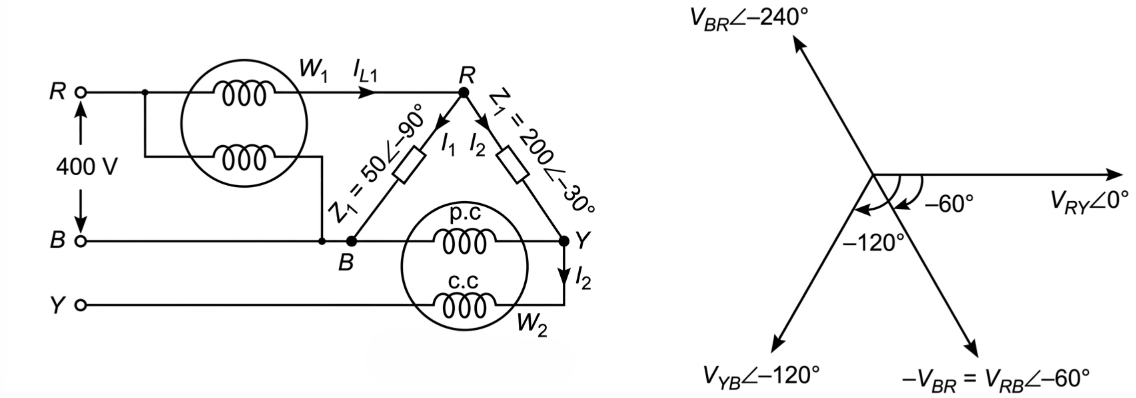

Fig. Solution Q2.24.1

Given:

$$Z_1 = 50\angle -90°\ \Omega, \qquad Z_2 = 200\angle -30°\ \Omega$$

Line voltages:

$$V_{RY} = 400\angle 0°, \quad V_{YB} = 400\angle -120°, \quad V_{BR} = 400\angle -240°$$

Hence,

$$V_{RB} = -V_{BR} = 400\angle -60°$$

Current through Z_1:

$$I_1 = \frac{V_{RB}}{Z_1} = \frac{400\angle -60°}{50\angle -90°} = 8\angle 30°$$

Current through Z_2:

$$I_2 = \frac{V_{RY}}{Z_2} = \frac{400\angle 0°}{200\angle -30°} = 2\angle 30°$$

Current through current coil of W_1:

$$I_{L1} = I_1 + I_2 = 8\angle 30° + 2\angle 30° = 10\angle 30°$$

For W_1:

$$V_{PC} = V_{RB} = 400\angle -60°$$

$$W_1 = 400 \times 10\cos(-60° - 30°) = 4000\cos(-90°) = 0$$

For W_2:

$$V_{YB} = 400\angle -120°, \qquad I_2 = 2\angle 30°$$

$$W_2 = 400 \times 2\cos(-120° - 30°) = 800\cos(-150°) = -692.8\text{ W}$$

Difference:

$$|W_1 - W_2| = |0 - (-692.8)|$$

$$\boxed{692.80\text{ W}}$$

Key Points

- In wattmeter calculations,

$$W = VI\cos\theta$$

where θ is the angle between pressure coil voltage and current coil current.

Mistakes to be avoided

- Read Question carefully, it says find "magnitude" of difference.

Q2.23.1 MCQ 2023 2M Ans: C

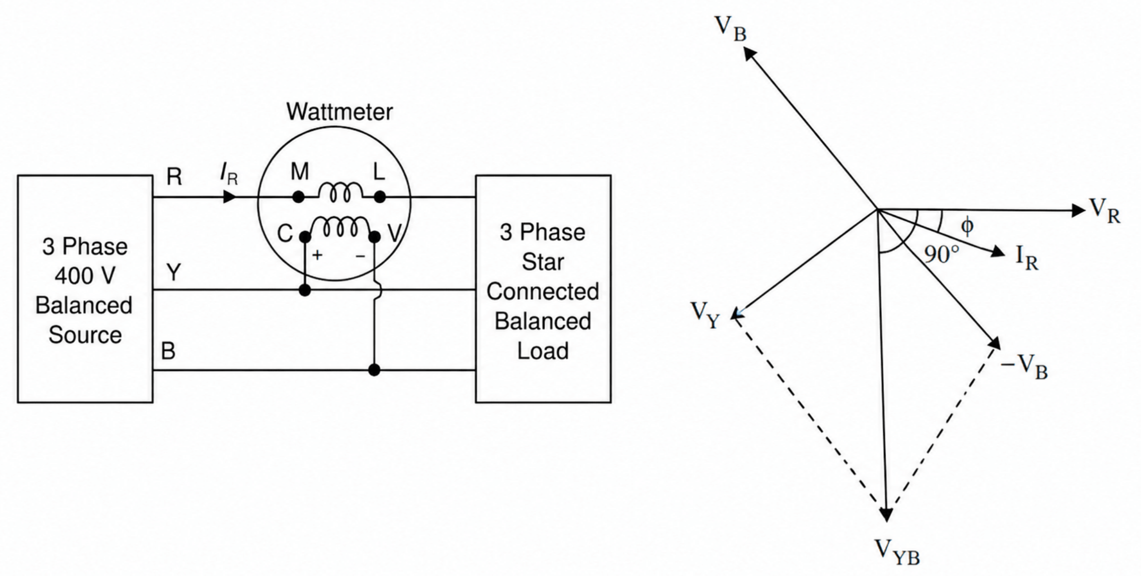

Fig. Solution Q2.23.1

Observation from wattmeter connection:
The current coil (CC) is connected in R-line, hence wattmeter measures:

$$I_{CC} = I_R$$

The pressure coil (PC) is connected between Y and B, therefore:

$$V_{PC} = V_{YB}$$

Thus wattmeter reading is:

$$W = V_{YB} I_R \cos(\text{angle between } V_{YB} \text{ and } I_R)$$

Why angle becomes $90^\circ - \phi$:
Assume phase sequence:

$$R \rightarrow Y \rightarrow B$$

Let,

$$V_R = V_p\angle 0^\circ, \quad V_Y = V_p\angle -120^\circ, \quad V_B = V_p\angle 120^\circ$$

Line voltage:

$$V_{YB} = V_Y - V_B = \sqrt{3}V_p\angle -90^\circ$$

Therefore, angle between V_{YB} and I_R becomes: $90^\circ - \phi$
Hence wattmeter reads:

$$W = V_L I_L \cos(90^\circ - \phi) = V_L I_L \sin\phi$$

Substituting values:

$$-400 = 400 \times 2 \times \sin\phi \Rightarrow -400 = 800\sin\phi$$

$$\sin\phi = -\frac{1}{2} \Rightarrow \phi = -30^\circ$$

Negative angle indicates leading power factor.
Therefore,

$$\text{PF} = \cos 30^\circ = 0.866$$

Hence, the correct option is

$$\boxed{\textbf{C.}\ 0.866 \text{ leading}}$$

Key Points

- In this connection, the current coil carries phase current I_R while the pressure coil measures line voltage V_{YB}. Since V_{YB} is shifted by 90° from phase voltage V_R, the angle in wattmeter becomes:

$$90^\circ - \phi$$

Hence wattmeter reads:

$$W = V_L I_L \cos(90^\circ - \phi) = V_L I_L \sin\phi$$

which is the expression for reactive power:

$$Q = V_L I_L \sin\phi$$

Therefore, this wattmeter connection measures **reactive power**.

Q2.20.1 NAT | 2020 | 1M | Ans: 1.732 | ● ○ ○

From KCL:

$$I_1 = I_2 + I_3$$

Given:

$$I_2 = 1\angle 10^\circ, \qquad I_3 = 1\angle 70^\circ$$

Thus,

$$I_1 = 1\angle 10^\circ + 1\angle 70^\circ$$

Converting into rectangular form:

$$I_1 = (\cos 10^\circ + j\sin 10^\circ) + (\cos 70^\circ + j\sin 70^\circ)$$

$$= 1.3268 + j1.1133$$

Magnitude:

$$|I_1| = \sqrt{(1.3268)^2 + (1.1133)^2} = \sqrt{3}$$

$$\boxed{1.732\ \text{A}}$$

Mistakes to be avoided

- Do not directly add magnitudes of phasor currents; add them in rectangular form.
- The phase angle 40° is irrelevant because an ammeter shows only the magnitude of current.

Q2.19.1 MCQ 2019 2M Ans: D

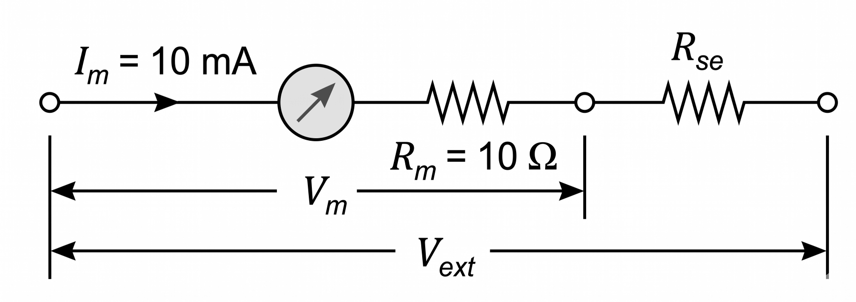

Fig. Solution Q2.19.1

Method 1: Direct voltage equation

Given:

$$R_m = 10\,\Omega, \qquad I_m = 10\,\text{mA} = 0.01\,\text{A}$$

Required voltmeter range:

$$V_{ext} = 100\,\text{V}$$

For full-scale deflection:

$$V_{ext} = I_m(R_m + R_{se})$$

Substituting,

$$100 = 0.01(10 + R_{se}) \Rightarrow 10 + R_{se} = \frac{100}{0.01} = 10000$$

$$R_{se} = 10000 - 10 = 9990\,\Omega$$

Hence,

$$\boxed{R_{se} = 9990\,\Omega}$$

Method 2: Multiplying factor

Voltage across original meter:

$$V_m = I_m R_m = 10\,\text{mA} \times 10\,\Omega = 0.1\,\text{V}$$

Thus original meter range is:

$$0 \longrightarrow 100\,\text{mV}$$

Required range:

$$0 \longrightarrow 100\,\text{V}$$

Multiplying factor:

$$m = \frac{V_{\text{ext}}}{V_m} = \frac{100}{0.1} = 1000$$

Series resistance:

$$R_{se} = R_m(m-1) = 10(1000-1) = 9990\,\Omega$$

Therefore, the correct option is

$$\boxed{\textbf{D.}\ 9990\,\Omega}$$

Key Points

- To convert a galvanometer into a voltmeter, a high series resistance is added.
- To convert a galvanometer into an ammeter, a low shunt resistance is connected in parallel.

Q2.18.1 MCQ 2018 1M Ans: D

Given that **one wattmeter reads half of the other** and both readings are **positive**, let

$$W_1 = \frac{1}{2} W_2$$

Let,

$$\phi = \text{power factor angle}$$

Method 1

For **Two-wattmeter method**,

$$\tan\phi = \sqrt{3}\,\frac{W_1 - W_2}{W_1 + W_2}$$

Let $W_1 = x$ and $W_2 = 2x$. Then,

$$\tan\phi = \sqrt{3}\,\frac{x - 2x}{x + 2x} = \sqrt{3}\,\frac{-x}{3x} = -\frac{1}{\sqrt{3}}$$

Hence,

$$\phi = 30^\circ$$

and therefore,

$$\text{PF} = \cos 30^\circ = 0.866$$

Method 2

For a balanced load,

$$W_1 = V_L I_L \cos(30^\circ + \phi), \qquad W_2 = V_L I_L \cos(30^\circ - \phi)$$

Given $W_1 = \frac{1}{2} W_2$,

$$\cos(30^\circ + \phi) = \frac{1}{2}\cos(30^\circ - \phi)$$

Using expansion and simplifying,

$$\tan\phi = \frac{1}{\sqrt{3}} \Rightarrow \phi = 30^\circ$$

Thus,

$$\text{PF} = \cos 30^\circ$$

Therefore, the correct option is

$$\boxed{\textbf{D. } 0.866}$$

Key Points

- If both wattmeter readings are positive, then:

$$\phi < 60^\circ$$

Mistakes to be avoided

- While using

$$\tan\phi = \sqrt{3}\,\frac{W_1 - W_2}{W_1 + W_2}$$

maintain correct sign convention.

Q2.18.2 MCQ 2018 2M Ans: A

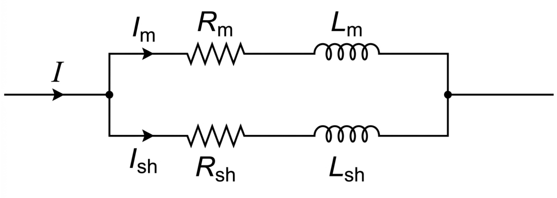

Fig. Solution Q2.18.2

Given: Moving iron ammeter range 0−1 A, $R_m = 50\,\text{m}\Omega$, $L_m = 0.1\,\text{mH}$. Required extended range: 0−10 A. For **frequency-independent** operation, the meter branch and shunt branch must satisfy

$$\frac{R_m}{L_m} = \frac{R_{sh}}{L_{sh}}$$

$$Time\ constant,\ \tau = \frac{L_m}{R_m} = \frac{0.1\times 10^{-3}}{50\times 10^{-3}} = 2\times 10^{-3}\,\text{s} = 2\,\text{ms}$$

For full-scale current,

$$I_m = 1\,\text{A}, \qquad I = 10\,\text{A} \Rightarrow I_{sh} = 9\,\text{A}$$

Since the voltage across meter branch and shunt branch is same,

$$I_m R_m = I_{sh} R_{sh}$$

$$1\times 50\,\text{m}\Omega = 9R_{sh} \Rightarrow R_{sh} = \frac{50}{9}\,\text{m}\Omega = 5.55\,\text{m}\Omega$$

Hence,

$$\boxed{\tau = 2\,\text{ms}, \qquad R_{sh} = 5.55\,\text{m}\Omega}$$

Therefore, the correct option is

$$\boxed{\textbf{A.}\ 2, 5.55}$$

Key Points

- Equal time constants mean both branches react equally to frequency change. Therefore, current division remains constant with frequency. Hence, the meter reads correctly at all frequencies.

Q2.17.1 MCQ 2017 2M Ans: D

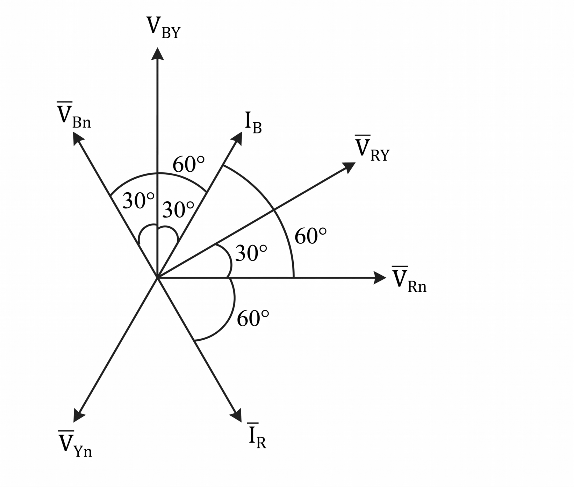

Fig. Solution Q2.17.1

Given:

$$S = 3464\ \text{VA}, \qquad V_L = 400\ \text{V}$$

Using,

$$S = \sqrt{3}\, V_L I_L \Rightarrow 3464 = \sqrt{3} \times 400 \times I_L \Rightarrow \boxed{I_L = 5\ \text{A}}$$

Hence,

$$I_R = I_Y = I_B = 5\ \text{A}$$

Initially, with switch at N: Each wattmeter reads phase power: 577.35 W, therefore

$$Total\ Power,\ P = 3 \times 577.35 = 1732\ \text{W}$$

$$P = \sqrt{3}\, V_L I_L \cos\phi \Rightarrow 1732 = \sqrt{3} \times 400 \times 5 \cos\phi$$

$$1732 = 3464 \cos\phi \Rightarrow \cos\phi = 0.5 \Rightarrow \boxed{\phi = 60^\circ}$$

Switch moved to position Y, now pressure coils use line voltages.

For W_1:

$$\angle(V_{RY}, I_R) = 90^\circ \Rightarrow W_1 = 400 \times 5 \cos 90^\circ = 0$$

For W_2:

When switch moves from N to Y, the pressure coil of W_2 gets short-circuited.
Hence,

$$V_{PC(2)} = 0 \Rightarrow W_2 = 0$$

For W_3:

$$\angle(V_{BY}, I_B) = 30^\circ \Rightarrow W_3 = 400 \times 5 \cos 30^\circ = 1732\ \text{W}$$

Therefore,

$$\boxed{W_1 = 0, \qquad W_2 = 0, \qquad W_3 = 1732\ \text{W}}$$

Hence, the correct option is

$$\boxed{\textbf{D. } W_1 = W_2 = 0 \text{ and } W_3 = 1732\ \text{W}}$$

Mistakes to be avoided

- Do not use phase voltage in place of line voltage in three-phase power formula.
- Carefully identify pressure coil connection after switch movement, do not only use phasor diagram to calculate power.

Q2.16.1 NAT 2016 2M Ans: 0.312 ●●○

Given,

$$v(t) = 100 + 10\sin\omega t - 5\sin 3\omega t$$

Moving-coil voltmeter (PMMC):
A moving-coil instrument responds to the **average value**.
Since,

$$\overline{\sin\omega t} = 0, \qquad \overline{\sin 3\omega t} = 0$$

Hence,

$$V_1 = 100\,\text{V}$$

Moving-iron voltmeter:
A moving-iron instrument responds to the **RMS value**.
Therefore,

$$V_2 = \sqrt{100^2 + \left(\frac{10}{\sqrt{2}}\right)^2 + \left(\frac{5}{\sqrt{2}}\right)^2}$$

$$= \sqrt{10000 + 50 + 12.5} = \sqrt{10062.5} = 100.312\,\text{V}$$

Hence,

$$V_2 - V_1 = 100.312 - 100 = 0.312\,\text{V}$$

$$\boxed{0.31}$$

Key Points

- PMMC instruments respond to the average value, while moving-iron instruments respond to RMS value.
- **Visual difference:** Moving-iron instrument scale is non-linear, whereas PMMC instrument scale is linear.

Q2.16.2 NAT 2016 2M Ans: 2 ●○○

Given:

Meter constant:

$$1200\ \text{rev/kWh}$$

This means,

$$1200\ \text{revolutions} = 1\,\text{kWh}$$

Given:

$$20\ \text{revolutions in } 30\,\text{s}$$

Energy consumed in 20 revolutions:

$$E = \frac{20}{1200} = \frac{1}{60} = 0.01667\,\text{kWh}$$

Time:

$$30\,\text{s} = \frac{30}{3600} = \frac{1}{120}\,\text{h}$$

Power:

$$P = \frac{E}{t} = \frac{1/60}{1/120} = 2\,\text{kW}$$

Hence,

$$\boxed{2}$$

Key Points

- Meter constant gives the number of revolutions per unit energy:

$$\text{Meter constant} = \frac{\text{revolutions}}{\text{energy}}$$

Mistakes to be avoided

- Be careful with units as time is given in seconds and Energy in KWh

Q2.15.1 | NAT | 2015 | 1M | Ans: 0.222

Given:

$$\text{Ammeter range} = 0 - 50\,\text{A}$$
$$V = 0.1\,\text{V at full scale}$$

Required new range: 0-500 A

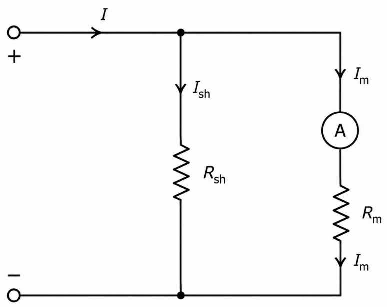

Fig. Solution Q2.15.1

Hence,

$$I_m = 50\,\text{A}, \qquad I = 500\,\text{A}$$

Shunt current:

$$I_{sh} = 500 - 50 = 450\,\text{A}$$

Since meter and shunt are in parallel,

$$V_{sh} = V_m = 0.1\,\text{V}$$

Therefore,

$$R_{sh} = \frac{V}{I_{sh}} = \frac{0.1}{450} = 2.22 \times 10^{-4}\,\Omega$$

Converting into milliohms,

$$R_{sh} = 0.222\,\text{m}\Omega$$

Hence,

$$\boxed{0.22}$$

Mistakes to be avoided

- Final answer is asked in mΩ, so read question carefully.

Q2.15.2 MCQ 2015 1M Ans: C ● ● ○

P. Permanent Magnet Moving Coil (PMMC)

For PMMC instrument,

$$T_d \propto BI$$

Torque depends on current direction.
For AC, current reverses every half cycle, hence torque also reverses every half cycle. Therefore,

$$T_{\text{avg}} = 0$$

Thus PMMC works only for:

$$\boxed{\text{DC only}}$$

Hence,

$$P \rightarrow 1$$

Q. Moving iron connected through Current Transformer (CT)

For moving iron instrument,

$$T_d \propto I^2$$

Hence,

$$(+I)^2 = (-I)^2$$

So MI instrument alone can measure AC and DC.
But CT works only on AC because:

$$e = N\frac{d\phi}{dt}$$

For DC,

$$\frac{d\phi}{dt} = 0$$

Thus,

$$\boxed{\text{MI + CT works only on AC}}$$

Hence,

$$Q \rightarrow 2$$

R. Rectifier instrument

A rectifier instrument uses:

$$\text{AC/DC} \rightarrow \text{Rectifier} \rightarrow \text{PMMC}$$

Rectifier converts AC into DC before feeding PMMC.
Thus it can measure:

$$\boxed{\text{AC and DC}}$$

Hence,

$$R \rightarrow 3$$

S. Electrodynamometer

For electrodynamometer,

$$T_d \propto I_1 I_2$$

For AC,

$$(-I_1)(-I_2) = I_1 I_2$$

Hence torque remains positive for both AC and DC.
Therefore,

$$\boxed{\text{AC and DC}}$$

Hence,

$$S \rightarrow 3$$

Final matching:

$$P \rightarrow 1, \quad Q \rightarrow 2, \quad R \rightarrow 3, \quad S \rightarrow 3$$

Therefore, the correct option is

$$\boxed{\textbf{C.}\ \text{P-1, Q-2, R-3, S-3}}$$

Key Points

- **CT** → Current Transformer. Used to step down high AC current to a measurable low current for protection and measurement. Works only with AC (requires changing magnetic flux).

Q2.15.3 MCQ | 2015 | 1M | Ans: A | ● ○ ○

Given:

$$P = 5\,\text{kW}, \qquad \text{PF} = 0.707 \Rightarrow \cos\phi = 0.707 \Rightarrow \phi = 45^\circ$$

Method 1

For two-wattmeter method,

$$W_1 + W_2 = P = 5$$

and

$$\tan\phi = \sqrt{3}\,\frac{W_1 - W_2}{W_1 + W_2}$$

Substituting $\phi = 45^\circ$,

$$1 = \sqrt{3}\,\frac{W_1 - W_2}{5} \Rightarrow W_1 - W_2 = \frac{5}{\sqrt{3}} = 2.887$$

Now,

$$W_1 + W_2 = 5, \qquad W_1 - W_2 = 2.887$$
$$2W_1 = 7.887 \Rightarrow W_1 = 3.944\,\text{kW}$$
$$W_2 = 5 - 3.944 = 1.056\,\text{kW}$$

Method 2

For a balanced load,

$$W_1 = V_L I_L \cos(30^\circ + \phi), \qquad W_2 = V_L I_L \cos(30^\circ - \phi)$$

With $\phi = 45^\circ$,

$$W_1 : W_2 = \cos 75^\circ : \cos 15^\circ = 0.2588 : 0.9659 = 1 : 3.732$$

Also,

$$W_1 + W_2 = 5$$

Let $W_1 = x$, then $W_2 = 3.732x$. So,

$$x + 3.732x = 5 \Rightarrow x = 1.056$$
$$W_1 = 1.056\,\text{kW}, \qquad W_2 = 3.944\,\text{kW}$$

$$\boxed{\textbf{A.}\ 3.94\text{ kW and }1.06\text{ kW}}$$

Q2.15.4 NAT 2015 2M Ans: 0.15 ● ● ○

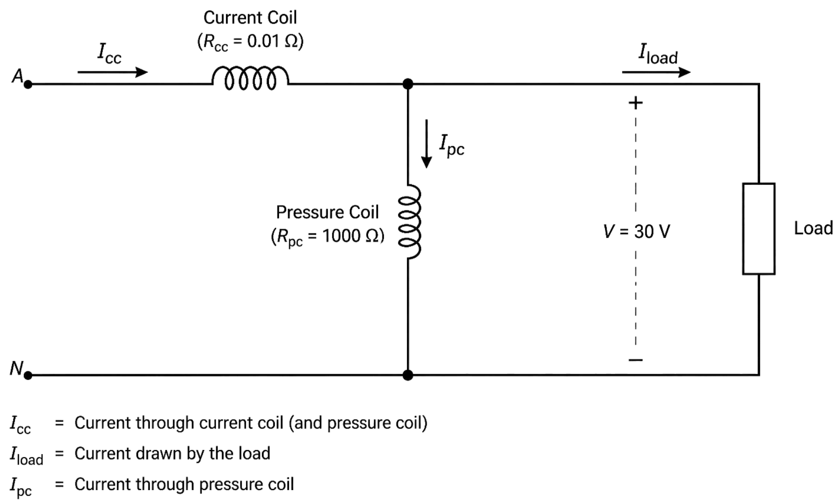

Fig. Solution Q2.15.4

From the figure, the current coil (CC) is in series with the load and the pressure coil (PC) is connected across the load only. Therefore, the wattmeter measures

$$P_m = P_{\text{load}} + P_{\text{PC}}$$

True load power:

$$P_{\text{true}} = VI\cos\phi = 30 \times 25 \times 0.8 = 600\text{ W}$$

Pressure coil power loss:

$$P_{\text{PC}} = \frac{V^2}{R_P} = \frac{30^2}{1000} = 0.9\text{ W}$$

Hence,

$$P_m = 600 + 0.9 = 600.9\text{ W}$$

Percentage error:

$$\%\text{ error} = \frac{600.9 - 600}{600} \times 100 = 0.15\%$$

$$\boxed{0.15}$$

Key Points

- Since the added loss is positive, the measured power is slightly higher than the true power.

Mistakes to be avoided

- Do not include current coil loss here; only pressure coil loss affects the reading.

Q2.14.1 NAT 2014 1M Ans: 2 ● ○ ○

Given:

$$\cos\phi = 0.2, \qquad V = 150\,\text{V}, \qquad I = 10\,\text{A}$$

Full-scale reading:

$$150 \text{ divisions}$$

Maximum measurable power:

$$P_{max} = VI\cos\phi = 150 \times 10 \times 0.2 = 300\,\text{W}$$

Thus, 150 divisions correspond to 300 W.
Therefore, multiplying factor:

$$\text{MF} = \frac{300}{150} = 2\,\text{W/division}$$

Hence,

$$\boxed{2}$$

Key Points

- LPF wattmeters are commonly used in the **open-circuit test of transformers** because transformer no-load power factor is very low.

Q2.14.2 NAT 2014 1M Ans: 0.8 ● ○ ○

Given:

$$W_1 = 100\,\text{W}, \qquad W_2 = 250\,\text{W}$$

For two-wattmeter method,

$$\tan\phi = \sqrt{3}\,\frac{W_1 - W_2}{W_1 + W_2}$$

Taking the larger wattmeter first,

$$\tan\phi = \sqrt{3}\,\frac{250 - 100}{250 + 100}$$

$$= 1.732 \times \frac{150}{350} = 0.742$$

Hence,

$$\phi = \tan^{-1}(0.742) \approx 36.6^\circ$$

Therefore,

$$\text{PF} = \cos\phi = \cos 36.6^\circ \approx 0.8$$

$$\boxed{0.8}$$

Q2.14.3 MCQ 2014 1M Ans: C ● ○ ○

Given:

$$W_1 = 2W_2$$

For two-wattmeter method,

$$\tan\phi = \sqrt{3}\,\frac{W_1 - W_2}{W_1 + W_2}$$

Substituting $W_1 = 2W_2$,

$$\tan\phi = \sqrt{3}\,\frac{2W_2 - W_2}{2W_2 + W_2} = \sqrt{3}\,\frac{W_2}{3W_2} = \frac{1}{\sqrt{3}}$$

Hence,

$$\phi = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = 30^\circ$$

Converting into radians,

$$30^\circ = \frac{\pi}{6}$$

Hence, the correct option is

$$\boxed{\textbf{C. } \phi = \frac{\pi}{6}}$$

Key Points

- In AC circuits, the **load impedance angle** is the same as the **power factor angle**:

$$Z = |Z|\angle\phi$$

Q2.14.4 MCQ 2014 2M Ans: A ● ● ○

For a moving-iron voltmeter, the reading is the RMS value.

Given,

$$v(t) = \begin{cases} 1 + \sin\omega t, & 0 \le \omega t < 6\pi \\ -1 + \sin\omega t, & 6\pi \le \omega t < 12\pi \end{cases}$$

Let

$$\theta = \omega t$$

Then,

$$V_{\text{rms}}^2 = \frac{1}{12\pi}\left[\int_0^{6\pi} (1 + \sin\theta)^2\, d\theta + \int_{6\pi}^{12\pi} (-1 + \sin\theta)^2\, d\theta\right]$$

Expanding,

$$(1 + \sin\theta)^2 = 1 + 2\sin\theta + \sin^2\theta$$

$$(-1 + \sin\theta)^2 = 1 - 2\sin\theta + \sin^2\theta$$

Therefore,

$$V_{\text{rms}}^2 = \frac{1}{12\pi}\left[\int_0^{6\pi} (1 + 2\sin\theta + \sin^2\theta)\, d\theta + \int_{6\pi}^{12\pi} (1 - 2\sin\theta + \sin^2\theta)\, d\theta\right]$$

Now,

$$\int_0^{6\pi} \sin\theta\, d\theta = \int_{6\pi}^{12\pi} \sin\theta\, d\theta = 0$$

Also,

$$\int_0^{6\pi} 1\, d\theta = 6\pi, \qquad \int_0^{6\pi} \sin^2\theta\, d\theta = 3\pi$$

Similarly,

$$\int_{6\pi}^{12\pi} 1\, d\theta = 6\pi, \qquad \int_{6\pi}^{12\pi} \sin^2\theta\, d\theta = 3\pi$$

Hence,

$$V_{\text{rms}}^2 = \frac{1}{12\pi}\left[(6\pi + 3\pi) + (6\pi + 3\pi)\right]$$
$$= \frac{18\pi}{12\pi} = \frac{3}{2}$$

Therefore,

$$V_{\text{rms}} = \sqrt{\frac{3}{2}}$$

$$\boxed{\textbf{A. } \sqrt{\frac{3}{2}}}$$

Key Points

- RMS value of any periodic signal is:

$$V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt}$$

Q2.14.5

NAT | 2014 | 2M | Ans: 10.14 | ● ● ○

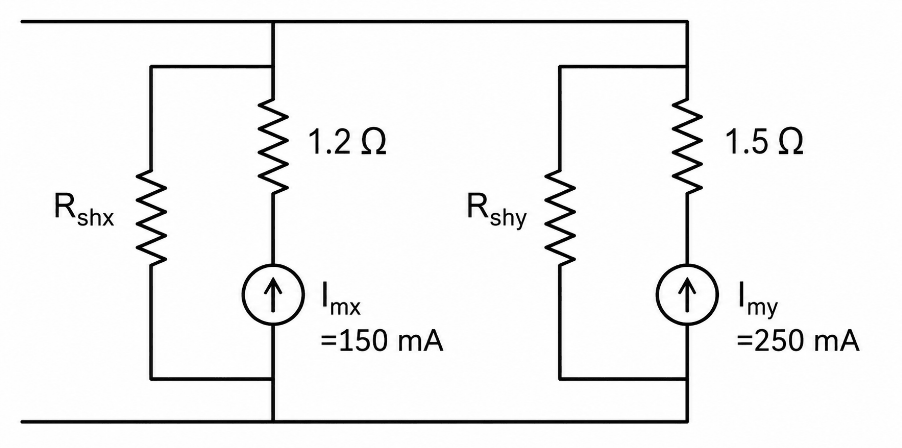

Fig. Solution Q2.14.5

Ammeter X:

$$R_X = 1.2\,\Omega, \qquad I_{mx} = 0.15\,\text{A}$$

At full-scale 15 A,

$$V_X = I_{mx}R_X = 0.15 \times 1.2 = 0.18\,\text{V}$$

So the equivalent resistance of extended ammeter X is

$$R_{\text{eq}X} = \frac{V_X}{15} = \frac{0.18}{15} = 0.012\,\Omega$$

Ammeter Y:

$$R_Y = 1.5\,\Omega, \qquad I_{my} = 0.25\,\text{A}$$

At full-scale 15 A,

$$V_Y = I_{my}R_Y = 0.25 \times 1.5 = 0.375\,\text{V}$$

So the equivalent resistance of extended ammeter Y is

$$R_{eqY} = \frac{V_Y}{15} = \frac{0.375}{15} = 0.025\,\Omega$$

Since the two extended ammeters are in **parallel**, current division gives

$$I_X = 15\frac{R_{eqY}}{R_{eqX} + R_{eqY}} = 15\frac{0.025}{0.012 + 0.025} = 10.135\text{ A}$$

Hence, ammeter X reads

$$\boxed{10.14}$$

Mistakes to be avoided

- Read the question carefully to know how the ammeters are connected.

Q2.14.6

NAT | 2014 | 1M | Ans: 57.7 | ● ○ ○

A moving-iron (MI) voltmeter measures the **RMS value** since

$$T_d \propto I^2$$

From the graph, the sawtooth waveform rises linearly from

$$0 \rightarrow 100\text{ V}$$

during

$$T = 20\text{ ms}$$

Hence,

$$v(t) = \frac{100}{T}t, \qquad 0 \le t \le T$$

RMS value:

$$V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt}$$

Substituting,

$$V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T \left(\frac{100}{T}t\right)^2 dt}$$

$$= \sqrt{\frac{10000}{T^3}\int_0^T t^2 dt}$$

Since,

$$\int_0^T t^2 dt = \frac{T^3}{3}$$

Therefore,

$$V_{\text{rms}} = \sqrt{\frac{10000}{3}} = \frac{100}{\sqrt{3}} = 57.7\text{ V}$$

Hence, MI voltmeter reading is

$$\boxed{57.7\text{ V}}$$

Q2.14.7 NAT 2014 2M Ans: 3.2 ● ● ○

Given for the **PMMC ammeter**:

$$n = 100, \qquad B = 0.2\,\mathrm{Wb/m^2}, \qquad A = 80\,\mathrm{mm^2} = 80 \times 10^{-6}\,\mathrm{m^2}$$

For the **electrodynamometer ammeter**:

$$\frac{dM}{d\theta} = 0.5\,\mathrm{mH/rad} = 0.5 \times 10^{-3}\,\mathrm{H/rad}$$

Since both meters have the same spring constant, their deflections are equal when

$$\theta_1 = \theta_2$$

PMMC meter:

$$T_d = nBAI, \qquad K\theta_1 = nBAI$$

Electrodynamometer meter:

$$T_d = I^2\frac{dM}{d\theta}, \qquad K\theta_2 = I^2\frac{dM}{d\theta}$$

For equal deflection,

$$K\theta_1 = K\theta_2 \Rightarrow nBAI = I^2\frac{dM}{d\theta} \Rightarrow I = \frac{nBA}{dM/d\theta}$$

Substituting,

$$I = \frac{100 \times 0.2 \times 80 \times 10^{-6}}{0.5 \times 10^{-3}}$$

$$I = \frac{1.6 \times 10^{-3}}{0.5 \times 10^{-3}} = 3.2\,\mathrm{A}$$

$$\boxed{3.2\,\mathrm{A}}$$

Key Points

- PMMC deflection grows linearly with current: $\theta \propto I$

  Electrodynamometer deflection grows quadratically with current: $\theta \propto I^2$

  At one particular current, these two curves intersect. For this question, that current is: $\boxed{3.2\,\mathrm{A}}$

Q2.13.1 MCQ 2013 1M Ans: A ● ○ ○

For an **ideal PMMC voltmeter**, the reading is proportional to the **average (DC) value**, not the RMS value.

Given,

$$v(t) = 14.14\sin(314t)$$

so the peak value is

$$V_m = 14.14\,\mathrm{V}$$

The diode is ideal, so the waveform becomes **half-wave rectified**. Also,

$$100\,\mathrm{k\Omega} \gg 1\,\mathrm{k\Omega}$$

hence the voltage division effect is negligible.

For a half-wave rectified sine wave,

$$V_{\text{avg}} = \frac{V_m}{\pi}$$

Therefore,

$$V_{\text{avg}} = \frac{14.14}{\pi} = 4.50\,\text{V}$$

Hence, the correct option is

$$\boxed{\textbf{A.}\ 4.46}$$

Key Points

- For the diode :

$$\text{Positive half-cycle} \longrightarrow \text{ON}$$

$$\text{Negative half-cycle} \longrightarrow \text{OFF}$$

- Since the input waveform is sinusoidal, we can directly use the standard half-wave rectified average value:

$$V_{\text{avg}} = \frac{V_m}{\pi}$$

Q2.13.2 MCQ 2013 1M Ans: D ● ○ ○

Since all are **moving-iron voltmeters**, they measure RMS magnitude.

Given,

$$X_C = -j1\,\Omega, \qquad X_L = j2\,\Omega$$

Since both are in series, the same current I flows through them.

Voltage across capacitor:

$$V_1 = I(-j1) \Rightarrow |V_1| = I$$

Hence,

$$V_1 = I$$

Voltage across inductor:

$$V_2 = I(j2) \Rightarrow |V_2| = 2I$$

Hence,

$$V_2 = 2I$$

Total impedance:

$$Z = -j1 + j2 = j1$$

Therefore,

$$V = IZ = I(j1) \Rightarrow |V| = I$$

Hence,

$$V = I$$

And,

$$V_2 - V_1 = 2I - I = I (= V)$$

Therefore, the correct option is

$$\boxed{\textbf{D.}\ V = V_2 - V_1}$$

Q2.12.1 MCQ 2012 1M Ans: C ● ● ○

Given,

$$v(t) = E_1 \sin \omega t + E_3 \sin 3\omega t, \qquad i(t) = I_1 \sin(\omega t - \phi_1) + I_3 \sin(3\omega t - \phi_3) + I_5 \sin 5\omega t$$

Average power is

$$P = \overline{v(t)i(t)}$$

On expanding $v(t)i(t)$, all cross-frequency terms such as

$$\sin \omega t \sin 3\omega t, \qquad \sin \omega t \sin 5\omega t, \qquad \sin 3\omega t \sin 5\omega t$$

have zero average over one cycle.
Instantaneous power:

$$p(t) = v(t)i(t)$$

Substituting,

$$p(t) = \left(E_1 \sin \omega t + E_3 \sin 3\omega t\right)\left[I_1 \sin(\omega t - \phi_1) + I_3 \sin(3\omega t - \phi_3) + I_5 \sin 5\omega t\right]$$

Expanding,

$$\begin{aligned} p(t) = {} & E_1 I_1 \sin \omega t \sin(\omega t - \phi_1) + E_1 I_3 \sin \omega t \sin(3\omega t - \phi_3) + E_1 I_5 \sin \omega t \sin 5\omega t \\ & + E_3 I_1 \sin 3\omega t \sin(\omega t - \phi_1) + E_3 I_3 \sin 3\omega t \sin(3\omega t - \phi_3) + E_3 I_5 \sin 3\omega t \sin 5\omega t \end{aligned}$$

Now use

$$\sin A \sin B = \frac{1}{2}\left[\cos(A - B) - \cos(A + B)\right]$$

First term:

$$E_1 I_1 \sin \omega t \sin(\omega t - \phi_1) = \frac{1}{2} E_1 I_1 \left[\cos \phi_1 - \cos(2\omega t - \phi_1)\right]$$

$$\Rightarrow \overline{E_1 I_1 \sin \omega t \sin(\omega t - \phi_1)} = \frac{1}{2} E_1 I_1 \cos \phi_1$$

Second term:

$$E_1 I_3 \sin \omega t \sin(3\omega t - \phi_3) = \frac{1}{2} E_1 I_3 \left[\cos(2\omega t - \phi_3) - \cos(4\omega t - \phi_3)\right]$$

$$\Rightarrow \text{average} = 0$$

Third term:

$$E_1 I_5 \sin \omega t \sin 5\omega t = \frac{1}{2} E_1 I_5 \left[\cos 4\omega t - \cos 6\omega t\right]$$

$$\Rightarrow \text{average} = 0$$

Fourth term:

$$E_3 I_1 \sin 3\omega t \sin(\omega t - \phi_1) = \frac{1}{2} E_3 I_1 \left[\cos(2\omega t + \phi_1) - \cos(4\omega t - \phi_1)\right]$$

$$\Rightarrow \text{average} = 0$$

Fifth term:

$$E_3 I_3 \sin 3\omega t \sin(3\omega t - \phi_3) = \frac{1}{2} E_3 I_3 \left[\cos \phi_3 - \cos(6\omega t - \phi_3)\right]$$

$$\Rightarrow \overline{E_3 I_3 \sin 3\omega t \sin(3\omega t - \phi_3)} = \frac{1}{2} E_3 I_3 \cos \phi_3$$

Sixth term:

$$E_3 I_5 \sin 3\omega t \sin 5\omega t = \frac{1}{2} E_3 I_5 \left[\cos 2\omega t - \cos 8\omega t\right]$$

$$\Rightarrow \text{average} = 0$$

Thus the average power is

$$P = \frac{1}{2}E_1 I_1 \cos\phi_1 + \frac{1}{2}E_3 I_3 \cos\phi_3$$

Therefore, the correct option is

$$\boxed{\textbf{C. } \frac{1}{2}\left(E_1 I_1 \cos\phi_1 + E_3 I_3 \cos\phi_3\right)}$$

Key Points

- For the same-frequency term,

$$E_1 I_1 \sin\omega t \sin(\omega t - \phi_1) = \frac{1}{2}E_1 I_1 \left[\cos\phi_1 - \cos(2\omega t - \phi_1)\right]$$

Hence, after taking average $\overline{\cos(2\omega t - \phi_1) = 0}$ over a complete cycle.

$$\overline{E_1 I_1 \sin\omega t \sin(\omega t - \phi_1)} = \frac{1}{2}E_1 I_1 \cos\phi_1$$

- For the cross-frequency term,

$$\frac{1}{2}E_1 I_3 \left[\cos(2\omega t - \phi_3) - \cos(4\omega t - \phi_3)\right]$$

both cosine terms have zero average over one cycle, so the average is

$$0$$

- Therefore, only same-frequency terms contribute to average power.

Mistakes to be avoided

- The 5^{th} harmonic current does not contribute because there is no corresponding 5^{th} harmonic voltage.
- Do not use RMS power formula directly here; the signals are expressed in peak form.

Q2.12.2 MCQ | 2012 | 2M | Ans: D | ● ● ○

For an analog voltmeter, the reading is proportional to the meter current:

$$I_m = \frac{V}{R_v + R_s}$$

Hence, for the same applied voltage,

$$\boxed{\text{Reading} \propto \frac{1}{R_v + R_s}}$$

Given:

$$R_{s1} = 20\,\text{k}\Omega, \quad V_1 = 440\,\text{V}$$

$$R_{s3} = 80\,\text{k}\Omega, \quad V_2 = 352\,\text{V}$$

So,

$$\frac{440}{352} = \frac{R_v + 80}{R_v + 20}$$

$$1.25 = \frac{R_v + 80}{R_v + 20} \Rightarrow 1.25(R_v + 20) = R_v + 80$$

$$1.25R_v + 25 = R_v + 80 \Rightarrow 0.25R_v = 55 \Rightarrow \boxed{R_v = 220\,\text{k}\Omega}$$

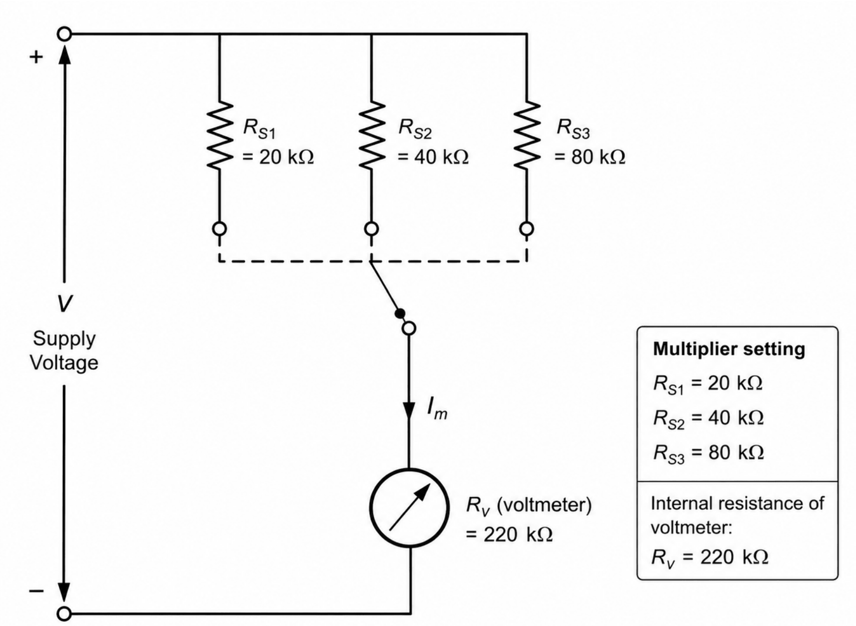

Fig. Solution Q2.12.2

Now,

$$440 = \frac{K}{220 + 20} \Rightarrow K = 440 \times 240 = 105600$$

For $R_{s2} = 40\,\text{k}\Omega$,

$$V_3 = \frac{K}{220 + 40} = \frac{105600}{260} = 406.15\,\text{V}$$

Therefore, the correct option is

$$\boxed{\textbf{D.}\ 406\,\text{V}}$$

Key Points

- **Intuition:**

  Increasing multiplier resistance:

  $$R_m \uparrow$$

  causes meter current

  $$I = \frac{V}{R_v + R_m}$$

  to decrease.

  Thus, the meter reads lower.

  Therefore:

  $$20\,\text{k}\Omega \rightarrow 440\,\text{V}$$

  $$80\,\text{k}\Omega \rightarrow 352\,\text{V}$$

  $$40\,\text{k}\Omega \rightarrow 406\,\text{V}$$

- The voltmeter becomes less sensitive as multiplier resistance increases.

Q2.12.3 MCQ 2012 1M Ans: A ● ○ ○

For a **PMMC meter**, the reading is proportional to the **average (DC) value**.

Hence,

$$V_{\text{avg}} = \frac{\text{area under one cycle}}{T}$$

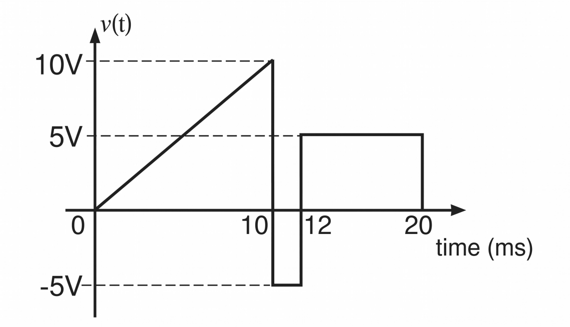

Fig. Solution Q2.12.3

From the waveform:

Segment 1: 0 to 10 ms

A triangle from 0 to 10 V, so

$$A_1 = \frac{1}{2} \times 10 \times 10 = 50 \text{ V} \cdot \text{ms}$$

Segment 2: 10 to 12 ms

A negative rectangle:

$$A_2 = -5 \times 2 = -10 \text{ V} \cdot \text{ms}$$

Segment 3: 12 to 20 ms

A positive rectangle:

$$A_3 = 5 \times 8 = 40 \text{ V} \cdot \text{ms}$$

Total area:

$$A = 50 - 10 + 40 = 80 \text{ V} \cdot \text{ms}$$

Total period:

$$T = 20 \text{ ms}$$

Therefore,

$$V_{\text{avg}} = \frac{80}{20} = 4 \text{ V}$$

Hence, the correct option is

$$\boxed{\textbf{A.}\ 4 \text{ V}}$$

Key Points

- The DC (average) value of a waveform is equal to the area under one complete cycle divided by the time period:

$$f_{\text{avg}} = \frac{1}{T} \int_0^T f(t)\, dt$$

Mistakes to be avoided

- Do not forget the minus sign for the negative area portion of the waveform.

Q2.11.1 MCQ 2011 1M Ans: B ● ○ ○

In a **Low Power Factor (LPF) wattmeter**, errors become significant because the pressure (voltage) coil circuit has inductance.

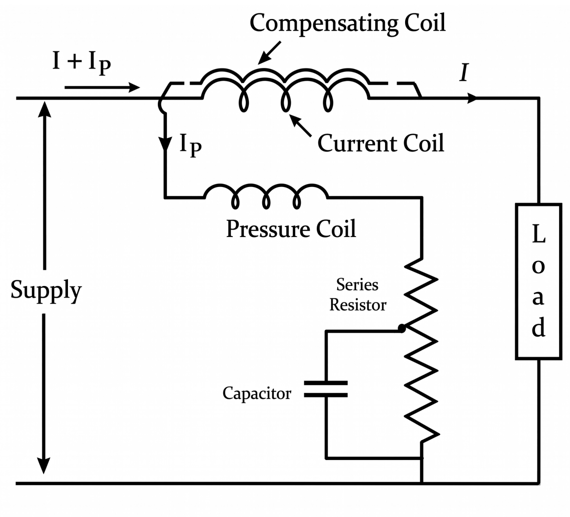

Fig. Solution Q2.11.1

Hence pressure coil current I_p does not remain exactly in phase with the applied voltage.
The pressure coil impedance is:

$$Z_p = R_p + jX_p$$

Thus, I_p lags the applied voltage, causing error in wattmeter reading, especially at low power factor.
To reduce this error, a **compensating coil** is used.
The compensating coil:
i) is connected in series with the pressure coil circuit,
ii) carries pressure coil current I_p,
iii) is wound coaxially over the main current coil,
iv) produces magnetic field opposite to the unwanted field due to pressure coil current.
Hence,

$$\text{Current coil field} \propto (I + I_p)$$

$$\text{Compensating coil field} \propto I_p$$

and acts in opposition.
Therefore, net field becomes:

$$(I + I_p) - I_p = I$$

Thus, the resultant field depends only on the load current I.
Hence, the error due to pressure coil current is neutralized.
Therefore, the compensating coil compensates the effect of:

Pressure (voltage) coil circuit impedance

and not the impedance of the current coil.
Statement (i):
"The compensating coil compensates the effect of impedance of the current coil."
This is **false**, because the compensating coil does not compensate current coil impedance.

$$\boxed{(i)\ \text{False}}$$

Statement (ii):

"The compensating coil compensates the effect of impedance of the voltage coil circuit."
This is **true**, because it neutralizes the phase error caused by:

$$R_p + jX_p$$

$$\boxed{(ii)\ \text{True}}$$

Therefore, the correct option is

$$\boxed{\textbf{B.}\ \text{(i) is False but (ii) is True}}$$

Key Points

- The series resistance and capacitor in the pressure coil circuit of an LPF wattmeter are used to make the pressure coil current I_p nearly in phase with the applied voltage.

  Pressure coil impedance is:$Z_p = R_p + jX_L$

  Hence, pressure coil current lags the applied voltage.

  The series resistance makes the pressure coil circuit more resistive:

  $$\tan\theta = \frac{X_L}{R}$$

  Increasing R reduces θ, so I_p becomes more in phase with voltage.

  The capacitor provides a leading effect:

  $$X_C = \frac{1}{\omega C}$$

  Thus,

  $$X_L - X_C \approx 0$$

  making

  $$I_p \approx \text{in phase with } V$$

Q2.10.1 MCQ 2010 1M Ans: D

Given:

$$\text{Original range} = 0 - 5\text{ A}, \qquad R_m = 0.2\,\Omega$$

Required new range:

$$0 - 25\text{ A}$$

To increase ammeter range, a **shunt resistance** is connected in parallel.

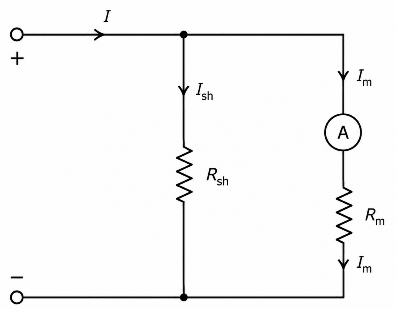

Fig. Solution Q2.10.1

At full-scale:

$$I_m = 5\,\text{A}, \qquad I = 25\,\text{A}$$

Hence shunt current:

$$I_{sh} = 25 - 5 = 20\,\text{A}$$

Since meter and shunt are in parallel, voltage across both is same.
Voltage across meter:

$$V_m = I_m R_m = 5 \times 0.2 = 1\,\text{V}$$

Therefore,

$$V_{sh} = 1\,\text{V}$$

Shunt resistance:

$$R_{sh} = \frac{V_{sh}}{I_{sh}} = \frac{1}{20} = 0.05\,\Omega$$

Therefore, the correct option is

$$\boxed{\textbf{D. } 0.05\,\Omega \text{ in parallel with the meter}}$$

Key Points

- **Increase ammeter range:**

  Add shunt resistance in parallel

- **Decrease ammeter range:**

  Add resistance in series

- For voltmeters, the logic is reversed:

$$\text{Increase voltmeter range} \longrightarrow \text{Series resistance}$$

$$\text{Decrease voltmeter range} \longrightarrow \text{Parallel resistance (shunt)}$$

Q2.10.2 MCQ 2010 1M Ans: D

A wattmeter measures

$$P = V_{PC} I_{CC} \cos\phi$$

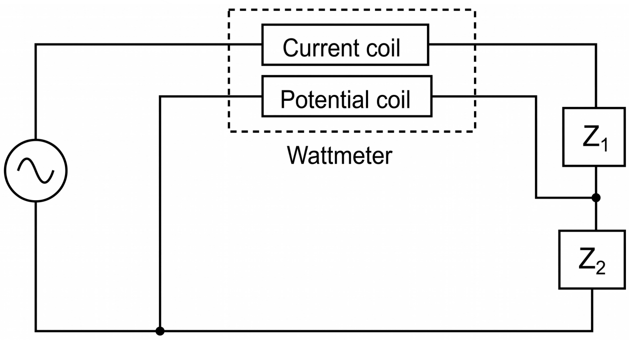

Fig. Solution Q2.10.2

where:

$$V_{PC} = \text{Voltage across the pressure (potential) coil}$$

$$I_{CC} = \text{Current through the current coil}$$

Here, the **current coil** is in series with $Z_1 + Z_2$ (no current enters pressure coil), hence

$$I_{CC} = I$$

The **pressure coil** is connected only across Z_2, therefore

$$V_{PC} = V_{Z_2}$$

Thus wattmeter reading becomes

$$W = V_{Z_2}\, I\, \cos\phi$$

which is exactly the power absorbed by Z_2 only.
Hence,

$$\boxed{\text{Wattmeter reads power absorbed by } Z_2 \text{ only}}$$

Therefore, the correct option is

$$\boxed{\textbf{D.}\ \text{Power consumed by } Z_2}$$

Mistakes to be avoided

- Properly observe the connection of the **pressure coil** and **current coil** before deciding what power the wattmeter measures.
- Do not assume the wattmeter always measures total power of the entire circuit.

Q2.09.1 MCQ 2009 2M Ans: C ● ● ○

For a wattmeter,

$$W = V_{PC} I_{CC} \cos\phi$$

where

$$V_{PC} = \text{voltage across pressure coil}, \qquad I_{CC} = \text{current through current coil}$$

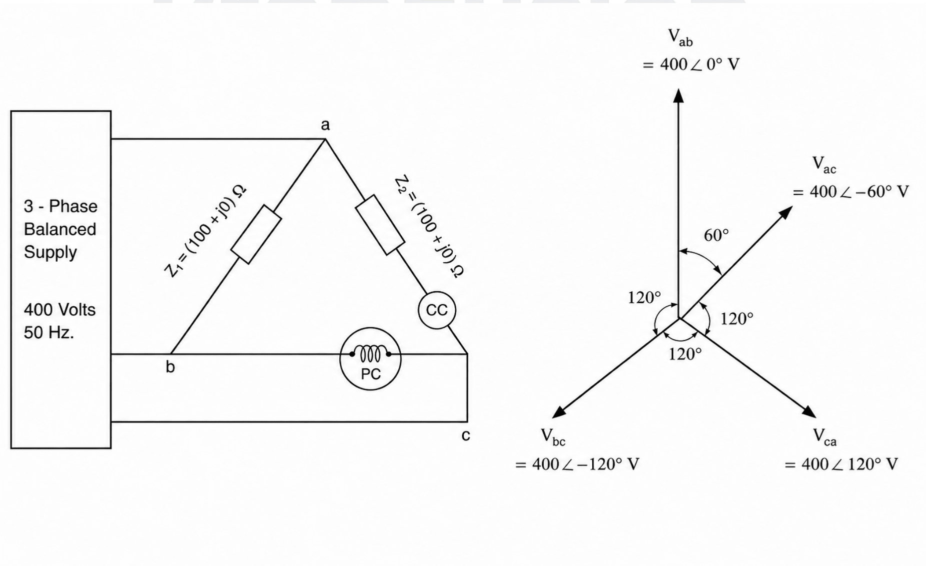

Fig. Solution Q2.09.1

From the circuit, the load is delta connected and

$$Z_1 = Z_2 = (100 + j0)\ \Omega$$

The source voltage is

$$V_{Line} = 400\,\text{V}$$

For a delta load,

$$V_{\text{phase}} = V_{Line}$$

Thus, branch voltage across Z_2 is

$$V_{ac} = 400\,\text{V}$$

Hence current through branch Z_2 (and through CC) is

$$I_{CC} = \frac{400}{100} = 4\,\text{A}$$

Since Z_2 is resistive, I_{CC} is in phase with V_{ac}.
The pressure coil is connected across $b - c$, so

$$V_{PC} = V_{bc}$$

Taking balanced sequence (V_{ab} as reference)

$$V_{ab} = 400\angle 0^\circ, \qquad V_{bc} = 400\angle -120^\circ, \qquad V_{ac} = 400\angle -60^\circ$$

Thus,

$$I_{CC} = 4\angle -60^\circ$$

Angle between V_{PC} and I_{CC} is

$$\phi = (-60^\circ) - (-120^\circ) = 60^\circ$$

Therefore,

$$W = 400 \times 4 \times \cos 60^\circ = 800\,\text{W}$$

Therefore, the correct option is

$$\boxed{\textbf{C.}\ 800\text{W}}$$

Mistakes to be avoided

- Do phase calculation properly.

Q2.09.2 MCQ 2009 1M Ans: B ● ○ ○

In a dynamometer type wattmeter, the pressure coil is connected across the supply voltage.

To measure true power,

$$P = VI\cos\phi$$

the pressure coil current must remain nearly in phase with the applied voltage.
If the pressure coil were inductive,

$$Z_p = R_p + jX_p$$

then pressure coil current I_p would lag the voltage, causing phase error and incorrect wattmeter reading.
To reduce this error, a large non-inductive resistance is connected in series with the pressure coil so that

$$R_p \gg X_p$$

Hence,

$$Z_p \approx R_p$$

Therefore, pressure coil current becomes nearly in phase with voltage.
Thus, the pressure coil behaves as:

$$\boxed{\text{Highly resistive}}$$

Hence, the correct option is

$$\boxed{\textbf{B.}\ \text{Highly resistive}}$$

Q2.06.1 MCQ | 2006 | 2M | Ans: B | ● ● ○

For the **immersion heater** (purely resistive), the energy meter reads 2.3 kWh in 1 hour. Hence initial power consumed is

$$P_1 = \frac{2.3\,\text{kWh}}{1\,\text{h}} = 2.3\,\text{kW} = 2300\,\text{W}$$

For a resistive load,

$$P = \frac{V_{\text{rms}}^2}{R}$$

With initial RMS voltage $V_1 = 230\,\text{V}$,

$$2300 = \frac{230^2}{R} \Rightarrow R = 23\,\Omega$$

Now the input is a symmetric square wave of $400\,\text{V}_{pp}$, so

$$V_p = \frac{400}{2} = 200\,\text{V}$$

For a symmetric square wave,

$$V_{\text{rms}} = V_p = 200\,\text{V}$$

Therefore new power is

$$P_2 = \frac{V_{\text{rms}}^2}{R} = \frac{200^2}{23} = 1739\,\text{W} = 1.739\,\text{kW}$$

Therefore, the correct option is

$$\boxed{\textbf{B.}\ 1.739}$$

Key Points

- For a purely resistive load:

$$P = \frac{V_{\text{rms}}^2}{R}$$

- Frequency does not affect power consumed by a pure resistor.

Mistakes to be avoided

- Do not confuse peak voltage with peak-to-peak voltage:

$$V_p = \frac{V_{pp}}{2}$$

Q2.06.2 MCQ | 2006 | 2M | Ans: C | ● ● ○

Given,

$$i(t) = -8 + 6\sqrt{2}\sin(\omega t + 30^\circ)$$

This has:

$$I_{\text{DC}} = -8\,\text{A}$$

and sinusoidal AC component

$$i_{\text{AC}} = 6\sqrt{2}\sin(\omega t + 30^\circ)$$

1. Centre-zero PMMC meter

A PMMC meter responds to the average value. The average of the sinusoidal term over one cycle is zero,

$$I_{\text{avg}} = -8 + 0 = -8\,\text{A}$$

Hence, PMMC reading is

$$\boxed{-8\,\text{A}}$$

2. True RMS meter
For a DC plus sinusoidal current,

$$I_{\text{rms}} = \sqrt{I_{\text{DC}}^2 + I_{\text{AC,rms}}^2}$$

Here,

$$I_{\text{AC,rms}} = \frac{6\sqrt{2}}{\sqrt{2}} = 6\,\text{A}$$

Therefore,

$$I_{\text{rms}} = \sqrt{(-8)^2 + 6^2} = \sqrt{64 + 36} = 10\,\text{A}$$

So true RMS meter reads

$$\boxed{10\,\text{A}}$$

3. Moving iron instrument
A moving iron instrument responds to RMS value, hence

$$\boxed{10\,\text{A}}$$

Therefore, the correct option is

$$\boxed{\textbf{C.}\ -8,\ 10,\ 10}$$

Key Points

- Both **True RMS meters** and **Moving Iron (MI)** instruments measure RMS values, but their working principles are different.
- **True RMS meter:**

  A true RMS meter computes:

$$X_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T x^2(t)\,dt}$$

  It performs:

$$\text{Square} \longrightarrow \text{Average} \longrightarrow \text{Square root}$$

  Hence, it gives accurate RMS value for almost any waveform
- **Moving Iron (MI) instrument:**

  Deflecting torque:

$$T_d \propto I^2$$

  Hence,

$$\theta \propto I_{\text{rms}}^2$$

  So MI instruments effectively respond to RMS value.
- MI instruments are not ideal RMS meters because of:

$$\text{Nonlinear scale, hysteresis, eddy current and frequency errors}$$

- A **centre-zero PMMC** can indicate both positive and negative DC values.

Q2.05.1 MCQ | 2005 | 2M | Ans: A | ● ● ○

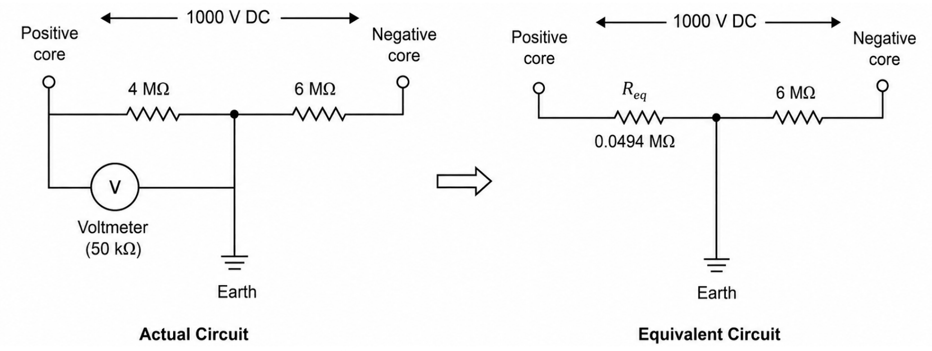

Fig. Solution Q2.05.1

Given:

$$V_s = 1000\,\text{V}, \qquad R_1 = 4\,\text{M}\Omega, \qquad R_2 = 6\,\text{M}\Omega, \qquad R_v = 50\,\text{k}\Omega = 0.05\,\text{M}\Omega$$

The voltmeter is connected between the positive core and earth, so it is in parallel with R_1.

Step 1: Equivalent resistance of R_1 and voltmeter

$$R_{eq} = R_1 \parallel R_v = \frac{4 \times 0.05}{4 + 0.05} = 0.0494\,\text{M}\Omega$$

Thus the circuit becomes R_{eq} in series with R_2.

Step 2: Total resistance

$$R_T = R_{eq} + R_2 = 0.0494 + 6 = 6.0494\,\text{M}\Omega$$

Step 3: Circuit current

$$I = \frac{V_s}{R_T} = \frac{1000}{6.0494 \times 10^6} = 165.3\,\mu\text{A}$$

Step 4: Voltmeter reading (across R_{eq})

$$V_v = IR_{eq} = 165.3 \times 10^{-6} \times 0.0494 \times 10^6 = 8.16\,\text{V}$$

Therefore, the correct option is

$$\boxed{\textbf{A.}\ 8\,\text{V}}$$

Q2.04.1 MCQ | 2004 | 2M | Ans: B | ● ● ○

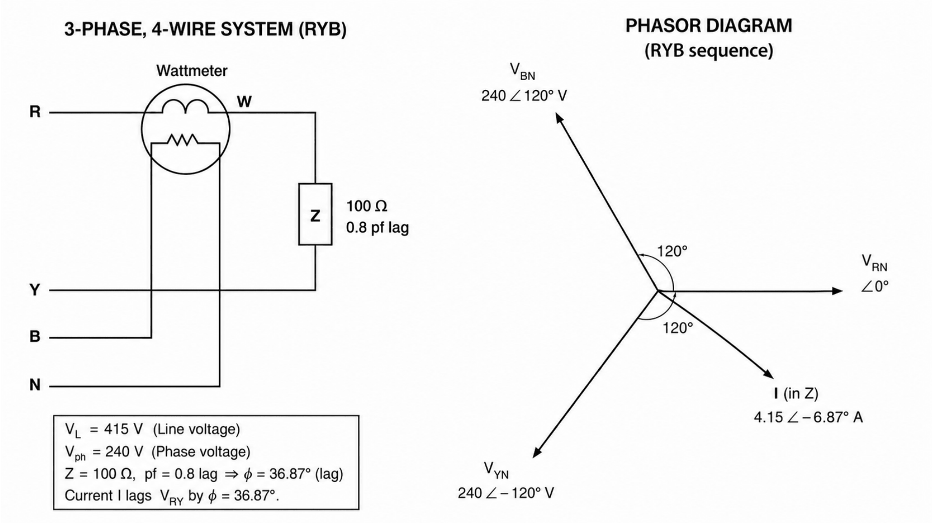

Fig. Solution Q2.04.1

Given:

$$V_L = 415\,\text{V}, \qquad Z = 100\,\Omega, \qquad \cos\phi = 0.8$$

Hence,

$$\phi = 36.87^\circ \ \text{(lagging)}$$

Step 1: Load current
Since the load is connected between R and Y,

$$V_{RY} = 415\angle 30^\circ \ \text{V}$$

$$I = \frac{415}{100} = 4.15\,\text{A}$$

So the current coil current is

$$I_{CC} = 4.15\angle(30^\circ - 36.87^\circ) = 4.15\angle(-6.87^\circ)$$

Step 2: Pressure coil voltage
The pressure coil is connected between B and N, so

$$V_{PC} = V_{BN} = 240\angle 120^\circ \ \text{V}$$

Step 3: Angle between V_{PC} and I_{CC}

$$\theta = 120^\circ - (-6.87^\circ) = 126.87^\circ$$

$$\cos\theta = \cos 126.87^\circ = -0.6$$

Step 4: Wattmeter reading

$$W = V_{PC} I_{CC} \cos\theta$$

$$W = 240 \times 4.15 \times (-0.6) = -597.6\,\text{W}$$

Therefore, the correct option is

$$\boxed{\textbf{B.} \ -597\,\text{W}}$$

Mistakes to be avoided

- Be careful with signs.

Q2.04.2 | MCQ | 2004 | 2M | Ans: B | ● ○ ○

Given:

$$T_d = 240\,\mu\text{Nm} = 240 \times 10^{-6}\,\text{Nm}, \qquad I = 10\,\text{A}$$

For a moving iron instrument,

$$T_d = \frac{1}{2} I^2 \frac{dL}{d\theta}$$

Hence,

$$\frac{dL}{d\theta} = \frac{2T_d}{I^2}$$

Substituting,

$$\frac{dL}{d\theta} = \frac{2(240 \times 10^{-6})}{10^2} = \frac{480 \times 10^{-6}}{100} = 4.8 \times 10^{-6}\,\text{H/rad}$$

Hence, the correct option is

$$\boxed{\textbf{B.}\ 4.8\,\mu\text{H/radian}}$$

Key Points

- Moving iron instruments work on the principle of change in self inductance with angular displacement.

Mistakes to be avoided

- The given deflection angle 120° is not needed in this problem.

Q2.04.3 | MCQ | 2004 | 2M | Ans: C | ● ○ ○

Given:

$$V = 250\,\text{V}, \qquad I = 15\,\text{A}$$

Meter constant:

$$14.4\ \text{A s/rev}$$

This means charge per revolution is

$$Q = 14.4\,\text{A}\cdot\text{s}$$

Energy consumed per revolution:

$$E_{\text{rev}} = VQ = 250 \times 14.4 = 3600\,\text{W}\cdot\text{s}$$

Since

$$1\,\text{Wh} = 3600\,\text{W}\cdot\text{s}$$

therefore,

$$E_{\text{rev}} = 1\,\text{Wh}$$

Hence,

$$1\,\text{rev} = 1\,\text{Wh}$$

Therefore,

$$1000\,\text{rev} = 1000\,\text{Wh} = 1\,\text{kWh}$$

So the meter constant is

$$\boxed{1000\,\text{rev/kWh}}$$

Hence, the correct option is

$$\boxed{\textbf{C.}\ 1000\,\text{rev/kWh}}$$

Key Points

- Important conversion:

$$1\ \text{Wh} = 3600\ \text{J} = 3600\ \text{W} \cdot \text{s}$$

- Meter constant:

$$\text{rev/kWh} = \frac{\text{Total energy in 1 kWh}}{\text{Energy per revolution}}$$

Mistakes to be avoided

- Do not forget that the question asks for:

$$\text{rev/kWh}$$

Q2.04.4 MCQ | 2004 | 2M | Ans: A | ● ○ ○

Given:

$$N = 100, \qquad B = 200\ \text{mT} = 0.2\ \text{T}$$
$$l = 10\ \text{mm} = 10 \times 10^{-3}\ \text{m}$$
$$d = 20\ \text{mm} = 20 \times 10^{-3}\ \text{m}$$
$$I = 50\ \text{mA} = 0.05\ \text{A}$$

For a PMMC instrument,

$$T = NBIA$$

where coil area is

$$A = l \times d$$

Thus,

$$A = (10 \times 10^{-3})(20 \times 10^{-3}) = 2 \times 10^{-4}\ \text{m}^2$$

Substituting,

$$T = 100 \times 0.2 \times 0.05 \times 2 \times 10^{-4}$$

Hence,

$$T = 2 \times 10^{-4}\ \text{Nm}$$

Converting into μNm,

$$T = 2 \times 10^{-4} \times 10^{6} = 200\ \mu\text{Nm}$$

Hence, the correct option is

$$\boxed{\textbf{A.}\ 200\ \mu\text{Nm}}$$

Q2.04.5 MCQ | 2004 | 1M | Ans: C | ● ○ ○

Given:

$$I_g = 10\ \text{mA} = 0.01\ \text{A}, \qquad R_g = 1000\ \Omega, \qquad R_{sh} = 100\ \Omega$$

Since galvanometer and shunt are in parallel,

$$I_g R_g = I_{sh} R_{sh}$$

Substituting,

$$0.01 \times 1000 = I_{sh} \times 100$$

$$10 = 100 I_{sh} \Rightarrow I_{sh} = 0.1\ \text{A}$$

Hence total measurable current is

$$I = I_g + I_{sh} = 0.01 + 0.1 = 0.11\text{ A}$$

Multiplying power:

$$m = \frac{I}{I_g} = \frac{0.11}{0.01} = 11$$

Hence, the correct option is

$$\boxed{\textbf{C. } 11}$$

Key Points

- Multiplying power of an ammeter is:

$$m = \frac{I}{I_g}$$

where:

$$I = \text{maximum measurable current}$$

Q2.04.6 MCQ 2004 1M Ans: C ● ● ○

Given:

$$R_{PC} = 1000\,\Omega, \quad R_{CC} = 0.02\,\Omega, \quad V_{\text{load}} = 200\text{ V}, \quad I_{\text{load}} = 20\text{ A}$$

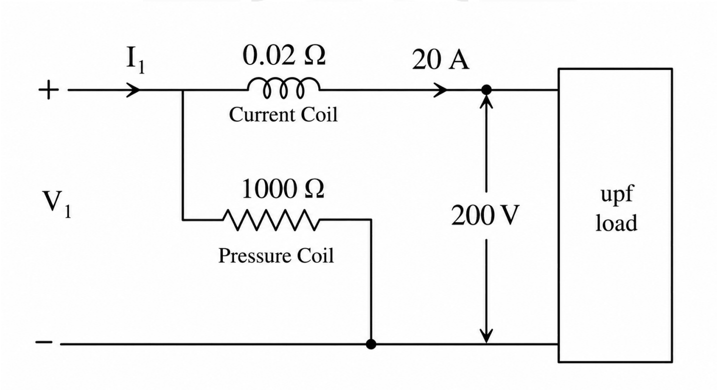

Fig. Solution Q2.04.6

From the figure, the pressure coil is connected before the current coil, so the pressure coil voltage is

$$V_1 = V_{\text{load}} + I_{\text{load}} R_{CC}$$

$$V_1 = 200 + (20)(0.02) = 200.4\text{ V}$$

Current through the pressure coil is

$$I_{PC} = \frac{V_1}{R_{PC}} = \frac{200.4}{1000} = 0.2004\text{ A}$$

Hence current through the current coil is

$$I_1 = I_{\text{load}} + I_{PC} = 20 + 0.2004 = 20.2004\text{ A}$$

However, the wattmeter reading is

$$P_m = V_1 I_1$$

Since pressure coil current contribution is negligible, error comes mainly from the current coil loss:

$$P_m = V_{\text{load}} I_{\text{load}} + I_{\text{load}}^2 R_{CC}$$

$$P_m = (200)(20) + 20^2(0.02) = 4000 + 8 = 4008\ \text{W}$$

True load power:

$$P_{\text{load}} = 200 \times 20 = 4000\ \text{W}$$

Percentage error:

$$\%\ \text{error} = \frac{4008 - 4000}{4000} \times 100 = 0.2\%$$

Thus, the measured power is **0.2% more** than the load power.
Hence, the correct option is

$$\boxed{\textbf{C.}\ 0.2\%\ \text{more}}$$

Key Points

- Current coil resistance is very small, but at high current its power loss may still cause noticeable error.

Q2.03.1 MCQ 2003 2M Ans: C ● ● ○

Given:

$$V = 220\ \text{V}, \qquad I = 5\ \text{A}$$

Flux angle adjusted to: 85°
Ideal flux angle: 90°
Hence phase error:

$$\alpha = 90^\circ - 85^\circ = 5^\circ$$

For an induction energy meter,

$$T_d \propto VI \sin\delta$$

where:

$$\delta = \text{angle between voltage flux and load current}$$

Under ideal condition,

$$\delta = 90^\circ - \phi$$

Therefore,

$$T_{\text{ideal}} \propto VI \sin(90^\circ - \phi)$$

$$T_{\text{ideal}} \propto VI \cos\phi$$

Because of phase error, actual flux angle becomes; 85°
Hence,

$$\delta = 85^\circ - \phi$$

Thus actual torque becomes:

$$T_{\text{actual}} \propto VI \sin(85^\circ - \phi)$$

Therefore, power error is:

$$\Delta P = VI\left[\sin(85^\circ - \phi) - \cos\phi\right]$$

Case 1: Power factor = 1
Hence,

$$\phi = 0^\circ$$

Substituting,

$$\Delta P = 220 \times 5\,[\sin 85^\circ - 1]$$

$$= 1100(-0.0038) = -4.18$$

Therefore,

$$\boxed{\Delta P \approx -4.2}$$

Case 2: Power factor $= 0.5$ **lagging**

$$\cos\phi = 0.5 \Rightarrow \phi = 60^\circ$$

Substituting,

$$\Delta P = 220 \times 5\,[\sin 25^\circ - 0.5]$$

$$= 1100(-0.0774) = -85.1$$

Therefore,

$$\boxed{\Delta P \approx -85.1}$$

Therefore, the correct option is

$$\boxed{\textbf{C.}\ -4\,mW, -85.1\,mW}$$

Key Points

- For ideal operation:

$$\delta = 90^\circ - \phi$$

hence,

$$T_d \propto VI\cos\phi$$

Mistakes to be avoided

- Be careful with sign of error:

$$\text{Error} = P_{\text{actual}} - P_{\text{ideal}}$$

- At low power factor, even small phase-angle error can produce large measurement error.

Q2.03.2

MCQ | 2003 | 2M | Ans: C | ● ● ○

Given,

$$L = \left(10 + 3\theta - \frac{\theta^2}{4}\right)\mu\text{H}$$

Control spring torque:

$$T_c = 25 \times 10^{-6}\theta$$

Current:

$$I = 5\,\text{A}$$

For a moving iron instrument,

$$T_d = \frac{1}{2}I^2\frac{dL}{d\theta}$$

At equilibrium,

$$T_d = T_c$$

Hence,

$$\frac{1}{2}I^2\frac{dL}{d\theta} = 25 \times 10^{-6}\theta$$

Differentiating inductance,

$$\frac{dL}{d\theta} = \left(3 - \frac{\theta}{2}\right)\mu\text{H/rad} = \left(3 - \frac{\theta}{2}\right) \times 10^{-6}\ \text{H/rad}$$

Substituting,

$$\frac{1}{2}(5)^2\left(3 - \frac{\theta}{2}\right)10^{-6} = 25 \times 10^{-6}\theta$$

$$12.5\left(3 - \frac{\theta}{2}\right) = 25\theta$$

$$37.5 - 6.25\theta = 25\theta \Rightarrow 37.5 = 31.25\theta \Rightarrow \theta = \frac{37.5}{31.25} = 1.2$$

$$\boxed{\theta = 1.2\,\text{rad}}$$

Hence, the correct option is

$$\boxed{\textbf{C. } 1.2}$$

Key Points

- The controlling torque in a Moving Iron (MI) instrument is:

$$T_c = K\theta$$

where:

$$K = \text{spring constant}, \qquad \theta = \text{pointer deflection}$$

Mistakes to be avoided

- The controlling torque is given in:

$$\text{N-m/rad}$$

or equivalently as:

$$T_c = K\theta$$

Read the units carefully before substituting values.

Q2.03.3 MCQ 2003 2M Ans: B ● ● ○

Given a **balanced star load**,

$$\cos\phi = 0.8 \Rightarrow \phi = 36.87^\circ$$

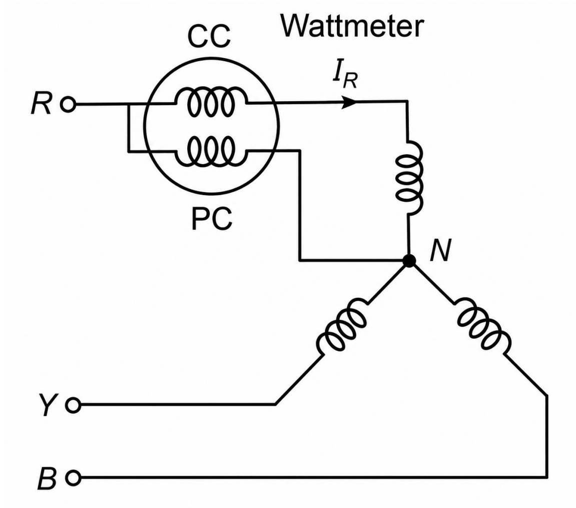

Fig. Solution Q2.03.3

Initially, the current coil is in R phase and the pressure coil is between R and neutral, so the wattmeter reads

$$W_1 = 400\,\text{W}$$

Hence,

$$W_1 = V_{ph} I \cos\phi = V_{ph} I (0.8) \Rightarrow V_{ph} I = \frac{400}{0.8} = 500$$

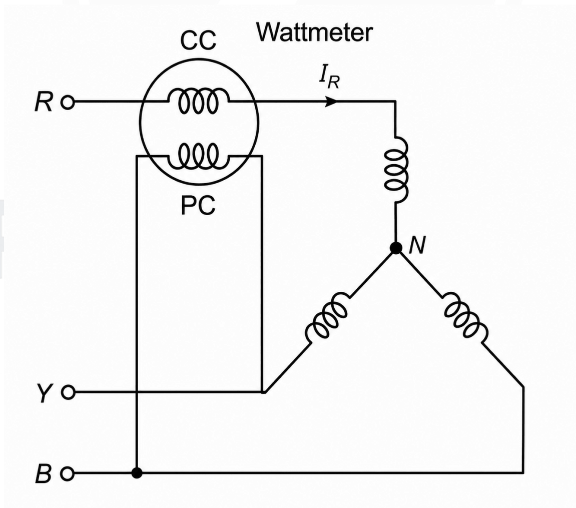

Fig. Solution Q2.03.3

Now the pressure coil is connected between B and Y, while the current coil remains in R.
For RYB sequence,

$$V_{BY} = \sqrt{3}\, V_{ph} \angle 90^\circ, \qquad I_R = I \angle(-36.87^\circ)$$

So the angle between V_{BY} and I_R is

$$90^\circ + 36.87^\circ = 126.87^\circ$$

Therefore,

$$W_2 = \sqrt{3}\, V_{ph} I \cos 126.87^\circ$$

Using $V_{ph} I = 500$,

$$W_2 = \sqrt{3} \times 500 \times (-0.6) = -519.6\,\text{W}$$

Thus the wattmeter gives reverse deflection, and the reading is

$$\boxed{519.6\,\text{W}}$$

Hence, the correct option is

$$\boxed{\textbf{B. } 519.6}$$

Q2.03.4 MCQ 2003 1M Ans: B ● ○ ○

The effect of stray magnetic fields depends on how they combine with the instrument's operating magnetic field.
If the stray field is **parallel** to the operating field, then

$$B_{\text{net}} = B_{\text{operating}} \pm B_{\text{stray}}$$

Thus the resultant magnetic field changes maximum, causing maximum change in torque and maximum error.
If the two fields are perpendicular,

$$B_{\text{net}} = \sqrt{B^2_{\text{operating}} + B^2_{\text{stray}}}$$

so the change in resultant field is comparatively smaller.
Hence, maximum effect occurs when:

$$\boxed{\text{Operating field} \parallel \text{Stray field}}$$

Therefore, the correct option is

$$\boxed{\textbf{B. } \text{parallel}}$$

Key Points

- Stray magnetic field error depends on both:

  Magnitude and direction of external magnetic field

- PMMC instruments are less affected by stray magnetic fields because they use strong permanent magnets.
- MI instruments are more affected because their operating magnetic field is comparatively weak.
- Stray magnetic field error can be reduced using:

  Steel shielding or sheet-steel lining

Q2.03.5 MCQ 2003 1M Ans: A ● ○ ○

The moving coil of a PMMC instrument is generally made of copper, whose resistance increases with temperature:

$$R_{\text{coil}} \uparrow \text{ as temperature increases}$$

Hence meter current changes:

$$I_m = \frac{V}{R_{\text{meter}}}$$

causing inaccurate readings.
To reduce this effect, a **swamping resistance** made of manganin is connected in series with the moving coil.
Since manganin has very low temperature coefficient,

$$R_{SW} \approx \text{constant}$$

and usually,

$$R_{SW} \gg R_{\text{coil}}$$

Therefore total meter resistance becomes:

$$R_{\text{meter}} = R_{SW} + R_{\text{coil}}$$

As temperature changes, variation in R_{coil} becomes negligible compared to the large swamping resistance, hence:

$$R_{\text{meter}} \approx \text{constant}$$

Thus meter current remains nearly constant and calibration remains stable.
Hence, the purpose of swamping resistance is:

$$\boxed{\text{to minimise the effect of temperature variation}}$$

Therefore, the correct option is

$$\boxed{\textbf{A. } \text{minimise the effect of temperature variation.}}$$

Key Points

- A manganin shunt is also connected across the meter branch.

  Since both R_{sh} and R_{SW} are made of manganin (very low temperature coefficient), the ratio:

$$\frac{I_{sh}}{I_m}$$

  remains nearly constant with temperature.

Q2.03.6 MCQ 2003 2M Ans: C ● ● ○

Given,

$$V_{\text{rms}} = 100\,\text{V}, \qquad I_{fs} = 1\,\text{mA}, \qquad R_m = 100\,\Omega$$

A PMMC instrument responds to the **average** value, so the AC input is first full-wave rectified.
For a sinusoidal input,

$$V_m = \sqrt{2}V_{\text{rms}} = \sqrt{2} \times 100 = 141.4\,\text{V}$$

Average value of a full-wave rectified sine wave:

$$V_{\text{avg}} = \frac{2V_m}{\pi} = \frac{2 \times 141.4}{\pi} = 90.03\,\text{V}$$

At full-scale deflection,

$$I_{fs} = \frac{V_{\text{avg}}}{R_s + R_m}$$

Hence,

$$R_s + R_m = \frac{90.03}{0.001} = 90030\,\Omega$$

Therefore,

$$R_s = 90030 - 100 = 89930\,\Omega = 89.93\,\text{k}\Omega$$

Hence, the correct option is

$$\boxed{\textbf{C. } 89.93\,\text{k}\Omega}$$

Q2.02.1 MCQ | 2002 | 1M | Ans: C | ● ○ ○

Given,

$$V_L = 100\,\text{V}, \qquad Z = 5\angle 60^\circ\,\Omega$$

Since the load is balanced and star connected,

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{100}{\sqrt{3}} = 57.7\,\text{V}$$

Phase current:

$$I_{ph} = \frac{V_{ph}}{|Z|} = \frac{57.7}{5} = 11.54\,\text{A}$$

Hence,

$$I_L = I_{ph} = 11.54\,\text{A}$$

For the two-wattmeter method,

$$W_1 = V_L I_L \cos(30^\circ + \phi), \qquad W_2 = V_L I_L \cos(30^\circ - \phi)$$

Here,

$$\phi = 60^\circ$$

Thus,

$$W_1 = 100 \times 11.54 \cos 90^\circ = 0$$

and

$$W_2 = 100 \times 11.54 \cos(-30^\circ) = 1154 \times 0.866 \approx 1000\,\text{W}$$

Hence, the correct option is

$$\boxed{\textbf{C. } W_1 = 0, \qquad W_2 = 1000\,\text{W}}$$

Key Points

- In star connection:

$$I_{Line} = I_{phase}$$

- In delta connection:

$$V_{Line} = V_{phase}$$

Q2.01.1 MCQ | 2001 | 1M | Ans: B | ● ○ ○

For a 3-phase, 3-wire system, the minimum number of wattmeters required to measure total power for both balanced and unbalanced loads is:

$$\boxed{2}$$

In a 3-phase, 3-wire system:

$$I_R + I_Y + I_B = 0$$

Thus only two line currents are independent, so only two wattmeters are sufficient to measure total power.
Using the two-wattmeter method:

$$P_{\text{total}} = W_1 + W_2$$

This method works for:

Balanced and unbalanced loads

and for:

Leading as well as lagging power factor

Therefore, two wattmeters are sufficient for any 3-phase, 3-wire system.

Hence, the correct option is

$$\boxed{\textbf{B.}\ 2}$$

Key Points

- According to **Blondel's theorem**, the number of wattmeters required to measure total power in an N-phase system is:

$$N \quad \text{or} \quad (N-1)$$

- If neutral wire is available:

$$\text{Number of wattmeters required} = N$$

- If neutral wire is not available:

$$\text{Number of wattmeters required} = N - 1$$

Mistakes to be avoided

- Carefully read whether the question asks for **balanced** or **unbalanced** load.
- Although one wattmeter can measure total power in a **balanced** load using special connections, the question here asks for the general minimum number required, which must also work for unbalanced loads.

Q2.01.2 MCQ 2001 1M Ans: D

Given,

$$P = 450\,\text{W}, \qquad t = 100\,\text{s}, \qquad N = 10$$

Energy consumed:

$$E = Pt = 450 \times 100 = 45000\,\text{J}$$

Since,

$$1\,\text{kWh} = 3.6 \times 10^6\,\text{J}$$

Energy in kWh:

$$E = \frac{45000}{3.6 \times 10^6} = 0.0125\,\text{kWh}$$

Thus,

$$10 \text{ revolutions} \equiv 0.0125\,\text{kWh}$$

Hence meter constant is

$$K = \frac{10}{0.0125} = 800\,\text{rev/kWh}$$

Therefore,

$$\boxed{800\,\text{rev/kWh}}$$

Hence, the correct option is

$$\boxed{\textbf{D.}\ 800}$$

Key Points

- Meter constant:

$$K = \frac{N \times 3600 \times 1000}{Pt}$$

where K = rev/kWh, N = number of revolutions, P = power in watts, t = time in seconds.

Q2.00.1 MCQ 2000 2M Ans: D ● ● ○

Explanation:

The two-wattmeter method measures total active power in a 3-phase, 3-wire system as:

$$P = W_1 + W_2$$

This relation is valid irrespective of whether the voltages are balanced or unbalanced.
For a 3-wire system:

$$I_R + I_Y + I_B = 0$$

The total instantaneous power is:

$$p = v_R i_R + v_Y i_Y + v_B i_B$$

Using:

$$i_Y = -(i_R + i_B)$$

Substituting,

$$\begin{aligned} p &= v_R i_R + v_Y[-(i_R + i_B)] + v_B i_B \\ &= (v_R - v_Y)i_R + (v_B - v_Y)i_B \end{aligned}$$

Thus,

$$p = V_{RY} i_R + V_{BY} i_B = \boxed{W_1 + W_2}$$

Hence the total power measured by the two-wattmeter method equals the true total power, independent of voltage unbalance.
Therefore, neither negative-sequence voltages, nor zero-sequence voltages affect the total power measured by the two-wattmeter method.
The correct answer is:

D. not affected by negative or zero sequence voltages

Key Points

- Any unbalanced 3-phase voltage/current set can be decomposed into:

Positive sequence + Negative sequence + Zero sequence

This is called **symmetrical components**.

Sequence	Magnitude	Phase Order	Example
Positive	Equal	RYB	Normal system
Negative	Equal	RBY	Reverse sequence
Zero	Equal	Same angle	All overlap

Q2.00.2 MCQ 2000 1M Ans: C ● ○ ○

A true RMS measuring circuit computes:

$$V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt}$$

The process is:

- Square the input:

$$v(t) \rightarrow v^2(t)$$

- Average over time:

$$\frac{1}{T}\int_0^T v^2(t)\,dt$$

- Take square root.

The slow response occurs because of the averaging stage.
To obtain a stable average value, the circuit uses an RC integrator / low-pass filter having time constant:

$$\tau = RC$$

This averaging smooths fluctuations, so sudden input changes take time to appear at the output.
Hence:

$$\boxed{\text{Averaging} \Rightarrow \text{Slow response}}$$

The correct answer is:

$$\boxed{\textbf{C.}\ \text{the averaging function built into the circuit}}$$

Q2.00.3 NAT 2000 5M Ans: * ● ● ○

For a balanced 3-phase load with phase sequence: RYB

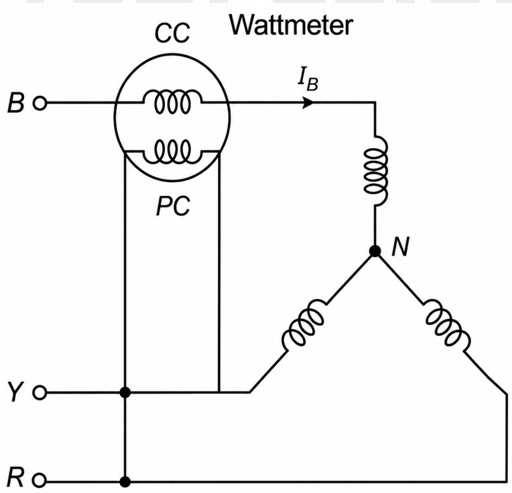

Fig. Solution Q2.00.3

Pressure coil (PC) is connected across: R-Y
Hence,

$$V_{PC} = V_{RY}$$

Current coil (CC) is connected in phase B
Hence,

$$I_{CC} = I_B$$

Need to show:

$$\text{Wattmeter reading} \propto \text{Reactive power}$$

Step 1: Phase and line voltages
For a balanced system:

$$V_R = V_{ph}\angle 0^\circ, \quad V_Y = V_{ph}\angle -120^\circ, \quad V_B = V_{ph}\angle 120^\circ$$

Line voltage:

$$V_{RY} = \sqrt{3}\,V_{ph}\angle 30^\circ$$

Thus,

$$V_{RY} = V_L\angle 30^\circ$$

where

$$V_L = \sqrt{3}\,V_{ph}$$

Step 2: Current in B phase
If load power factor angle is ϕ, then:

$$I_B = I\angle(120^\circ - \phi)$$

Step 3: Angle between V_{RY} and I_B

$$V_{RY} = V_L\angle 30^\circ, \qquad I_B = I\angle(120^\circ - \phi)$$

Hence,

$$\theta = (120^\circ - \phi) - 30^\circ = 90^\circ - \phi$$

Step 4: Wattmeter reading
Wattmeter measures:

$$W = VI\cos\theta$$

Substituting,

$$W = V_L I\cos(90^\circ - \phi)$$

Using,

$$\cos(90^\circ - \phi) = \sin\phi$$

we get:

$$W = V_L I\sin\phi$$

Step 5: Compare with reactive power
Reactive power of a balanced 3-phase load is:

$$Q = \sqrt{3}\,V_L I\sin\phi$$

Therefore,

$$Q = \sqrt{3}\,W \Rightarrow W = \frac{Q}{\sqrt{3}}$$

Hence,

$$\boxed{W \propto Q}$$

Thus, the wattmeter indication is proportional to the reactive power of the balanced 3-phase load.

Q2.99.1 MCQ | 1999 | 1M | Ans: C | ● ○ ○

1: Ideal wattmeter operation

In a dynamometer wattmeter:

i) Current coil (CC) carries load current I

ii) Pressure coil (PC) carries current proportional to voltage $I_p \propto V$

Ideally,

$$I_p \parallel V$$

Then wattmeter reads:

$$W = VI\cos\phi$$

which is the true power.

2: Practical situation

Pressure coil has:

$$\text{Resistance} + \text{inductance}$$

Hence pressure coil current lags applied voltage by angle α

Thus,

$$I_p = V\angle(-\alpha)$$

So wattmeter actually measures:

$$W' = VI\cos(\phi + \alpha)$$

instead of:

$$W = VI\cos\phi$$

Error:

$$= VI[\cos(\phi + \alpha) - \cos\phi]$$

3: Why error becomes large at low power factor

At low power factor:

$$\phi \to 90^\circ$$

Hence $\cos\phi$ is already very small.

So even a small phase error α causes large percentage error.

Example:

$$\cos 80^\circ = 0.174$$

and,

$$\cos 85^\circ = 0.087$$

which gives nearly:

$$50\% \text{ error}$$

Thus large errors occur because:

$$\boxed{\text{Pressure coil current is not in phase with voltage}}$$

The correct answer is:

$$\boxed{\textbf{C.}\ \text{pressure coil currents are not in phase}}$$

Q2.99.2 NAT 1999 5M Ans: * ● ● ●

Given,

$$V_{\text{rms}} = 230\,\text{V}, \qquad f = 50\,\text{Hz}, \qquad R = 46\,\Omega$$

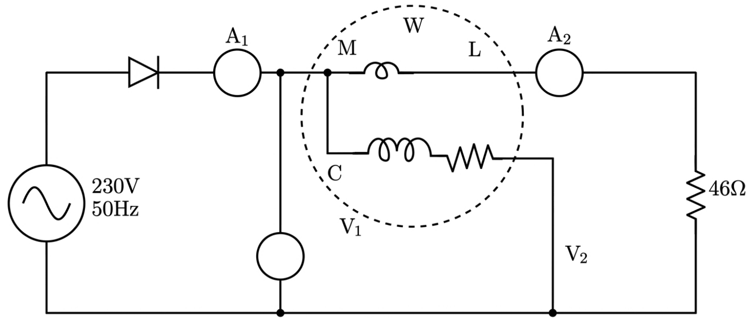

Fig. Solution Q2.99.2

The diagram shows half-wave rectification, so each instrument must be interpreted according to its principle.

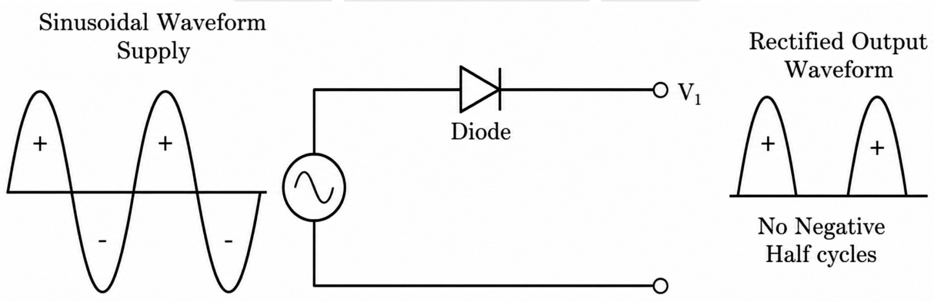

Fig. Solution Q2.99.2

Peak voltage:

$$V_m = \sqrt{2}\,V_{\text{rms}} = 1.414 \times 230 = 325\,\text{V}$$

Peak current:

$$I_m = \frac{V_m}{R} = \frac{325}{46} = 7.07\,\text{A}$$

1. Reading of A_1 (moving-iron ammeter)
Moving-iron instruments respond to **RMS value**. For a half-wave rectified sine wave,

$$I_{\text{rms}} = \frac{I_m}{2} = \frac{7.07}{2} = 3.535\,\text{A}$$

Hence,

$$A_1 = 3.54\,\text{A}$$

2. Reading of A_2 (full-wave rectifier type ammeter)
Rectifier-type instruments respond to **average value** and are calibrated to indicate RMS for a sine wave.
Average current for half-wave rectified waveform:

$$I_{\text{avg}} = \frac{I_m}{\pi} = \frac{7.07}{\pi} = 2.25\,\text{A}$$

Indicated reading:

$$A_2 = 1.11\,I_{\text{avg}} = 1.11 \times 2.25 = 2.50\,\text{A}$$

3. Reading of V_1 (peak-response voltmeter)
Peak-response voltmeter measures the **peak value**:

$$V_1 = V_m = 325\,\text{V}$$

4. Reading of V_2 (PMMC voltmeter)
PMMC measures:

Average DC value

Average current:

$$I_{\text{avg}} = 2.25\,\text{A}$$

Therefore,

$$V_2 = I_{\text{avg}} R = 2.25 \times 46 = 103.5\,\text{V}$$

Hence,

$$V_2 = 103.5\,\text{V}$$

Important:
Do not use:

$$V_2 = A_2 \times 46$$

because A_2 is a calibrated indication, not the actual average current.
5. Reading of wattmeter W
The electrodynamometer wattmeter measures **average power**:

$$W = I_{\text{rms}}^2 R = (3.535)^2 \times 46 = 575\,\text{W}$$

Therefore,

$$\boxed{A_1 = 3.54\,\text{A}, \quad A_2 = 2.50\,\text{A}, \quad V_1 = 325\,\text{V}, \quad V_2 = 103.5\,\text{V}, \quad W = 575\,\text{W}}$$

Q2.98.1 MCQ 1998 1M Ans: C ● ○ ○

Given,

$$S = 1000\,\Omega/\text{V}, \qquad V_{fs} = 100\,\text{V}$$

Total voltmeter resistance:

$$R_v = S \times V_{fs} = 1000 \times 100 = 100000\,\Omega = 100\,\text{k}\Omega$$

The meter reads half of full-scale value, hence applied voltage is:

$$V = \frac{100}{2} = 50\,\text{V}$$

Using Ohm's law,

$$I = \frac{V}{R_v} = \frac{50}{100000} = 0.0005\,\text{A} = 0.5\,\text{mA}$$

Hence, the correct option is

$$\boxed{\textbf{C.}\ 0.5\,\text{mA}}$$

Key Points

- Shortcut:

$$S = 1000\,\Omega/\text{V} \Rightarrow I_{fs} = \frac{1}{1000} = 1\,\text{mA}$$

At half-scale reading:

$$I = 0.5\,\text{mA}$$

Q2.98.2 NAT 1998 5M Ans: -404.8 ● ● ○

Given,

$$V_L = 400\,\text{V}, \qquad \text{phase sequence } RYB$$

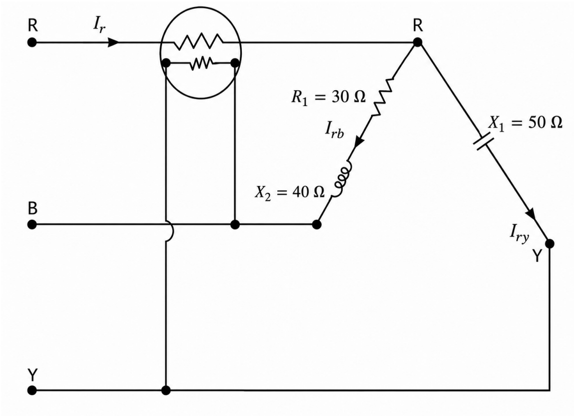

Fig. Solution Q2.98.2

Load impedances:

$$Z_{RB} = 30 + j40 = 50\angle 53.13^\circ, \qquad Z_{RY} = -j50 = 50\angle(-90^\circ)$$

Line voltages:

$$V_{RB} = 400\angle(-30^\circ), \qquad V_{RY} = 400\angle 30^\circ$$

Current through branch RB:

$$I_{RB} = \frac{V_{RB}}{Z_{RB}} = \frac{400\angle(-30^\circ)}{50\angle 53.13^\circ} = 8\angle(-83.13^\circ) = 0.96 - j7.94$$

Current through branch RY:

$$I_{RY} = \frac{V_{RY}}{-j50} = \frac{400\angle 30^\circ}{50\angle(-90^\circ)} = 8\angle 120^\circ = -4 + j6.93$$

Current through wattmeter current coil:

$$I_R = I_{RB} + I_{RY} = (0.96 - j7.94) + (-4 + j6.93) = -3.04 - j1.01 = 3.20\angle(-161.6^\circ)$$

Pressure coil voltage:

$$V_{PC} = V_{RY} = 400\angle 30^\circ$$

Angle between V_{PC} and I_R:

$$\phi = 30^\circ - (-161.6^\circ) = 191.6^\circ$$

Wattmeter reading:

$$W = V_{PC} I_R \cos\phi = 400 \times 3.20 \times \cos 191.6^\circ = -404.8\,\text{W}$$

$$\boxed{-404.8}$$

Key Points

- Negative wattmeter reading indicates:

  Reverse deflection

- Negative sign occurs when current coil and pressure coil polarities produce:

$$\cos\phi < 0$$

Mistakes to be avoided

- During complex number calculations, sign and angle mistakes are very common.

Q2.97.1 MCQ 1997 1M Ans: A ● ○ ○

In a dynamometer type wattmeter:

- Current coil carries load current:

$$i = I_m \sin\omega t$$

- Pressure coil carries current proportional to voltage:

$$v = V_m \sin(\omega t + \phi)$$

Deflecting torque:

$$T_d \propto vi$$

Thus,

$$T_d \propto V_m I_m \sin\omega t \sin(\omega t + \phi)$$

Using:

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

we get:

$$T_d \propto \frac{1}{2} V_m I_m [\cos\phi - \cos(2\omega t + \phi)]$$

This contains:

$$\text{Constant term} \propto VI\cos\phi, \quad \text{which represents active power.}$$
$$\text{Oscillating term} \propto \cos(2\omega t + \phi), \quad \text{whose average over one cycle is zero.}$$

Hence average torque:

$$T_{\text{avg}} \propto VI\cos\phi$$

Therefore, a dynamometer wattmeter measures:

$$\boxed{\text{Average active power}}$$

The correct answer is:

$$\boxed{\textbf{A.}\ \text{average value of active power}}$$

Q2.96.1 MCQ 1996 1M Ans: B ● ○ ○

In a PMMC (Permanent Magnet Moving Coil) instrument, deflecting torque is:

$$T_d = NBIA$$

Strong permanent magnets produce high magnetic flux density(B): Hence large torque is produced even for small current.
Also, the moving system (coil + pointer) is very light.

Therefore:

$$\frac{\text{Torque}}{\text{Weight of moving parts}}$$

is high.
This gives:

$$\text{High sensitivity, quick response and better accuracy}$$

Hence, an important advantage of PMMC instruments is:

$$\boxed{\text{High torque/weight ratio}}$$

Why other options are wrong:

A. Free from friction error ×
PMMC instruments use pivots and bearings, so friction error is reduced but not completely eliminated.

C. Low torque/weight ratio ×
This is opposite to the actual PMMC characteristic.

D. Can be used on both AC and DC ×
PMMC instruments work only on DC because for AC torque reverses every half cycle.

Hence average torque becomes zero.

$$\boxed{\textbf{B.}\ \text{has high (torque/weight of moving parts) ratio}}$$

Q2.96.2 MCQ 1996 1M Ans: C ● ○ ○

A uniform scale means:

$$\text{Equal change in input} \Rightarrow \text{Equal change in deflection}$$

i.e.,

$$\theta \propto X$$

For a PMMC instrument, deflecting torque is:

$$T_d = NBIA$$

Hence,

$$T_d \propto I$$

Controlling torque:

$$T_c = K\theta$$

At equilibrium:

$$T_d = T_c \Rightarrow I \propto \theta \Rightarrow \theta \propto I$$

Thus PMMC instruments have:

$$\boxed{\text{Uniform (linear) scale}}$$

Why other options are wrong:

A. Moving iron ×
For moving iron instruments:

$$T_d \propto I^2 \Rightarrow \theta \propto I^2$$

Hence scale is nonlinear and crowded near zero.

B. Induction ×
Induction instruments are mainly used for energy meters.

D. Dynamometer ×
Scale is generally non-uniform because torque depends on product of currents.

The correct answer is:

$$\boxed{\textbf{C.}\ \text{moving coil permanent magnet}}$$

Q2.96.3 MCQ 1996 1M Ans: B ● ○ ○

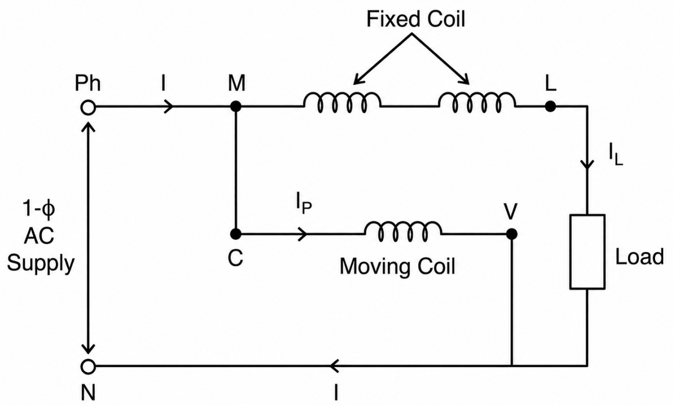

Fig. Solution Q2.96.3

A dynamometer wattmeter has:

- Fixed coil (Current coil, CC)

  Connected in series with the load, hence:

$$I_{CC} = I_{\text{load}}$$

- Moving coil (Pressure/Potential coil, PC)

  Connected across the supply through a high resistance, hence:

$$I_{PC} \propto V$$

The interaction of:

$$\text{Magnetic field due to CC} \propto I$$

and

$$\text{Current in PC} \propto V$$

produces deflecting torque:

$$T_d \propto VI\cos\phi$$

Thus the wattmeter measures power.
Why other options are wrong:

A. In series with fixed coil ×

Then the moving coil would carry load current instead of voltage-proportional current.

C. In series with load $\times$
This is the connection of the current coil.

D. Across the load
Practically load and supply voltages are nearly same, but standard wattmeter terminology states:

$$\boxed{\text{Pressure coil is connected across the supply}}$$

The correct answer is:

$$\boxed{\textbf{B.}\ \text{across the supply}}$$

Key Points

- The two fixed coils are connected in series with the load and carry full load current:

$$I_{CC} = I_L$$

Q2.96.4 MCQ 1996 1M Ans: A ● ○ ○

In an induction type ammeter, deflecting torque is produced due to interaction between:
i) induced eddy currents
ii) alternating magnetic flux
Deflecting torque:

$$T_d \propto \phi_1\phi_2 \sin\delta$$

For a fixed frequency:

$$\delta = \text{constant}$$

Also, magnetic flux is proportional to current:

$$\phi \propto I$$

Therefore,

$$\boxed{T_d \propto I^2}$$

The correct answer is:

$$\boxed{\textbf{A.}\ current^2}$$

Q2.95.1 NAT 1995 1M Ans: 150 ● ○ ○

Given two PMMC voltmeters:
Voltmeter 1:

$$\text{FOM}_1 = 10\,\text{k}\Omega/\text{V}, \qquad V_1 = 100\,\text{V}$$

Hence,

$$R_1 = 100 \times 10\,\text{k}\Omega = 1000\,\text{k}\Omega = 1\,\text{M}\Omega$$

Voltmeter 2:

$$\text{FOM}_2 = 20\,\text{k}\Omega/\text{V}, \qquad V_2 = 100\,\text{V}$$

Hence,

$$R_2 = 100 \times 20\,\text{k}\Omega = 2000\,\text{k}\Omega = 2\,\text{M}\Omega$$

When connected in series,

$$R_T = R_1 + R_2 = 1 + 2 = 3\,\text{M}\Omega$$

For a PMMC voltmeter,

$$I_{fs} = \frac{1}{\text{FOM}}$$

So,

$$I_{fs1} = \frac{1}{10\,\text{k}\Omega/\text{V}} = 100\,\mu\text{A}$$

$$I_{fs2} = \frac{1}{20\,\text{k}\Omega/\text{V}} = 50\,\mu\text{A}$$

Since the series current is the same, the lower full-scale current limits the combination:

$$I_{\max} = 50\,\mu\text{A}$$

Therefore, maximum measurable voltage is

$$V_{\max} = I_{\max} R_T = 50 \times 10^{-6} \times 3 \times 10^{6} = 150\,\text{V}$$

$$\boxed{150}$$

Q2.94.1 MCQ 1994 1M Ans: D ● ○ ○

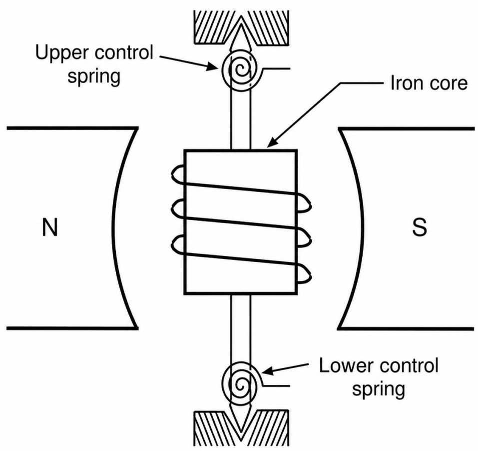

Fig. Solution Q2.94.1

A PMMC instrument has two control springs which:
i) provide controlling torque
ii) carry current to the moving coil
Current path:

$$\text{Terminal} \rightarrow \text{Top spring} \rightarrow \text{Moving coil} \rightarrow \text{Bottom spring} \rightarrow \text{Terminal}$$

If the bottom spring snaps, the moving coil circuit becomes open.
Hence,

$$I_{\text{coil}} = 0$$

Therefore deflecting torque:

$$T_d = NBIA$$

becomes:

$$T_d = 0$$

Thus no torque acts on the pointer, and it returns to zero position.
Hence meter reading becomes:

$$\boxed{0\,\text{mA}}$$

Therefore,

$$\boxed{\textbf{D.}\ \text{zero}}$$

Q2.93.1 MCQ 1993 1M Ans: C ● ○ ○

Given,

$$GF = 2, \qquad R = 120\,\Omega, \qquad \epsilon = 10^{-5}$$

Gauge factor relation:

$$GF = \frac{\Delta R/R}{\epsilon}$$

Hence,

$$\Delta R = GF \times R \times \epsilon$$

Substituting,

$$\Delta R = 2 \times 120 \times 10^{-5}$$

$$= 240 \times 10^{-5} = 2.4 \times 10^{-3}\,\Omega$$

Therefore,

$$\boxed{\Delta R = 2.4 \times 10^{-3}\,\Omega}$$

Hence, the correct option is

$$\boxed{\textbf{C.}\ \Delta R = 2.4 \times 10^{-3}\,\Omega}$$

Key Points

- Gauge factor (GF) of a strain gauge is defined as:

$$GF = \frac{\Delta R/R}{\epsilon}$$

- It represents:

$$\text{Fractional change in resistance per unit strain}$$

- Higher gauge factor $\Rightarrow$ Higher sensitivity

Q2.92.1 MCQ 1992 1M Ans: D ● ○ ○

A moving iron (MI) instrument works on:

$$T_d \propto \phi^2$$

where:

$$\phi = \text{magnetic flux produced by the coil current}$$

Since torque depends on ϕ^2
MI instruments can operate on both AC and DC.
Now suppose a stray DC magnetic field B_s appears along the meter axis.
This stray field either adds to or opposes the instrument magnetic field B_i.

Hence total magnetic field becomes:

$$B_{\text{total}} = B_i \pm B_s$$

Therefore torque becomes:

$$T_d \propto (B_i \pm B_s)^2$$

Expanding,

$$(B_i \pm B_s)^2 = B_i^2 + B_s^2 \pm 2B_iB_s$$

If the stray field aids the instrument field:

$$(B_i + B_s)^2$$

torque increases, hence meter reading increases.
If the stray field opposes the instrument field:

$$(B_i - B_s)^2$$

torque decreases, hence meter reading decreases.
Hence,

$$\boxed{\textbf{D.}\text{ either decreased or increased depending on the di- rection of the DC field}}$$

Q2.91.1 NAT 1991 1M Ans: 85 ● ○ ○

Given,

$$R = 10\,\Omega$$

The current waveform is:

$$12\text{ A for } T, \qquad 5\text{ A for next } T$$

and repeats every

$$2T$$

A DC voltmeter measures the **average** voltage across the resistor.
Average current:

$$I_{\text{avg}} = \frac{12T + 5T}{2T} = 8.5\text{ A}$$

Hence average voltage:

$$V_{\text{avg}} = I_{\text{avg}}R = 8.5 \times 10 = 85\text{ V}$$

$$\boxed{85\text{ V}}$$

DEBUG INFO

Chapter II

Analog Measuring Instruments

Total Questions : 65

MCQ : 49

MSQ : 0

NAT : 16

SOLUTIONS

DC And AC Bridges

Topics

1. Wheatstone Bridge
2. AC Bridges: Maxwell, Hay, Schering, DeSauty, Wien Bridge for measuring Inductance and Capacitance

Q3.26.1 NAT 2026 2M Ans: 750 ● ○ ○

Given,

$$P = 2000\,\Omega, \qquad Q = 500\,\Omega, \qquad R = 3000\,\Omega$$

Battery voltage:

$$E = 50\,\text{V}$$

Battery internal resistance:

$$r = 1\,\Omega$$

Galvanometer resistance:

$$G = 50\,\Omega$$

For a balanced Wheatstone bridge,

$$\frac{Q}{P} = \frac{S}{R}$$

Hence,

$$\frac{500}{2000} = \frac{S}{3000} \Rightarrow 2000S = 500 \times 3000 \Rightarrow S = 750\,\Omega$$

At balance, no current flows through the galvanometer, so G has no effect. The battery internal resistance r also does not affect the balance condition.

$$\boxed{750}$$

Q3.25.1 NAT 2025 2M Ans: 1.96 ● ● ○

Given,

$$R = 50\,\Omega, \qquad P = 10\,\Omega, \qquad Q = 1\,\text{k}\Omega, \qquad S = 5\,\text{k}\Omega, \qquad V_s = 10\,\text{V}$$

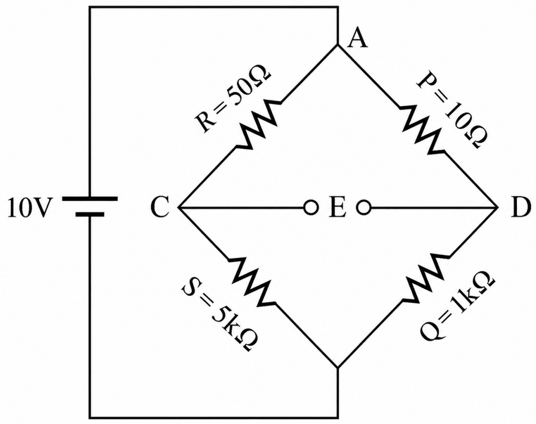

Fig. Solution Q3.25.1

The output voltage across the galvanometer terminals is

$$E = V_C - V_D$$

where

$$V_C = 10\frac{S}{R+S}, \qquad V_D = 10\frac{Q}{P+Q}$$

Initially,

$$E = 10\left(\frac{5000}{5050} - \frac{1000}{1010}\right) \approx 0$$

so the bridge is balanced.
Now increase R from $50\,\Omega$ to $51\,\Omega$:

$$V_C' = 10\frac{5000}{5051}, \qquad V_D' = 10\frac{1000}{1010}$$

Hence,

$$E' = 10\left(\frac{5000}{5051} - \frac{1000}{1010}\right) \approx -0.00196\,\text{V}$$

So,

$$|E'| = 1.96\,\text{mV}$$

Since $\Delta R = 1\,\Omega$,

$$\frac{\Delta E}{\Delta R} = \frac{1.96}{1} = 1.96\,\text{mV}/\Omega$$

$$\boxed{1.96}$$

Key Points

- There is a unit change in resistance: so the resistance can become:

$$50 \rightarrow 51\,\Omega \qquad or \qquad 50 \rightarrow 49\,\Omega$$

- The magnitude of sensitivity remains same in both cases, only the sign of output voltage changes.
- Hence both answers $+1.96\,\text{mV}/\Omega$ and $-1.96\,\text{mV}/\Omega$ are valid according to the final answer key.

Q3.21.1 MCQ 2021 1M Ans: B ● ○ ○

Different AC bridges are used for measurement of different electrical quantities:

Bridge	**Measures**
Schering bridge	Capacitance and dielectric loss
Maxwell bridge	Inductance
Kelvin bridge	Very low resistance
Wien bridge	Frequency

Hence, the bridge used for measurement of inductance is:

$$\boxed{\textbf{B.}\ \text{Maxwell bridge}}$$

Q3.15.1 MCQ 2015 2M Ans: A ● ● ○

Given circuit,

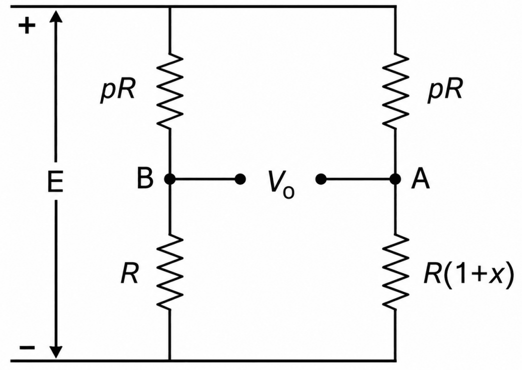

Fig. Solution Q3.15.1

Need to maximize:

$$|V_0| = |V_A - V_B|$$

Supply voltage = E

Step 1: Find node voltage V_A

Using voltage division in the right branch:

Upper resistance = pR

Lower resistance = R(1+x)

Hence,

$$V_A = E\frac{R(1+x)}{pR + R(1+x)} = E\frac{1+x}{p+1+x}$$

Step 2: Find node voltage V_B

Similarly for the left branch:

Upper resistance = pR

Lower resistance = R

Thus,

$$V_B = E\frac{R}{pR+R} = E\frac{1}{p+1}$$

Step 3: Find bridge output voltage

$$V_0 = V_A - V_B$$

Substituting,

$$V_0 = E\left(\frac{1+x}{p+1+x} - \frac{1}{p+1}\right)$$

Taking LCM,

$$V_0 = E\frac{(1+x)(p+1) - (p+1+x)}{(p+1)(p+1+x)}$$

Simplifying numerator,

$$(1+x)(p+1) - (p+1+x) = px$$

Hence,

$$V_0 = E\frac{px}{(p+1)(p+1+x)}$$

Since E and x are constants, maximize:

$$f(p) = \frac{p}{(p+1)(p+1+x)}$$

Step 4: Differentiate

$$f(p) = \frac{p}{p^2 + p(2+x) + (1+x)}$$

Using quotient rule,

$$f'(p) = \frac{[p^2 + p(2+x) + (1+x)] - p(2p+2+x)}{[p^2 + p(2+x) + (1+x)]^2}$$

For maximum value:

$$f'(p) = 0$$

Thus,

$$p^2 + p(2+x) + (1+x) = p(2p+2+x)$$

Simplifying,

$$1 + x = p^2$$

Therefore,

$$\boxed{\textbf{A. } \sqrt{1+x}}$$

Mistakes to be avoided

- Do not forget that only positive value is physically meaningful:

$$p = \sqrt{1+x}$$

Q3.14.1 | NAT | 2014 | 2M | Ans: 0.3 | ● ○ ○

At balance,

$$\frac{Z_2}{Z_1} = \frac{Z_4}{Z_3}$$

Here,

$$Z_1 = R_1 = 35\,\text{k}\Omega, \qquad Z_2 = R_2 = 105\,\text{k}\Omega$$
$$Z_3 = \frac{1}{j\omega C_1}, \qquad Z_4 = \frac{1}{j\omega C_2}$$

Hence,

$$\frac{R_2}{R_1} = \frac{Z_4}{Z_3} = \frac{C_1}{C_2}$$

So,

$$\frac{105}{35} = \frac{C_1}{0.1} \Rightarrow 3 = \frac{C_1}{0.1} \Rightarrow C_1 = 0.3\,\mu\text{F}$$

$$\boxed{0.3}$$

Q3.14.2 | NAT | 2014 | 2M | Ans: 141.4 | ● ● ●

Given source:

$$v(t) = 100 \sin \omega t$$

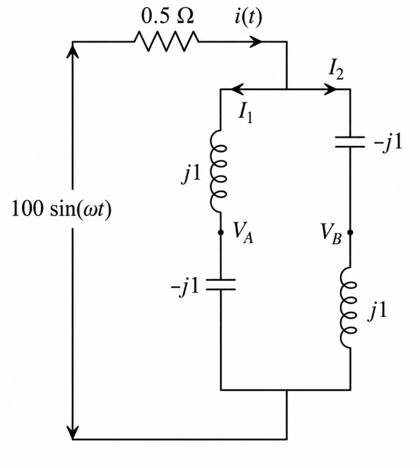

Fig. Solution Q3.14.2

Hence RMS source voltage:

$$V_s = \frac{100}{\sqrt{2}} = 70.71\,\text{V}$$

Series resistance:

$$R = 0.5\,\Omega$$

The bridge has two branches:

$$Z_1 = j1 + (-j1) = 0, \qquad Z_2 = -j1 + j1 = 0$$

Thus the bridge network behaves as a short circuit, so the current through the circuit is very large, but the voltmeter measures the potential difference between the midpoints of the two branches.
Taking current in each branch:

$$I_1 = I_2 = \frac{100/2}{0.5\sqrt{2}} = 70.71\,\text{A}$$

Voltage drop across the inductor:

$$V_L = I_b(j1) = j70.71$$

Voltage drop across the capacitor:

$$V_C = I_b(-j1) = -j70.71$$

So midpoint potentials are

$$V_A = -j70.71, \qquad V_B = +j70.71$$

Hence voltmeter voltage:

$$V_V = V_A - V_B = -j70.71 - j70.71 = -j141.42$$

Therefore,

$$|V_V| = 141.42\,\text{V}$$

$$\boxed{141.4}$$

Key Points

- Even though:

$$\frac{-j}{j} = \frac{j}{-j}$$

the bridge does not give zero output.

- The midpoint voltages are:

$$V_A = -j70.7, \qquad V_B = +j70.7$$

- These voltages have equal magnitude but are 180° out of phase
- Hence, $V_A \neq V_B$
- The bridge arms contain purely imaginary impedances with opposite signs, causing opposite phasor voltages.
- The balance condition of Wheatstone/AC bridge assumes finite impedances in all four arms and zero current through detector.

Q3.13.1 MCQ 2013 2M Ans: C ● ● ○

Given,

$$R_s = 300\,\Omega, \qquad R_1 = R_2 = R_3 = 300\,\Omega$$

Maximum permissible current through strain gauge:

$$I_{\max} = 20\,\text{mA}$$

Resistance increases by 1%
Hence,

$$R_s' = 300(1 + 0.01) = 303\,\Omega$$

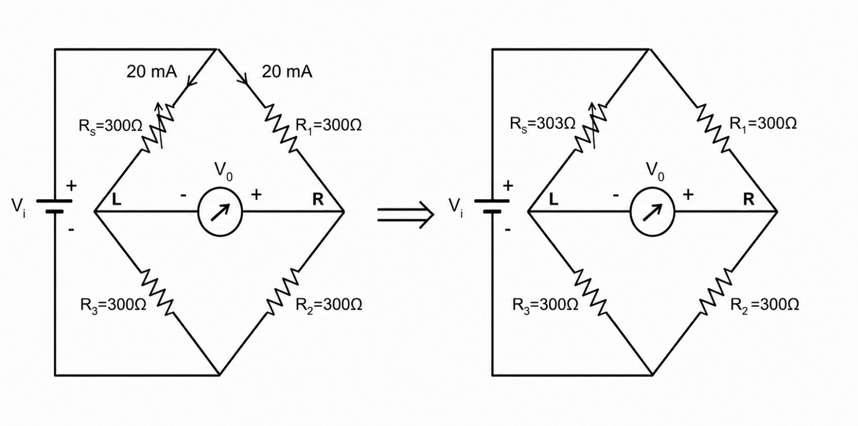

Fig. Solution Q3.13.1

Step 1: Find excitation voltage
Initially the bridge is balanced, so all resistances are:

$$300\,\Omega$$

Current through strain gauge arm:

$$I = \frac{V_i}{300 + 300} = \frac{V_i}{600}$$

Given:

$$I = 20\,\text{mA} = 0.02\,\text{A}$$

Thus,

$$0.02 = \frac{V_i}{600} \Rightarrow V_i = 12\,\text{V}$$

Step 2: Find midpoint voltages after strain

Left midpoint voltage:

$$V_L = 12 \times \frac{300}{303 + 300} = 12 \times \frac{300}{603} = 5.970\,\text{V}$$

Right midpoint voltage:

$$V_R = 12 \times \frac{300}{300 + 300} = 6\,\text{V}$$

Step 3: Bridge output voltage

$$V_0 = V_R - V_L = 6 - 5.970 = 0.02985\,\text{V}$$

Converting to mV:

$$V_0 = 29.85\,\text{mV}$$

$$\boxed{\textbf{C.}\ 29.85\,\text{mV}}$$

Mistakes to be avoided

- Carefully see the voltmeter connection and identify where the positive node and negative node are.

Q3.12.1

MCQ | 2012 | 1M | Ans: A

Different AC bridges are used for measurement of different electrical quantities:

Bridge	Used for / Measures
Heaviside-Campbell bridge	Mutual inductance
Schering bridge	Capacitance and dielectric loss
De Sauty bridge	Comparison of lossless capacitances
Wien bridge	Frequency measurement

Hence, the bridge used for measurement of mutual inductance is:

$$\boxed{\textbf{A. Heaviside-Campbell bridge}}$$

Q3.11.1

MCQ | 2011 | 1M | Ans: C

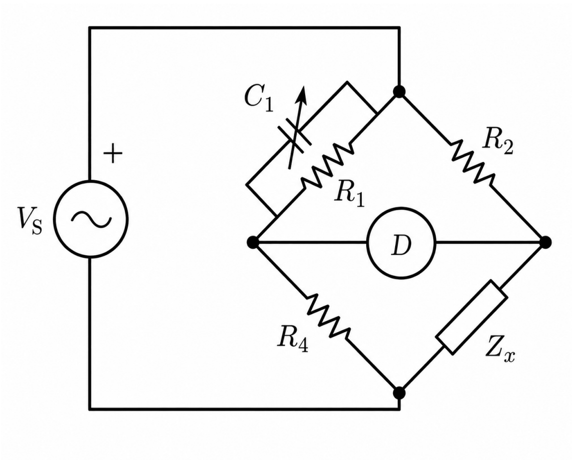

Fig. Solution Q3.11.1

The given circuit is a **Maxwell bridge** (Maxwell inductance-capacitance bridge), identified by:

Variable capacitor, C_1
in parallel with, R_1
Resistive ratio arms, R_2, R_4
Unknown impedance, Z_x

A Maxwell bridge is used for measurement of:

$$\text{Inductance of low-Q coils}$$

It is most suitable for:

$$1 < Q < 10$$

Hence:

$$\boxed{\textbf{C.}\ \text{low-Q inductor}}$$

Q3.10.1 MCQ 2010 2M Ans: A ● ● ○

This is a **Maxwell inductance-capacitance bridge**.

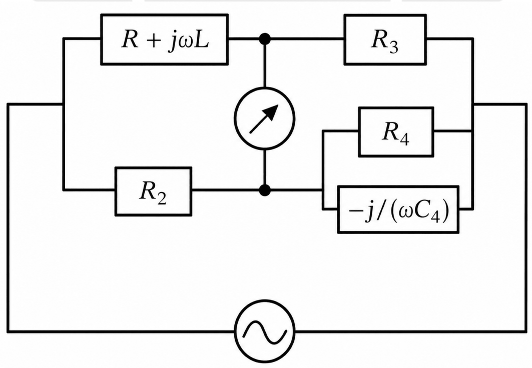

Fig. Solution Q3.10.1

Unknown inductive arm:

$$Z_1 = R + j\omega L$$

Other arms:

$$Z_2 = R_2, \qquad Z_3 = R_3, \qquad Z_4 = R_4 \parallel \frac{1}{j\omega C_4}$$

For balance,

$$\frac{Z_2}{Z_1} = \frac{Z_4}{Z_3}$$

Now,

$$Z_4 = \frac{R_4}{1 + j\omega C_4 R_4}$$

Hence,

$$\frac{R_2}{R + j\omega L} = \frac{R_4}{R_3} \cdot \frac{1}{1 + j\omega C_4 R_4}$$

So,

$$R + j\omega L = \frac{R_2 R_3}{R_4}(1 + j\omega C_4 R_4)$$

Comparing real and imaginary parts,

$$R = \frac{R_2 R_3}{R_4}$$

and

$$L = C_4 R_2 R_3$$

Hence, the correct option is

$$\boxed{\textbf{A. } R = \frac{R_2 R_3}{R_4}, \qquad L = C_4 R_2 R_3}$$

Key Points

- Maxwell bridge uses a:

  Standard capacitor instead of a standard inductor

- This is advantageous because, Precision capacitors are cheaper and more stable and High-quality standard inductors are difficult to manufacture.

Q3.08.1 MCQ 2008 1M Ans: A

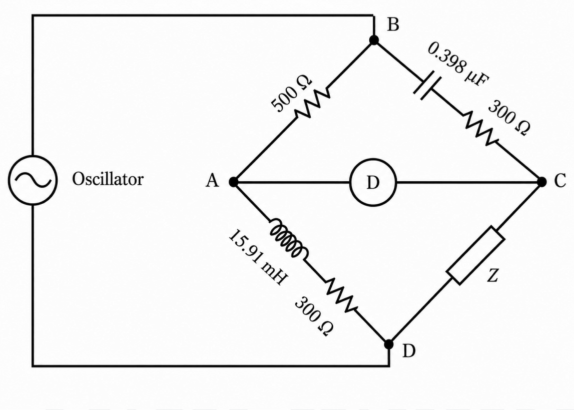

Fig. Solution Q3.08.1

Given bridge balance condition,

$$\frac{Z_{AB}}{Z_{AD}} = \frac{Z_{BC}}{Z_{CD}} \Rightarrow Z_{CD} = \frac{Z_{AB} Z_{BC}}{Z_{AD}}$$

Now,

$$Z_{AB} = 500\,\Omega$$

$$Z_{AD} = 300 + j\omega L = 300 + jX_L$$

with

$$L = 15.91\,\text{mH}, \qquad f = 2\,\text{kHz}$$

so

$$\omega = 2\pi f = 4000\pi, \qquad X_L = \omega L \approx 4000\pi(15.91 \times 10^{-3}) \approx 200\,\Omega$$

hence

$$Z_{AD} = 300 + j200$$

Also,

$$Z_{BC} = 300 - \frac{j}{\omega C} = 300 - jX_C$$

so

$$X_C = \frac{1}{\omega C} \approx \frac{1}{4000\pi(0.398\times10^{-6})} \approx 200\,\Omega$$

hence

$$Z_{BC} = 300 - j200$$

Therefore,

$$Z_{CD} = \frac{500(300 - j200)}{300 + j200}$$

Multiplying by the conjugate,

$$Z_{CD} = 500\frac{(300 - j200)^2}{300^2 + 200^2}$$

Since the product is purely real,

$$Z_{CD} = 260 + j0$$

$$\boxed{\textbf{A.}\ (260 + j0)\,\Omega}$$

Q3.07.1 MCQ 2007 1M Ans: A ● ● ○

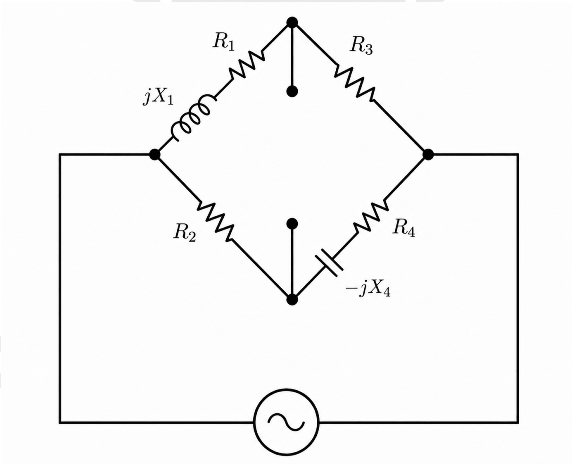

Fig. Solution Q3.07.1

The bridge arms are:

$$Z_1 = R_1 + jX_1, \qquad Z_2 = R_2, \qquad Z_3 = R_3, \qquad Z_4 = R_4 - jX_4$$

At balance,

$$Z_1 Z_4 = Z_2 Z_3$$

So,

$$(R_1 + jX_1)(R_4 - jX_4) = R_2 R_3$$

Expanding,

$$R_1R_4 + X_1X_4 + j(X_1R_4 - R_1X_4) = R_2R_3$$

Equating real and imaginary parts,

$$R_1R_4 + X_1X_4 = R_2R_3, \qquad X_1R_4 = R_1X_4$$

The imaginary equation must be satisfied first, so the phase balance is set by adjusting R_4. After that, R_1 is

adjusted to obtain final balance.
The solution where any one or both of R_1 or R_4 is/are 0 is also possible as the branch will not get shorted due to inductor and capacitor.
Hence, the correct sequence is:

$$\boxed{\text{First adjust } R_4,\ \text{then adjust } R_1}$$

Why others are wrong

B. First adjust R_2, then adjust R_3 ×
Both appear only in the real-balance term R_2R_3. They cannot satisfy

$$X_1R_4 = R_1X_4$$

so phase balance is ignored.

C. First adjust R_2, then adjust R_4 ×
The branch resistance R_2 should not be zero while satisfying the real part equation/

D. First adjust R_4, then adjust R_2 ×
The first step is correct, but the second step the branch resistance R_2 should not be zero while satisfying the real part equation.
Therefore, the correct option is

$$\boxed{\textbf{A.}\text{ First adjust } R_4,\ \text{then adjust } R1}$$

Q3.03.1 MCQ 2003 2M Ans: A ● ○ ○

Different bridges are used for measurement of different electrical quantities:

Group	Measurement	Correct Bridge
P	Resistance in milliohm range	Kelvin double bridge
Q	Low values of capacitance	Schering bridge
R	Comparison of nearly equal resistances	Carey-Foster bridge
S	Inductance of high-Q coil	Hay's bridge

Therefore, the correct option is

$$\boxed{\textbf{A.}\ P-2,\ Q-3,\ R-6,\ S-5}$$

DEBUG INFO

Chapter III

DC And AC Bridges

Total Questions : 13

MCQ : 9

MSQ : 0

NAT : 4

SOLUTIONS

IV

Measurement of Resistance

Topics

1. Kelvin Double Bridge
2. V-A, A-V method
3. Loss of Charge Method
4. Ohmmeter, Potentiometer

Q4.02.1 MCQ | 2002 | 1M | Ans: A

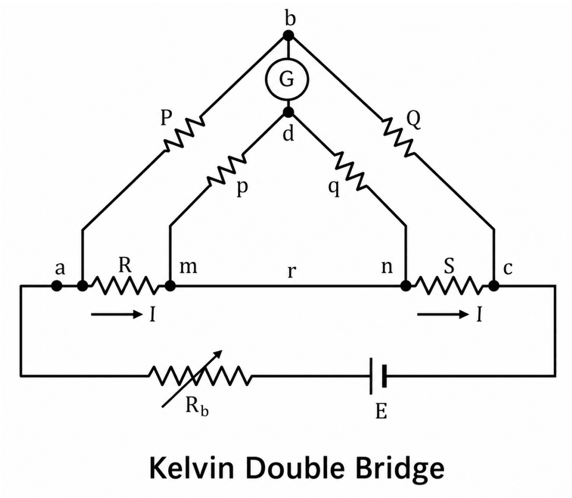

Fig. Solution Q4.02.1

Kelvin double bridge is a modified Wheatstone bridge used for measuring:

$$\boxed{\text{Very low resistances}}$$

typically in:

$$\text{milliohm } (m\Omega) \quad \text{and} \quad \text{micro-ohm } (\mu\Omega) \text{ range}$$

It eliminates errors caused by Lead resistance and Contact resistance which become significant while measuring very low resistances.
Therefore,

$$\boxed{\textbf{A. Resistance of very low value}}$$

Q4.97.1 MCQ | 1997 | 1M | Ans: B

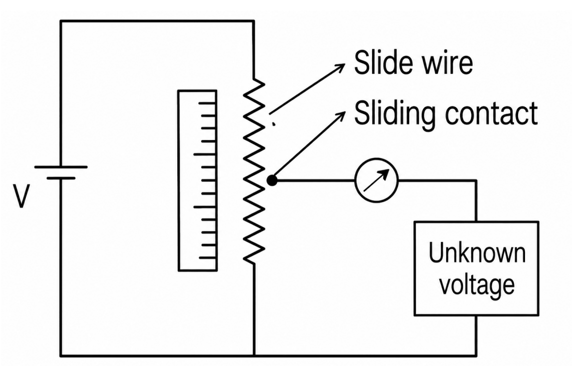

Fig. Solution Q4.97.1

A potentiometer works on the **null balance principle**.
It measures an unknown voltage by comparing it with a known voltage and adjusting until:

$$\text{Unknown voltage} = \text{Known voltage}$$

At balance:

$$I_G = 0$$

where I_G is galvanometer current.
Since measurement is obtained under zero-current condition, a potentiometer is called a:
Deflection instruments depend on pointer movement proportional to the measured quantity. Potentiometer works at zero deflection.
A potentiometer may be analog or digital; its defining feature is the null method.
Therefore,

$$\boxed{\textbf{B. Null type instrument}}$$

Q4.97.2 MCQ 1997 1M Ans: B

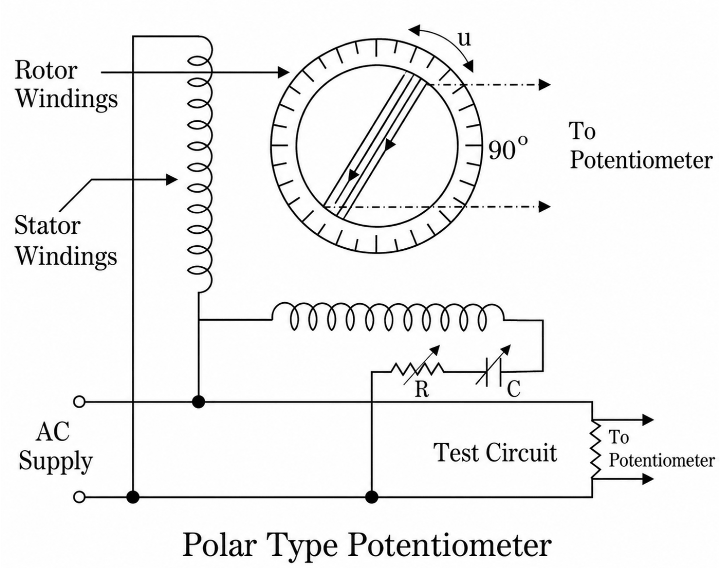

Fig. Solution Q4.97.2

A polar type AC potentiometer is standardized using a **transfer instrument**.
A transfer instrument should:
(i) work accurately on both AC and DC.
(ii) give the same indication for equal RMS values of AC and DC.
Thermal instruments satisfy these conditions because they operate on

$$\text{Heating effect} \propto I^2 R,$$

which is independent of waveform.

Property	**Electrostatic**	**Thermal**	**Dynamometer**	**PMMC**
Works on AC	✓	✓	✓	×
Works on DC	✓	✓	✓	✓
Measures true RMS	×	✓	Approx.	×
Depends on waveform	Yes	No	Yes	–
Frequency effect	Present	Negligible	Present	–
Phase error	No	No	Yes	–
Suitable for AC–DC transfer	×	✓ Best	Possible	×
Principle	Electrostatic force	I^2R heating	EM torque	Magnetic interaction
Main use	High voltage	RMS	Power measurement	DC measurement
Preferred for AC potentiometer standardization	×	✓	×	×

Therefore,

$\boxed{\textbf{B.}\text{ a thermal instrument}}$

Key Points

Feature	**Polar Type AC Potentiometer**	**Normal DC Potentiometer**
Measured quantity	AC magnitude + phase angle	DC magnitude only
Standardization	Requires AC & DC standardization	Only DC standardization
Balancing condition	Magnitude + phase balance	Only magnitude balance
Detector used	AC detector / vibration galvanometer	DC galvanometer
Application	AC impedance, phase, pf measurement	DC emf, resistance, current

Q4.94.1

MCQ | 1994 | 2M | Ans: B | ● ○ ○

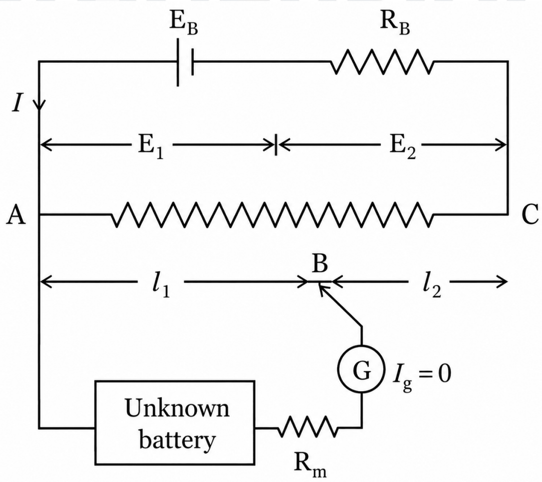

Fig. Solution Q4.94.1

For a potentiometer operating under null balance condition:

$$\frac{E_2}{E_1} = \frac{l_2}{l_1}$$

where:

$$E_1 = \text{known (standard) cell emf}, \qquad E_2 = \text{unknown/test cell emf}$$

$$l_1 = \text{balancing length corresponding to } E_1$$

$$l_2 = \text{balancing length corresponding to } E_2$$

Given:

$$E_1 = 1.18\,\text{V}, \qquad l_1 = 600\,\text{mm}, \qquad l_2 = 680\,\text{mm}$$

Substituting,

$$\frac{E_2}{1.18} = \frac{680}{600}$$

Hence,

$$E_2 = 1.18 \times \frac{680}{600}$$

$$= 1.18 \times 1.1333 = 1.337\,\text{V}$$

Therefore,

$$E_2 \approx 1.34\,\text{V}$$

Hence, the correct option is:

$$\boxed{\textbf{B. } 1.34\,\text{V}}$$

Mistakes to be avoided

- Do not use the nominal potentiometer range directly for calculation.
- Think of it this way:

$$2\,\text{V over } 800\,\text{mm} \Rightarrow \text{instrument specification} \qquad 1.18\,\text{V at } 600\,\text{mm} \Rightarrow \text{actual calibrated condition}$$

- During measurement, the calibrated standard cell condition overrides the nominal range.
- Hence use:

$$\boxed{\frac{1.18}{600} \text{ and not } \frac{2}{800}}$$

Q4.92.1 MCQ 1992 1M Ans: C ● ○ ○

In a DC potentiometer, unwanted thermal emf (thermoelectric emf) may be generated at junctions of dissimilar metals due to temperature differences.

Let:

$$E_x = \text{unknown emf}, \qquad E_t = \text{stray thermal emf}$$

Initially,

$$E_1 = E_x + E_t$$

After reversing the polarities of both the battery and galvanometer connections,

$$E_2 = E_x - E_t$$

Taking the average,

$$\frac{E_1 + E_2}{2} = \frac{(E_x + E_t) + (E_x - E_t)}{2} = \frac{2E_x}{2} = E_x$$

Thus,

$$\boxed{\text{Thermal emf cancels out}}$$

Therefore, the correct option is:

$\boxed{\textbf{C.}\text{ Stray thermal emf's}}$

DEBUG INFO

Chapter IV

Measurement of Resistance

Total Questions : 5

MCQ : 5

MSQ : 0

NAT : 0

SOLUTIONS

V

Electronic Measurement and Instrument Transformer

Topics

1. CRO
2. Current Transformer CT, Potential Transformer PT
3. Digital Voltmeter - DVM
4. Q-Meter

Q5.26.1 MCQ 2026 2M Ans: D

Given,

$$I_A = 572.812 + j50.115, \qquad I_B = -254.525 - j459.175, \qquad I_C = -207.083 + j444.091$$

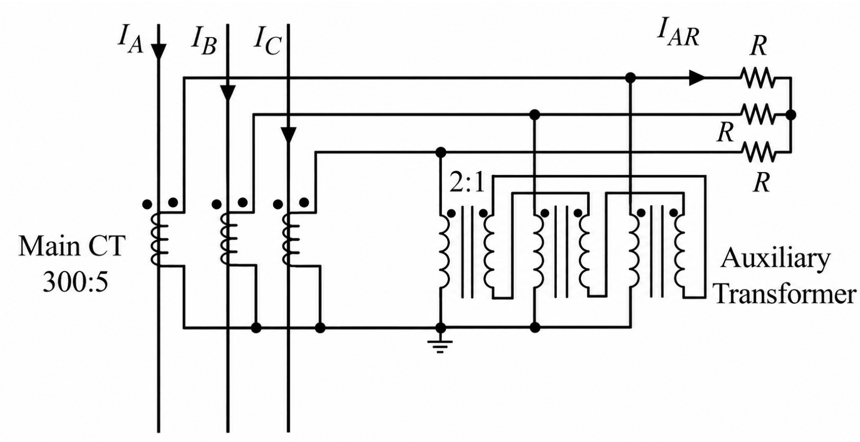

Fig. Solution Q5.26.1

Main CT ratio:

$$300 : 5 \Rightarrow \text{secondary current} = \frac{1}{60} \times \text{primary current}$$

Hence, the **secondary currents of Main CT** are:

$$I_a = \frac{I_A}{60} = 9.583\angle 5^\circ, \qquad I_b = \frac{I_B}{60} = 8.75\angle(-119^\circ), \qquad I_c = \frac{I_C}{60} = 8.167\angle 115^\circ$$

Zero-sequence current:

$$I_{a0} = \frac{I_a + I_b + I_c}{3}$$

Now,

$$I_a + I_b + I_c = 1.943\angle 16.01^\circ$$

So,

$$I_{a0} = 0.648\angle 16.01^\circ \approx 0.65\angle 16^\circ$$

For the auxiliary transformer,

$$I_{AR} = I_a - I_{a0}$$

Thus,

$$I_{AR} = 9.583\angle 5^\circ - 0.65\angle 16^\circ \approx 8.954\angle 4.105^\circ \text{ A}$$

$$\boxed{\textbf{D. } 8.954\angle 4.105^\circ \text{ A}}$$

Key Points

- Zero-sequence current:

$$I_0 = \frac{I_a + I_b + I_c}{3}$$

- For a $Yn\Delta$ auxiliary transformer, the delta side blocks zero-sequence current.
- In symmetrical components:

$$I_a = I_{a1} + I_{a2} + I_{a0}$$

so the final delta-side current is:

$$I_{AR} = I_{a1} + I_{a2}$$

Mistakes to be avoided

- Do not assume

$$I_A + I_B + I_C = 0$$

unless the system is perfectly balanced.

Q5.24.1 MCQ 2024 1M Ans: C ● ○ ○

Comparison between CT and PT:

Feature	CT	PT	Reason
Primary connected in parallel to grid	×	✓	PT measures voltage
Open-circuited secondary not desirable	✓	×	CT secondary should never be open
Primary current is line current	✓	×	CT primary carries actual line current
Secondary burden affects primary current	×	✓	PT behaves like a voltage transformer

Hence, the correct option is:

C. X matches with Q and R; Y matches with P and S.

Key Points

- CT secondary should never be open-circuited.

$$I_s = 0 \Rightarrow \phi \uparrow \Rightarrow E_s \uparrow$$

where:

$$I_s = \text{secondary current}, \quad \phi = \text{core flux}, \quad E_s = \text{secondary induced emf}$$

- Hence dangerously high voltage appears across CT secondary.
- PT behaves like a normal transformer.
- Increasing secondary burden increases:

$$I_s \uparrow \Rightarrow I_p \uparrow$$

where:

$$I_p = \text{primary current}$$

- Therefore PT primary current depends on secondary loading.

Q5.17.1 MCQ 2017 1M Ans: D ● ○ ○

For a stationary closed Lissajous pattern,

$$\frac{f_h}{f_v} = \frac{\text{Number of vertical tangencies}}{\text{Number of horizontal tangencies}}$$

Given,

$$N_H = 3, \qquad N_V = 2, \qquad f_h = 3\,\text{kHz}$$

Thus,

$$\frac{3}{f_v} = \frac{2}{3}$$

Hence,

$$f_v = 3 \times \frac{3}{2} = 4.5\,\text{kHz}$$

Therefore,

$$\boxed{\textbf{D.}\ 4.5\,\text{kHz}}$$

Q5.17.2 MCQ 2017 1M Ans: A ● ○ ○

In a CRO:

Delay due to slope and level detector:

$$100\,\text{ns}$$

Delay due to start-stop sweep generator:

$$50\,\text{ns}$$

Hence total delay before horizontal sweep starts:

$$100 + 50 = 150\,\text{ns}$$

Therefore the input signal in the Y-channel must be delayed by the same amount so that the initial portion of the waveform is not lost.

Thus the required delay line is:

$$150\,\text{ns}$$

inserted in:

$$\boxed{\text{Y-channel}}$$

Therefore, the correct option is:

$$\boxed{\textbf{A.}\ 150\,\text{ns inserted in Y-channel}}$$

Key Points

- Delay line in CRO delays the Y-signal so that the X-sweep gets enough time to start.
- This ensures the leading edge (initial portion) of the waveform becomes visible on the screen.

Q5.14.1 MCQ 2014 1M Ans: A ● ○ ○

Given, S_1 (triangular wave) is applied to the Y-plates and S_2 (square wave) is applied to the X-plates.

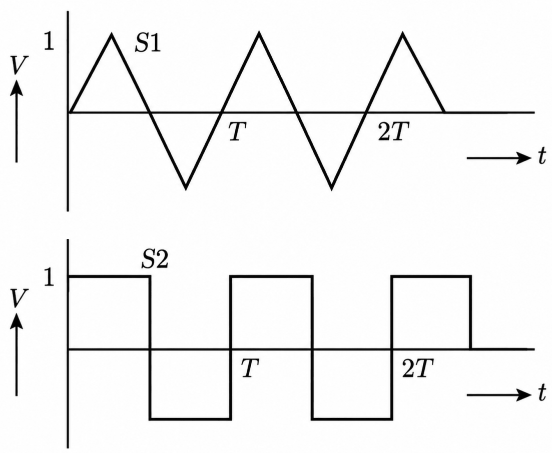

Fig. Solution Q5.14.1

Thus,

$$X = S_2, \qquad Y = S_1$$

The beam position over one cycle is:

Time	X from S_2	Y from S_1	Coordinate
$t = 0$	$+1$	0	$(+1, 0)$
$t = \frac{T}{4}$	$+1$	$+1$	$(+1, +1)$
$t = \frac{T}{2}^-$	$+1$	0	$(+1, 0)$
$t = \frac{T}{2}^+$	-1	0	$(-1, 0)$
$t = \frac{3T}{4}$	-1	-1	$(-1, -1)$
$t = T^-$	-1	0	$(-1, 0)$
$t = T^+$	$+1$	0	$(+1, 0)$

So the beam moves vertically upward at $X = +1$, returns to $(+1, 0)$, jumps horizontally to $(-1, 0)$, moves vertically downward, and then jumps back.

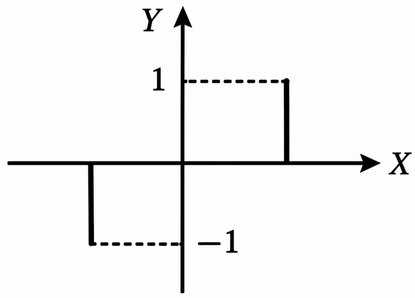

Fig. Solution Q5.14.1

This pattern matches **Option A**.

$\boxed{\text{A}}$

Mistakes to be avoided

- Follow waveform progression with time; wrong time ordering gives incorrect CRO pattern and lead answer to Option D.

Q5.14.2 MCQ 2014 1M Ans: B ● ○ ○

In a CRO/oscilloscope, the linear sweep (time-base signal) is applied to the horizontal deflection plates.

The purpose of the sweep voltage is to move the electron beam uniformly from left to right across the screen.

Thus,

$$V_x \propto t$$

Hence the horizontal direction represents time axis

$\boxed{\textbf{B. horizontal axis}}$

Key Points

- Time-base sweep $\longrightarrow$ X-plates (horizontal)
- Input signal $\longrightarrow$ Y-plates (vertical)
- Therefore CRO displays:

$$\text{Voltage vs Time}$$

Q5.11.1 MCQ | 2011 | 2M | Ans: C | ● ● ○

Given:

Accuracy specification:

$$\pm(0.2\% \text{ of reading} + 10 \text{ counts})$$

Reading = 100 V
Full-scale range = 200 V
DMM = $4\frac{1}{2}$-digit

Method 1: Using Resolution of DMM

Error due to percentage of reading:

$$0.2\% \text{ of } 100\,\text{V} = \frac{0.2}{100} \times 100 = 0.2\,\text{V}$$

A $4\frac{1}{2}$-digit DMM on 200 V range generally displays:

$$199.99\,\text{V}$$

Hence resolution (least count):

$$0.01\,\text{V}$$

Therefore,

$$10 \text{ counts} = 10 \times 0.01 = 0.1\,\text{V}$$

Thus total error:

$$\pm(0.2 + 0.1) = \pm 0.3\,\text{V}$$

Percentage error:

$$\frac{0.3}{100} \times 100 = 0.3\%$$

Method 2: Using Maximum Count

For a $4\frac{1}{2}$-digit DMM:

$$\text{Maximum count} = 19999$$

Full-scale range:

$$200\,\text{V}$$

Hence,

$$1 \text{ count} = \frac{200}{19999} \approx 0.01\,\text{V}$$

Therefore,

$$10 \text{ counts} = 10 \times \frac{200}{19999} \approx 0.100005\,\text{V} \approx 0.1\,\text{V}$$

Percentage error component:

$$0.2\% \text{ of } 100\,\text{V} = 0.2\,\text{V}$$

Thus total error:

$$\pm(0.1 + 0.2) = \pm 0.3\ \text{V}$$

Percentage error:

$$\frac{0.3}{100} \times 100 = 0.3\%$$

Therefore,

$$\boxed{\textbf{C.}\ \pm 0.3\%}$$

Key Points

- A $4\frac{1}{2}$-digit DMM means:

$$\text{First digit: } 0 \text{ or } 1, \qquad \text{remaining digits: } 0 \text{ to } 9$$

- Common DMM counts:

DMM Type	Maximum Count
$3\frac{1}{2}$-digit	1999
$4\frac{1}{2}$-digit	19999
$5\frac{1}{2}$-digit	199999

Q5.11.2 MCQ 2011 1M Ans: C ● ○ ○

In a dual-trace CRO, both input channels share the same vertical amplifier through an electronic switch (multiplexer). In **ALTERNATE** mode:

- Channel 1 is displayed during one sweep.
- Channel 2 is displayed during the next sweep.

Hence the multiplexer must switch channels once every sweep cycle.

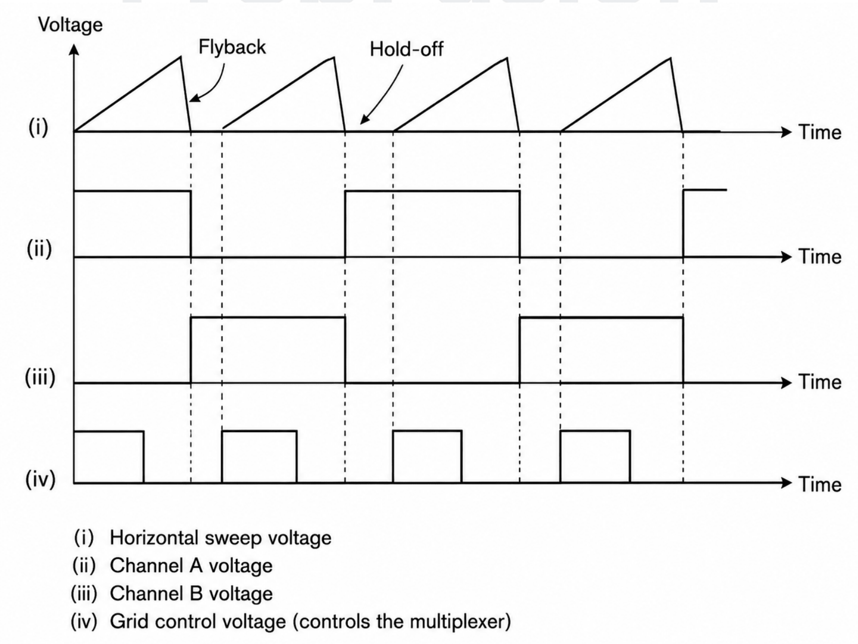

Fig. Solution Q5.11.2

The figure represents the waveform timing in a dual-trace oscilloscope operating in ALTERNATE mode. It shows how the oscilloscope alternately displays Channel A and Channel B during successive sweeps. Therefore, the switching frequency must be equal to the sweep frequency.
Therefore, the correct option is:

C. the frequency of the time-base (sweep) oscillator

Key Points

- In **Alternate mode**, channel switching occurs once per sweep.
- In **Chop mode**, channel switching occurs at very high frequency independent of sweep
- Alternate mode is preferred for high-frequency signals.
- Chop mode is preferred for low-frequency signals.

Q5.09.1 | MCQ | 2009 | 2M | Ans: A | ● ○ ○

Given:
The DMM is an average-reading type and indicates 10 V
For a symmetric triangular wave:

$$V_{avg} = \frac{V_m}{2}$$

Since the average-reading meter reads 10 V,

$$\frac{V_m}{2} = 10$$

Hence,

$$V_m = 20\,\text{V}$$

For a triangular waveform:

$$V_{rms} = \frac{V_m}{\sqrt{3}}$$

Substituting $V_m = 20\,\text{V}$,

$$V_{rms} = \frac{20}{\sqrt{3}}\,\text{V}$$

Therefore, the correct option is:

$$\boxed{\textbf{A. } \frac{20}{\sqrt{3}}\,\text{V}}$$

Q5.09.2 | MCQ | 2009 | 1M | Ans: D | ● ○ ○

In the X–Y mode of a CRO:

- one signal is applied to the X-plates
- the other signal is applied to the Y-plates

If both signals are sinusoidal and have the same frequency, a Lissajous figure is obtained.
For:

$$x = A\sin\omega t, \qquad y = B\sin(\omega t + \phi)$$

the shape depends on the phase difference ϕ.

Phase Difference	Pattern
$\phi = 0°$	Straight line
$0° < \phi < 90°$	Tilted ellipse
$\phi = 90°$	Circle (if amplitudes equal)
$90° < \phi < 180°$	Ellipse with opposite tilt

The figure changes:

$$\text{ellipse} \rightarrow \text{circle} \rightarrow \text{ellipse}$$

with changing orientation, indicating a varying phase difference between two sinusoidal signals of the same frequency.
Hence the correct option is:

D. There is a constant but small phase difference between the signals

Key Points

- A stationary Lissajous figure implies:

$$f_1 = f_2$$

If $f_1 \neq f_2$, the pattern continuously drifts and becomes non-stationary.

Q5.08.1 MCQ 2008 2M Ans: D ● ● ○

Given,

$$p(\omega_1 t) = A \sin \omega_1 t$$

applied to the X-input and the unknown signal $q(\omega_2 t)$ applied to the Y-input.
The CRO is in X–Y mode and shows a Lissajous figure.

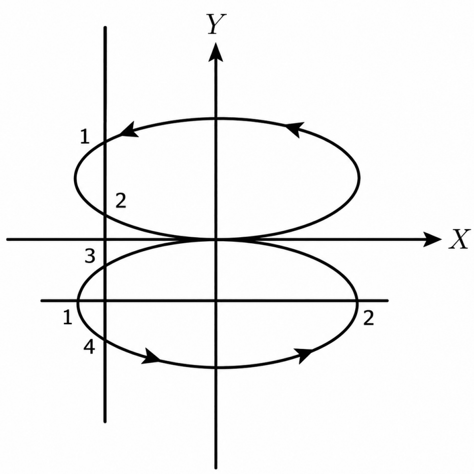

Fig. Solution Q5.08.1

For Lissajous figures,

$$\frac{f_x}{f_y} = \frac{\text{Number of tangencies to vertical sides}}{\text{Number of tangencies to horizontal sides}}$$

From the figure, the horizontal lines cut the curve at 2 points and vertical line at 4 points. Hence,

$$\frac{f_x}{f_y} = \frac{4}{2} \Rightarrow f_y = \frac{f_x}{2}$$

Therefore,

$$\omega_2 = \frac{\omega_1}{2}$$

The figure is symmetric about both axes and corresponds to a signal in quadrature phase. Since a cosine wave is a sine wave shifted by 90°,

$$q(\omega_2 t) = A\cos\omega_2 t$$

Therefore, the correct option is

$$\boxed{\textbf{D. } q(\omega_2 t) = A\cos\omega_2 t, \qquad \omega_2 = \frac{\omega_1}{2}}$$

Mistakes to be avoided

- Draw vertical and horizontal line(to find number of tangencies) at the place where you will get maximum points of intersection.
- Frequency ratio in Lissajous figures is inverse of what many students intuitively assume.
- Symmetry alone is not sufficient; also check loop orientation and number of lobes.

Q5.07.1

MCQ | 2007 | 1M | Ans: B | ● ● ○

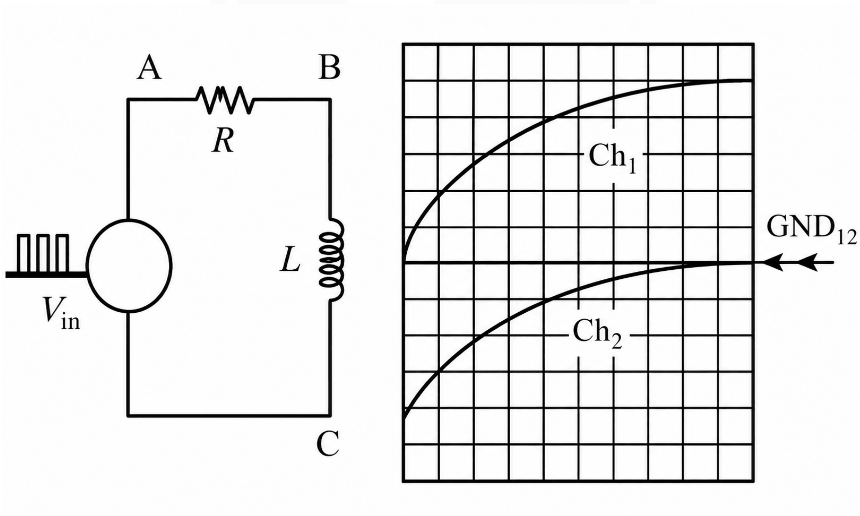

Fig. Solution Q5.07.1

The circuit is a series RL circuit driven by a square wave input.
A non-isolated CRO has both channel grounds internally connected, so:

$$G_1 \equiv G_2$$

Hence both ground clips must be connected to the same circuit point. From the given waveforms, the common reference point is B. Therefore,

$$G_1 = G_2 = B$$

Now, channel 1 waveform starts from zero and rises exponentially, which is the voltage across the resistor in an RL transient. Hence the signal probe for Ch1 must be at A, so that

$$V_{AB} = V_R$$

Channel 2 waveform starts from high (in negative) and decays exponentially to zero, which is the voltage across the inductor. The ground is already at B, is the reason for getting negative value. The signal probe for Ch2 must be at C, so that

$$V_{BC} = V_L$$

Therefore the correct connection is:

$$\boxed{\textbf{B. } A,\ B,\ C,\ B}$$

Q5.06.1 MCQ 2006 2M Ans: A ● ● ○

Given:

$$K_n = \frac{200}{1} = 200, \qquad I_P = 160\,\text{A}$$

Initial ratio error:

$$e_{r1} = -0.5\%$$

Initial phase angle error:

$$\delta_1 = 30'$$

Initial secondary turns:

$$N_{s1} = 200$$

After reducing one turn:

$$N_{s2} = 199$$

Let the actual transformation ratio be K_a.
For a current transformer,

$$K_a = n\left(1 + \frac{I_e}{I_P}\right)$$

where n is the turns ratio and I_e is the excitation current component.

Step 1: Find initial actual ratio

Using the ratio error relation,

$$e_{r1} = \frac{K_n - K_{a1}}{K_{a1}} \times 100$$

So,

$$-0.5 = \frac{200 - K_{a1}}{K_{a1}} \times 100 \Rightarrow K_{a1} = 201$$

Step 2: Find new actual ratio after reducing one turn

Since actual ratio is proportional to secondary turns,

$$K_{a2} = K_{a1} \cdot \frac{N_{s2}}{N_{s1}} = 201 \cdot \frac{199}{200} \approx 200$$

Step 3: Find new ratio error

$$e_{r2} = \frac{K_n - K_{a2}}{K_{a2}} \times 100 = \frac{200 - 200}{200} \times 100 = 0\%$$

Step 4: Phase angle error

A small change of one secondary turn mainly corrects the ratio error, while the phase angle error remains practically unchanged.

Hence,

$$\delta_2 \approx 30'$$

Therefore,

$$\boxed{\textbf{A.}\ 0.0,\ 30'}$$

Key Points

- Reducing secondary turns decreases turns ratio and is used for ratio-error compensation.
- Small reduction in turns usually has negligible effect on phase angle error.

Mistakes to be avoided

- Do not ignore sign convention of ratio error.

Q5.06.2 MCQ 2006 1M Ans: A ● ● ○

A sampling wattmeter computes:

$$P_{\text{avg}} = \frac{1}{T}\int_0^T v(t)i(t)\,dt$$

Given square-wave voltage,

$$V_{pp} = 10\,\text{V} \Rightarrow V_m = 5\,\text{V}$$

So,

$$v(t) = \begin{cases} -5, & 0 < t < \dfrac{T}{2} \\ +5, & \dfrac{T}{2} < t < T \end{cases}$$

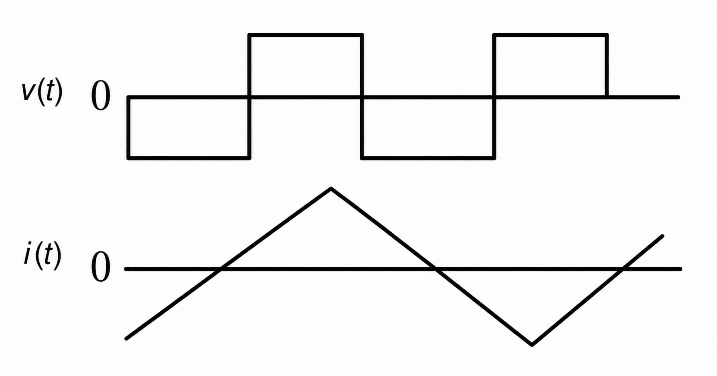

Fig. Solution Q5.06.2

Hence,

$$P_{\text{avg}} = \frac{1}{T}\left[\int_0^{T/2} -5\,i(t)\,dt + \int_{T/2}^{T} (5)\,i(t)\,dt\right]$$

$$P_{\text{avg}} = \frac{5}{T}\left[\int_0^{T/2} -i(t)\,dt + \int_{T/2}^{T} i(t)\,dt\right]$$

From the given triangular waveform, the positive area during the first half cycle equals the negative area during the second half cycle, so

$$\int_0^{T/2} i(t)\,dt = \int_{T/2}^{T} i(t)\,dt$$

Therefore,

$$P_{\text{avg}} = \frac{5}{T}(A - A) = 0$$

$$\boxed{\textbf{A. } 0\,\text{W}}$$

Mistakes to be avoided

- Be careful with signs and simplification of expression.

Q5.06.3 MCQ 2006 1M Ans: C ● ○ ○

Given calibration signal:

$$f = 1\,\text{kHz} \Rightarrow T = \frac{1}{f} = 1\,\text{ms}$$

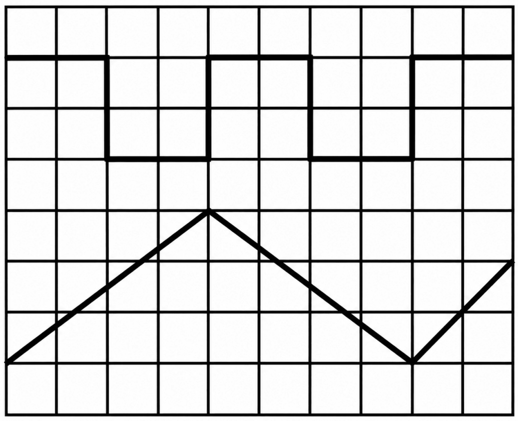

Fig. Solution Q5.06.3

From the upper square-wave trace, one complete cycle occupies 4 horizontal divisions. Hence,

$$\text{Time/div} = \frac{1\,\text{ms}}{4} = 0.25\,\text{ms/div}$$

The calibration pulse amplitude is:

$$V_{pp} = 5\,\text{V}$$

From the figure, this spans 2 vertical divisions. Hence,

$$\text{Voltage/div} = \frac{5}{2} = 2.5\,\text{V/div}$$

For the lower triangular waveform, peak-to-peak height is 3 vertical divisions. So,

$$V_{pp} = 3 \times 2.5 = 7.5\,\text{V}$$

One cycle of the triangular wave occupies 8 horizontal divisions. Hence,

$$T = 8 \times 0.25 = 2\,\text{ms}$$

Therefore,

$$\boxed{\textbf{C. } 7.5\,\text{V},\ 2\,\text{ms}}$$

Q5.05.1 MCQ 2005 2M Ans: B ● ● ○

Given,

$$x(t) = P\sin(4t + 30^\circ)$$

So the angular frequency of X-input is

$$\omega_x = 4\,\text{rad/s}$$

For a stationary Lissajous figure,

$$\frac{f_x}{f_y} = \frac{\text{number of tangencies on vertical sides}}{\text{number of tangencies on horizontal sides}}$$

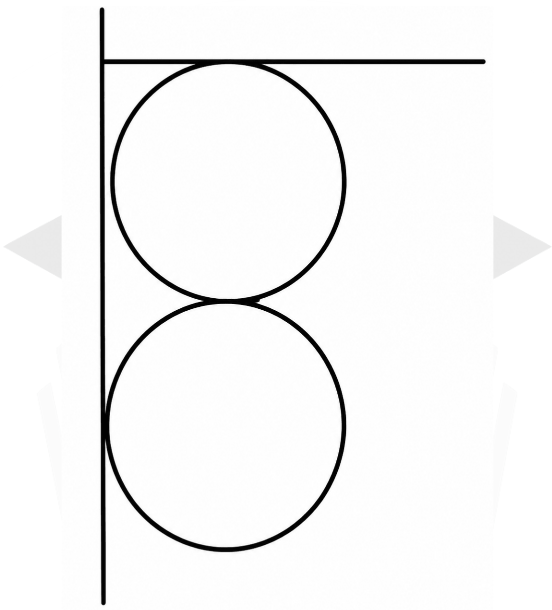

Fig. Solution Q5.05.1

From the given figure-8 pattern,

$$\frac{f_x}{f_y} = \frac{4}{2}$$

Hence,

$$f_y = \frac{f_x}{2} \Rightarrow \omega_y = \frac{\omega_x}{2} = \frac{4}{2} = 2\,\text{rad/s}$$

So the Y-input must have angular frequency 2 rad/s, and from the given options the matching signal is:

$$y(t) = Q\sin(2t + 15^\circ)$$

$$\boxed{\textbf{B. } Q\sin(2t + 15^\circ)}$$

Mistakes to be avoided

- Proper imagination of "a vertical figure-of-8 display" is important.

Q5.05.2 MCQ 2005 1M Ans: C ● ○ ○

A Q-meter is used to measure:

$$\text{Quality factor } (Q)$$

of coils and resonant circuits.
It operates on the principle of **series resonance**.
At resonance:

$$X_L = X_C$$

Hence the circuit impedance becomes minimum and current becomes maximum.
The quality factor is given by:

$$Q = \frac{V_C}{V}$$

where:

$$V_C = \text{voltage across capacitor}, \qquad V = \text{applied voltage}$$

Thus the Q-meter measures voltage magnification in a series resonant circuit.
Therefore, the correct option is:

$$\boxed{\textbf{C. Series resonance}}$$

Key Points

- Q-meter measurements become inaccurate for:

$$\text{very low-Q coils, highly lossy inductors, and coils with large distributed capacitance}$$

Q5.04.1 MCQ 2004 2M Ans: A ● ● ○

Given,

$$f = 50\,\text{Hz}, \qquad N_s = 500, \qquad I_s = 5\,\text{A}, \qquad R_b = 1\,\Omega, \qquad AT_m = 200$$

Since the burden is purely resistive, the secondary current is in phase with the secondary voltage and the magnetizing ampere-turns are perpendicular to the secondary ampere-turns.
Secondary ampere-turns:

$$AT_s = N_s I_s = 500 \times 5 = 2500$$

For a current transformer,

$$\tan\theta = \frac{AT_m}{AT_s}$$

Hence,

$$\tan\theta = \frac{200}{2500} = 0.08 \Rightarrow \theta = \tan^{-1}(0.08) \approx 4.6^\circ$$

Therefore, the correct option is

$$\boxed{\textbf{A.}\ 4.6^\circ}$$

Key Points

- For CT :

$$\tan\theta = \frac{\text{vertical component}}{\text{horizontal component}}$$

where,

$$I_c = \text{core-loss component}, \qquad I_m = \text{magnetizing component}$$

- Therefore,

$$\tan\theta = \frac{I_m}{nI_s + I_c}$$

- For small angle and pure resistive load, we can prove

$$\tan\theta \approx \frac{I_m}{nI_s}$$

Q5.04.2 MCQ 2004 2M Ans: B ● ○ ○

Given,

$$f = 50\,\text{Hz}, \qquad N_s = 500, \qquad I_s = 5\,\text{A}, \qquad R = 1\,\Omega$$

Secondary voltage:

$$V_s = I_s R = 5 \times 1 = 5\,\text{V}$$

For sinusoidal flux, transformer emf equation is:

$$E_{\text{rms}} = 4.44 f N \phi_m$$

Neglecting leakage and the load is purely resistive, So;

$$E_s \approx V_s$$

Therefore,

$$5 = 4.44 \times 50 \times 500 \times \phi_m$$

Hence,

$$\phi_m = \frac{5}{4.44 \times 50 \times 500}$$

$$\phi_m = 4.5 \times 10^{-5}\,\text{Wb}$$

$$\boxed{\textbf{B. } 45\,\mu\text{Wb}}$$

Key Points

- In a CT, secondary voltage is small and since the secondary turns are large,

$$E = 4.44 f N \phi_m$$

even a small voltage requires only a very small flux.

Q5.04.3 MCQ 2004 2M Ans: B ● ● ○

Given:

$$V_s = 10\angle 0^\circ\,\text{V}, \qquad f = 100\,\text{kHz}$$

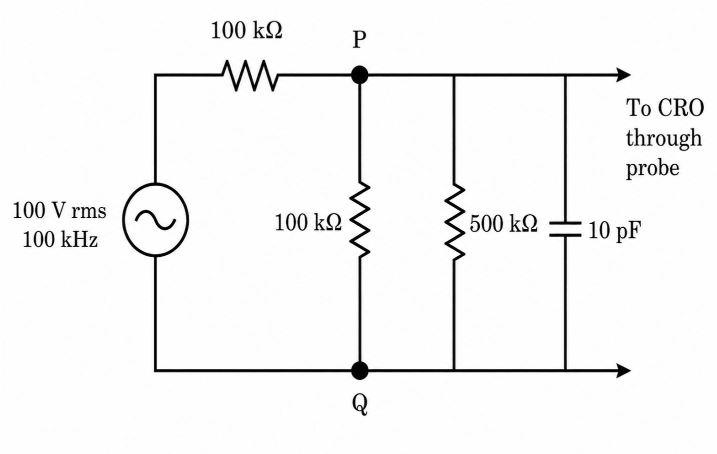

Fig. Solution Q5.04.3

Probe impedance:

$$500\,k\Omega \parallel 10\,pF$$

Capacitive reactance of probe:

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 100 \times 10^3 \times 10 \times 10^{-12}} = 159.15\,k\Omega$$

Applying KCL at node P,

$$\frac{V_P - 10}{100 \times 10^3} + \frac{V_P}{100 \times 10^3} + \frac{V_P}{500 \times 10^3} + \frac{V_P}{-j159.15 \times 10^3} = 0$$

Multiplying throughout by 10^3,

$$\frac{V_P - 10}{100} + \frac{V_P}{100} + \frac{V_P}{500} + \frac{V_P}{-j159.15} = 0$$

Rearranging,

$$\left(\frac{1}{100} + \frac{1}{100} + \frac{1}{500} + \frac{1}{-j159.15}\right) V_P = \frac{10}{100}$$

Now,

$$\frac{1}{100} = 0.01, \qquad \frac{1}{500} = 0.002, \qquad \frac{1}{-j159.15} = j0.006283$$

Hence,

$$(0.022 + j0.006283)V_p = 0.1$$

Therefore,

$$V_p = \frac{0.1}{0.022 + j0.006283}$$

$$V_p = 4.37\angle - 15.94^\circ\ \text{V}$$

Therefore, the correct option is

$$\boxed{\textbf{B.}\ 4.37\ \text{V}}$$

Mistakes to be avoided

- One can use KVL by simplifying the circuit instead of KCL, however answer will come but calculations might be tough in actual GATE paper, so try to understand circuit and what to apply where.

Q5.03.1 MCQ 2003 1M Ans: D ● ○ ○

Given,

$$V_x = V_{xm}\sin\omega t, \qquad V_y = V_{ym}\sin(\omega t + \phi)$$

applied to the X- and Y-plates of a CRO. The displayed Lissajous pattern depends on the phase difference ϕ.

- Same-frequency sinusoidal signals produce Lissajous figures.
- Shape depends on phase difference.
- $\phi = 0°$ gives a straight line with positive slope.
- $\phi = 180°$ gives a straight line with negative slope.
- $\phi = 90°$ gives a circle/ellipse with clockwise rotation.
- $\phi = 270°$ gives a circle/ellipse with anti-clockwise rotation.
- Intermediate phase differences produce ellipses.

Hence, the correct option is

$$\boxed{\textbf{D. } P{-}1,\ Q{-}5,\ R{-}6,\ S{-}4}$$

Q5.03.2 MCQ 2003 2M Ans: B ● ● ○

Given,

Rated CT:

$$\frac{500\text{ A}}{5\text{ A}}$$

Nominal ratio:

$$K_n = \frac{500}{5} = 100$$

Since it is a bar-primary CT,

$$N_p = 1$$

Hence,

$$N_s = 100$$

Secondary current:

$$I_s = 5\text{ A}$$

Magnetizing ampere-turns:

$$AT_m = 250$$

Since the burden is purely resistive,

$$I_c = 0$$

Magnetizing current referred to the secondary side:

$$I_m = \frac{AT_m}{N_s} = \frac{250}{100} = 2.5\text{ A}$$

I_m is perpendicular to I_s (Since flux lags induced emf by 90°);
The actual primary current referred to the secondary side is:

$$I_p' = \sqrt{I_s^2 + I_m^2}$$

$$I_p' = \sqrt{5^2 + 2.5^2} = \sqrt{31.25} = 5.59\text{ A}$$

Hence the actual transformation ratio is:

$$K_a = \frac{I'_p}{I_s} K_n$$

$$K_a = \frac{5.59}{5} \times 100 = 111.8$$

Therefore, percentage ratio error:

$$e_r = \frac{K_n - K_a}{K_a} \times 100$$

$$e_r = \frac{100 - 111.8}{111.8} \times 100 = -10.56\%$$

$$\boxed{\textbf{B.}\ -10.56}$$

Mistakes to be avoided

- Be careful with the sign of ratio error. Options have both positive and negative values.

Q5.03.3 MCQ 2003 1M Ans: A ● ○ ○

Given,

$$Q = 120, \qquad C_1 = 300\,\text{pF}$$

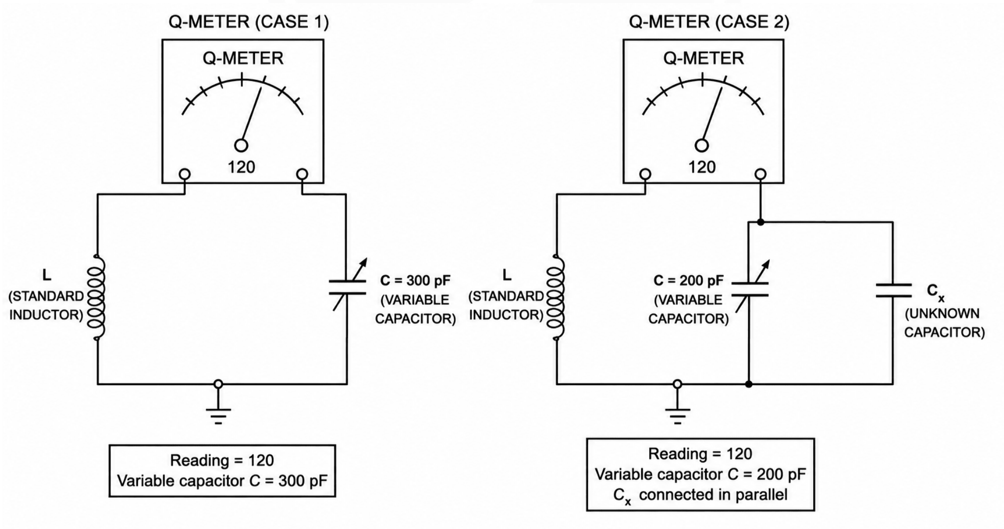

Fig. Solution Q5.03.3

When an unknown **lossless** capacitor C_x is connected in parallel with the variable capacitor, the same Q-meter reading is obtained after readjusting the variable capacitor to

$$C_2 = 200\,\text{pF}$$

Since the Q-meter operates at the same resonant condition, the total resonating capacitance remains unchanged. Initially,

$$C_{\text{total}} = 300\,\text{pF}$$

After connecting C_x,

$$C_{\text{total}} = 200 + C_x$$

Equating the two:

$$200 + C_x = 300$$

Hence,

$$C_x = 100\,\text{pF}$$

$$\boxed{\textbf{A.}\ 100}$$

Q5.02.1 MCQ 2002 1M Ans: D ● ○ ○

Given,

$$V_x = \sin(\omega t), \qquad V_y = \sin(\omega t)$$

Since the two signals are identical and in phase,

$$V_y = V_x$$

In X–Y mode,

$$x = V_x, \qquad y = V_y$$

Hence,

$$y = x$$

This is the equation of a straight line passing through the origin with slope

$$m = 1$$

Therefore,

$$\tan\theta = 1 \Rightarrow \theta = 45^\circ$$

So the trace observed on the CRO screen is a straight line inclined at 45° to the X-axis.

$$\boxed{\textbf{D.}\ \text{A straight line inclined at } 45^\circ \text{ with respect to the x-axis}}$$

Q5.00.1 MCQ 2000 1M Ans: C ● ○ ○

Instrument transformers (CTs and PTs) are not ideal transformers.
They possess:
a) winding resistance
b) leakage reactance
c) magnetizing reactance
d) core-loss resistance
These are represented by the transformer's:

open-circuit parameters and short-circuit parameters

These non-ideal parameters cause:
i) ratio (magnitude) error
ii) phase angle error
Hence, instrument transformer errors arise due to the open-circuit and short-circuit parameters of the transformer.

C. Open and short circuit parameters of the instrument transformers

Q5.99.1 MCQ 1999 2M Ans: A ● ● ○

In a dual-slope ADC:

- The unknown input voltage V_{in} is integrated for a fixed time T_1.
- A known reference voltage V_{ref} of opposite polarity is then applied.
- The time taken for the integrator output to return to zero is the deintegration time T_2.

For a dual-slope ADC,

$$\frac{V_{in}}{V_{ref}} = \frac{T_2}{T_1}$$

Hence,

$$V_{in} = V_{ref}\frac{T_2}{T_1}$$

Given,

$$V_{ref} = 100\,\text{mV}$$

$$T_1 = 300\,\text{ms}, \qquad T_2 = 370.2\,\text{ms}$$

Therefore,

$$V_{in} = 100 \times \frac{370.2}{300}$$

$$V_{in} = 123.4\,\text{mV}$$

A $3\frac{1}{2}$-digit DVM can display values up to 199.9

$$\boxed{\textbf{A. } 123.4}$$

Mistakes to be avoided

- Always compare the calculated value with the DVM/DMM display range. If the computed value exceeds the maximum displayable count, the instrument will indicate over-range (overflow) rather than the calculated value.

Q5.98.2 MCQ 1998 2M Ans: C ● ● ○

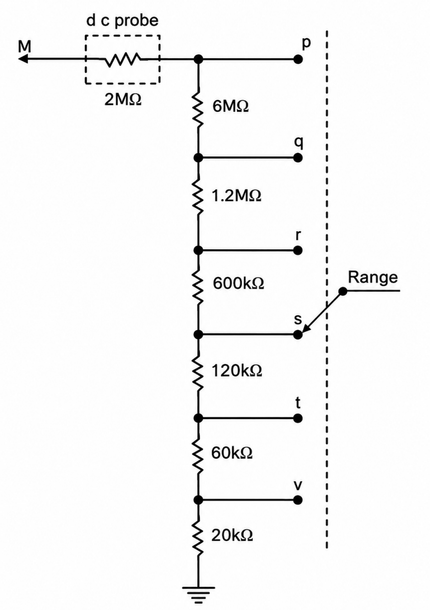

Fig. Solution Q5.98.2

Given,

$$V_M = 12\,\text{V}$$

with the switch at position q.
Total resistance from M to ground:

$$R_{\text{total}} = 2 + 6 + 12 + 0.6 + 0.12 + 0.06 + 0.02 = 10\,\text{M}\Omega$$

For switch position q, resistance from q to ground:

$$R_q = 1.2 + 0.6 + 0.12 + 0.06 + 0.02 = 2\,\text{M}\Omega$$

So the internal meter voltage at full-scale is:

$$V_{\text{meter}} = 12\left(\frac{2}{10}\right) = 2.4\,\text{V}$$

Now for switch position s,

$$R_s = 120\,k\Omega + 60\,k\Omega + 20\,k\Omega = 200\,k\Omega = 0.2\,\text{M}\Omega$$

At full-scale, the meter still requires 2.4 V, so

$$2.4 = V_M\left(\frac{0.2}{10}\right)$$

$$2.4 = 0.02V_M \Rightarrow V_M = \frac{2.4}{0.02} = 120\,\text{V}$$

$$\boxed{\textbf{C.}\ 120\,\text{V}}$$

Mistakes to be avoided

- Be careful with units while adding resistances. Convert all values to a common unit ($M\Omega$, $k\Omega$, etc.) before performing calculations.

Q5.98.2 MCQ 1998 2M Ans: A ● ● ○

For an electrostatic deflection CRT:

- V_A = accelerating voltage
- V_D = deflecting voltage
- d = separation between deflection plates
- L_s = distance from centre of the plates to screen
- L = length of deflection plates
- m = mass of electron

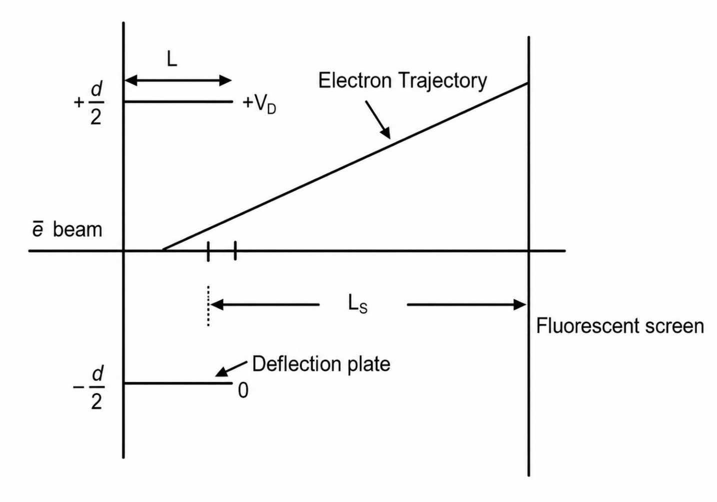

Fig. Solution Q5.98.2

When an electron is accelerated through a voltage V_A,

$$\frac{1}{2}mv_x^2 = eV_A \Rightarrow v_x^2 = \frac{2eV_A}{m}$$

The time spent inside the deflection plates is

$$t = \frac{L}{v_x}$$

The electric field between the plates is

$$E = \frac{V_D}{d}$$

Therefore, the vertical acceleration is

$$a_y = \frac{eE}{m} = \frac{eV_D}{md}$$

The vertical velocity acquired by the electron while traversing the plates is

$$v_y = a_y t = \left(\frac{eV_D}{md}\right)\left(\frac{L}{v_x}\right)$$

Hence, the deflection angle is

$$\tan\theta = \frac{v_y}{v_x} = \frac{eV_D L}{md\, v_x^2}$$

We get,

$$\tan\theta = \frac{V_D L}{2dV_A}$$

For a CRT, the electron trajectory appears to originate from the centre of the deflection plates. Therefore, the screen deflection is

$$D = L_s \tan\theta$$

Substituting for $\tan\theta$,

$$D = L_s\left(\frac{V_D L}{2dV_A}\right) = \frac{LL_sV_D}{2dV_A}$$

Hence, the deflection sensitivity is

$$S = \frac{D}{V_D} = \frac{LL_s}{2dV_A}$$

Therefore,

$$S \propto \frac{LL_s}{dV_A}$$

Hence,

$$\boxed{\textbf{A. } \frac{LL_s}{dV_A}}$$

Q5.96.1 NAT 1996 3M Ans: * ● ● ○

Given $f_x = 100\,\text{Hz}$.

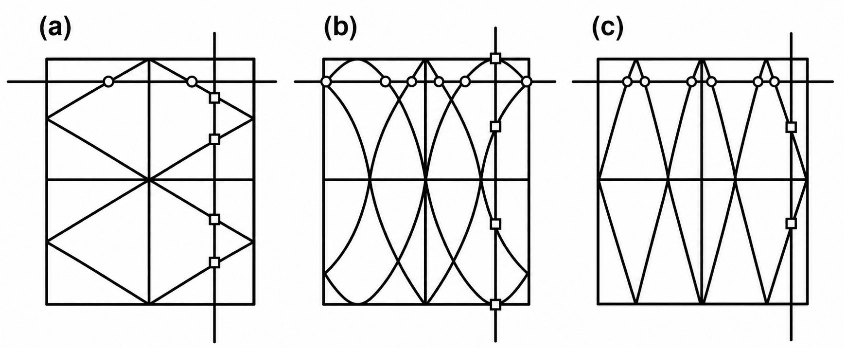

Fig. Solution Q5.96.1

For a stationary Lissajous figure,

$$\frac{f_x}{f_y} = \frac{\text{Number of vertical tangencies}}{\text{Number of horizontal tangencies}}$$

(a) From the figure,

$$N_V = 4, \qquad N_H = 2$$

Hence,

$$\frac{100}{f_y} = \frac{4}{2} = 2 \Rightarrow f_y = 50\,\text{Hz}$$

(b) From the figure,

$$N_V = 4, \qquad N_H = 6$$

Hence,

$$\frac{100}{f_y} = \frac{4}{6} = \frac{2}{3} \Rightarrow f_y = 150\,\text{Hz}$$

(c) From the figure,

$$N_V = 2, \qquad N_H = 6$$

Hence,

$$\frac{100}{f_y} = \frac{2}{6} = \frac{1}{3} \Rightarrow f_y = 300\,\text{Hz}$$

Therefore,

$$\boxed{(a) \rightarrow 50\,\text{Hz (P)}, \quad (b) \rightarrow 150\,\text{Hz (S)}, \quad (c) \rightarrow 300\,\text{Hz (U)}}$$

Q5.95.1 MCQ | 1995 | 1M | Ans: B | ● ○ ○

The sweep generator of a CRO produces a sawtooth (ramp) voltage $V_s(t)$.
Normally, this voltage is applied to the horizontal plates.
If the sweep output is connected to the input channel, then:

$$V_x = V_s(t) \qquad and \qquad V_y = V_s(t)$$

Thus,

$$V_y = V_x$$

Hence,

$$y = x$$

This represents a straight line passing through the origin.
Since the X- and Y-channel sensitivities are equal, the line is inclined at 45°.

$$\boxed{\textbf{B. Diagonal line}}$$

Q5.95.2 MCQ | 1995 | 1M | Ans: A | ● ○ ○

Given,

$$v(t) = 1 + \sin(314t)\ \text{V}$$

Since,

$$314 = 2\pi \times 50$$

the signal can be written as

$$v(t) = 1 + \sin(2\pi 50t)$$

Thus,

$$\text{DC component} = 1\ \text{V}$$

and

$$\text{AC component} = \sin(2\pi 50t)$$

The integration time is:

$$T_{\text{int}} = 300\,\text{ms} \qquad \text{and} \qquad \text{signal frequency, } f = 50\,\text{Hz}$$

Hence the signal period is:

$$T = \frac{1}{50} = 20\,\text{ms}$$

Number of cycles integrated:

$$\frac{300}{20} = 15$$

Thus the ADC integrates over exactly 15 complete cycles.
A dual-slope ADC measures the average value of the input during the integration interval.
The average value of a sine wave over an integral number of cycles is:

$$\frac{1}{T}\int_0^T \sin(\omega t)\, dt = 0$$

Therefore,

$$V_{\text{avg}} = 1 + \text{Average}\big(\sin(314t)\big) = 1 + 0 = 1\ \text{V}$$

$$\boxed{\textbf{A. } 1.000}$$

Key Points

- A dual-slope ADC measures the average value of the input over the integration interval; if the integration time contains an integer number of AC cycles, the average AC component is zero.

Mistakes to be avoided

- Always check whether the integration time contains an integer number of cycles. If it does, the average value of the sinusoidal component is zero; otherwise it must be evaluated explicitly.

Q5.94.1 MCQ | 1994 | 1M | Ans: C | ● ○ ○

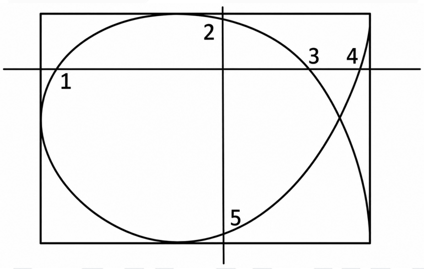

Fig. Solution Q5.94.1

For a **stationary Lissajous figure**,

$$\frac{f_x}{f_y} = \frac{\text{Number of tangencies to vertical sides}}{\text{Number of tangencies to horizontal sides}}$$

From the figure, tangencies to the vertical sides are (2 and 5):

$$N_v = 2$$

Similarly, tangencies to the horizontal sides are (1, 3 and 4):

$$N_h = 3$$

Therefore,

$$\frac{f_x}{f_y} = \frac{N_v}{N_h} = \frac{2}{3}$$

$$\boxed{\textbf{C. } 2:3}$$

Q5.93.1 MCQ 1993 1M Ans: B ● ○ ○

Given,

$$v(t) = 5\sin(314t + 45^\circ)$$

Comparing with

$$v(t) = V_m \sin(\omega t + \phi)$$

we get

$$\omega = 314\,\text{rad/s}$$

Hence,

$$f = \frac{\omega}{2\pi} = \frac{314}{2\pi} \approx 50\,\text{Hz}$$

Therefore,

$$T = \frac{1}{f} = \frac{1}{50} = 20\,\text{ms}$$

The CRO screen has 10 horizontal divisions and the time-base setting is 5 ms/div.
Hence total time displayed on the screen is

$$10 \times 5 = 50\,\text{ms}$$

Number of cycles displayed:

$$N = \frac{\text{Total screen time}}{\text{Time period}} = \frac{50}{20} = 2.5$$

Therefore,

$$\boxed{\textbf{B. } 2.5 \text{ cycles}}$$

Q5.92.1 NAT 1992 2M Ans: 0.3 ● ● ○

Given,

$$f = 50\,\text{Hz}$$

The first integration is carried out for 10 periods.
Hence,

$$T_1 = 10\left(\frac{1}{f}\right) = 10\left(\frac{1}{50}\right) = 0.2\,\text{s}$$

Reference voltage: $V_{\text{ref}} = 2\,\text{V}$ *and* Input voltage: $V_{\text{in}} = 1\,\text{V}$

For a dual-slope ADC,

$$\frac{V_{\text{in}}}{V_{\text{ref}}} = \frac{T_2}{T_1}$$

Therefore,

$$T_2 = T_1 \frac{V_{\text{in}}}{V_{\text{ref}}} = 0.2 \times \frac{1}{2} = 0.1\,\text{s}$$

Hence the total conversion time is

$$T_{\text{conv}} = T_1 + T_2 = 0.2 + 0.1 = 0.3\,\text{s}$$

$$\boxed{0.3}$$

Mistakes to be avoided

- Do not report only the deintegration time T_2; the conversion time is:

$$T_{\text{conv}} = T_1 + T_2$$

DEBUG INFO

Chapter V

Electronic Measurement and Instrument Transformer

Total Questions : 34

MCQ : 32

MSQ : 0

NAT : 2